

VZ class project April 17 2026

```
knitr::opts_chunk$set(warning = FALSE, message = FALSE)

library(rethinking)
library(dplyr)
library(lubridate)

project_dir <- "C:/Users/vezuc/OneDrive - Florida International University/PhD/Classes/Bayesian Statistics/2026 Spring/Week 10/Project 10"
```

Load prepared data

```
track_centers_env <- readRDS(file.path(project_dir, "track_centers_env.rds"))
gps_env <- readRDS(file.path(project_dir, "gps_env.rds"))
```

Bayesian Models

```
### Bayesian track-level model of movement

## set factor levels
month_levels <- if (is.factor(track_centers_env$month)) {
  levels(track_centers_env$month)
} else {
  unique(as.character(track_centers_env$month))
}

year_levels <- sort(unique(c(
  as.character(track_centers_env$year),
  as.character(gps_env$year)
```

```

)))

## Track-level Gaussian models

track_bayes <- track_centers_env %>%
  filter(
    !is.na(center_y),
    !is.na(center_x),
    !is.na(month),
    !is.na(year),
    !is.na(tide_height_m),
    !is.na(moon_illumination)
  ) %>%
  mutate(
    month = factor(as.character(month), levels = month_levels),
    year = factor(as.character(year), levels = year_levels),
    month_id = as.integer(month),
    year_id = as.integer(year),
    tide_s = as.numeric(scale(tide_height_m)),
    moon_s = as.numeric(scale(moon_illumination)),
    center_y_s = as.numeric(scale(center_y)),
    center_x_s = as.numeric(scale(center_x))
  )

d_track_y_scaled <- list(
  N = nrow(track_bayes),
  Y = track_bayes$center_y_s,
  month = track_bayes$month_id,
  tide = track_bayes$tide_s,
  moon = track_bayes$moon_s,
  year = track_bayes$year_id,
  n_month = nlevels(track_bayes$month),
  n_year = nlevels(track_bayes$year)
)

d_track_x_scaled <- list(
  N = nrow(track_bayes),
  Y = track_bayes$center_x_s,
  month = track_bayes$month_id,
  tide = track_bayes$tide_s,
  moon = track_bayes$moon_s,

```

```

year = track_bayes$year_id,
n_month = nlevels(track_bayes$month),
n_year = nlevels(track_bayes$year)
)

m_track_y <- ulam(
  alist(
    Y ~ dnorm(mu, sigma),
    mu <- a + a_month[month] + b_tide * tide + b_moon * moon + a_year[year],

    a ~ dnorm(0, 1),
    a_month[month] ~ dnorm(0, sigma_month),
    a_year[year] ~ dnorm(0, sigma_year),

    b_tide ~ dnorm(0, 1),
    b_moon ~ dnorm(0, 1),

    sigma ~ dexp(1),
    sigma_month ~ dexp(1),
    sigma_year ~ dexp(1)
  ),
  data = d_track_y_scaled,
  chains = 4, cores = 4, iter = 2000
)

```

Running MCMC with 4 parallel chains, with 1 thread(s) per chain...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)

Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 4 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)

```

Chain 2 Iteration: 300 / 2000 [15%] (Warmup)
 Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
 Chain 3 Iteration: 300 / 2000 [15%] (Warmup)
 Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
 Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
 Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
 Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
 Chain 3 Iteration: 400 / 2000 [20%] (Warmup)
 Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
 Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
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 Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 3 Iteration: 500 / 2000 [25%] (Warmup)
 Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 3 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 2 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 3 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 2 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 3 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 2 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 3 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 2 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 2 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 3 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 2 Iteration: 1200 / 2000 [60%] (Sampling)
 Chain 3 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
 Chain 2 Iteration: 1300 / 2000 [65%] (Sampling)
 Chain 3 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 2 Iteration: 1400 / 2000 [70%] (Sampling)

Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
 Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
 Chain 3 Iteration: 1200 / 2000 [60%] (Sampling)
 Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
 Chain 2 Iteration: 1500 / 2000 [75%] (Sampling)
 Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
 Chain 2 Iteration: 1600 / 2000 [80%] (Sampling)
 Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
 Chain 3 Iteration: 1300 / 2000 [65%] (Sampling)
 Chain 4 Iteration: 1500 / 2000 [75%] (Sampling)
 Chain 2 Iteration: 1700 / 2000 [85%] (Sampling)
 Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
 Chain 3 Iteration: 1400 / 2000 [70%] (Sampling)
 Chain 4 Iteration: 1600 / 2000 [80%] (Sampling)
 Chain 2 Iteration: 1800 / 2000 [90%] (Sampling)
 Chain 4 Iteration: 1700 / 2000 [85%] (Sampling)
 Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
 Chain 2 Iteration: 1900 / 2000 [95%] (Sampling)
 Chain 3 Iteration: 1500 / 2000 [75%] (Sampling)
 Chain 4 Iteration: 1800 / 2000 [90%] (Sampling)
 Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
 Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
 Chain 3 Iteration: 1600 / 2000 [80%] (Sampling)
 Chain 2 finished in 3.9 seconds.
 Chain 4 Iteration: 1900 / 2000 [95%] (Sampling)
 Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
 Chain 3 Iteration: 1700 / 2000 [85%] (Sampling)
 Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
 Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
 Chain 4 finished in 4.2 seconds.
 Chain 3 Iteration: 1800 / 2000 [90%] (Sampling)
 Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
 Chain 1 finished in 4.5 seconds.
 Chain 3 Iteration: 1900 / 2000 [95%] (Sampling)
 Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
 Chain 3 finished in 4.8 seconds.

All 4 chains finished successfully.
 Mean chain execution time: 4.3 seconds.
 Total execution time: 5.0 seconds.

```

m_track_x <- ulam(
  alist(
    Y ~ dnorm(mu, sigma),
    mu <- a + a_month[month] + b_tide * tide + b_moon * moon + a_year[year],

    a ~ dnorm(0, 1),
    a_month[month] ~ dnorm(0, sigma_month),
    a_year[year] ~ dnorm(0, sigma_year),

    b_tide ~ dnorm(0, 1),
    b_moon ~ dnorm(0, 1),

    sigma ~ dexp(1),
    sigma_month ~ dexp(1),
    sigma_year ~ dexp(1)
  ),
  data = d_track_x_scaled,
  chains = 4, cores = 4, iter = 2000
)

```

Running MCMC with 4 parallel chains, with 1 thread(s) per chain...

```

Chain 1 Iteration:    1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration:    1 / 2000 [ 0%] (Warmup)
Chain 3 Iteration:    1 / 2000 [ 0%] (Warmup)
Chain 4 Iteration:    1 / 2000 [ 0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration:   100 / 2000 [ 5%] (Warmup)
Chain 3 Iteration:   100 / 2000 [ 5%] (Warmup)
Chain 4 Iteration:   100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration:   200 / 2000 [10%] (Warmup)
Chain 2 Iteration:   200 / 2000 [10%] (Warmup)
Chain 3 Iteration:   200 / 2000 [10%] (Warmup)
Chain 4 Iteration:   200 / 2000 [10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [15%] (Warmup)
Chain 2 Iteration:   300 / 2000 [15%] (Warmup)
Chain 3 Iteration:   300 / 2000 [15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [20%] (Warmup)
Chain 2 Iteration:   400 / 2000 [20%] (Warmup)
Chain 3 Iteration:   400 / 2000 [20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [25%] (Warmup)
Chain 2 Iteration:   500 / 2000 [25%] (Warmup)

```

Chain 3 Iteration: 500 / 2000 [25%] (Warmup)
 Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 3 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 3 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 3 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 3 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 2 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 2 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 3 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 4 Iteration: 300 / 2000 [15%] (Warmup)
 Chain 3 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 3 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 2 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 3 Iteration: 1200 / 2000 [60%] (Sampling)
 Chain 2 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 2 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
 Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 3 Iteration: 1300 / 2000 [65%] (Sampling)
 Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 2 Iteration: 1200 / 2000 [60%] (Sampling)
 Chain 3 Iteration: 1400 / 2000 [70%] (Sampling)
 Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
 Chain 2 Iteration: 1300 / 2000 [65%] (Sampling)
 Chain 2 Iteration: 1400 / 2000 [70%] (Sampling)
 Chain 3 Iteration: 1500 / 2000 [75%] (Sampling)
 Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 2 Iteration: 1500 / 2000 [75%] (Sampling)
 Chain 3 Iteration: 1600 / 2000 [80%] (Sampling)
 Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 2 Iteration: 1600 / 2000 [80%] (Sampling)
 Chain 3 Iteration: 1700 / 2000 [85%] (Sampling)
 Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
 Chain 2 Iteration: 1700 / 2000 [85%] (Sampling)
 Chain 3 Iteration: 1800 / 2000 [90%] (Sampling)
 Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)

```

Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration:  900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2 finished in 3.3 seconds.
Chain 3 finished in 3.2 seconds.
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 finished in 3.9 seconds.
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 4.4 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 3.7 seconds.
 Total execution time: 4.5 seconds.

```
### 2-second GPS cell-based model
```

```

cell_size <- 25

gps_use <- gps_env %>%
  filter(
    !is.na(file_name),
    !is.na(x_true_m),

```

```

    !is.na(y_true_m),
    !is.na(month),
    !is.na(year),
    !is.na(tide_height_m),
    !is.na(moon_illumination)
  ) %>%
  mutate(
    month = factor(as.character(month), levels = month_levels),
    year = factor(as.character(year), levels = year_levels),
    x_cell = floor(x_true_m / cell_size),
    y_cell = floor(y_true_m / cell_size)
  )

# Track effort = total number of 2-sec fixes in each track
track_effort <- gps_use %>%
  count(file_name, name = "n_fixes_track") %>%
  mutate(log_effort = log(n_fixes_track))

# Aggregate number of 2-sec fixes per track x cell
gps_cells <- gps_use %>%
  group_by(
    file_name, year, month,
    tide_height_m, moon_illumination,
    x_cell, y_cell
  ) %>%
  summarise(
    n_fixes_cell = n(),
    .groups = "drop"
  ) %>%
  left_join(track_effort, by = "file_name") %>%
  mutate(
    seconds_in_cell = n_fixes_cell * 2,
    cell_id = as.integer(factor(paste(x_cell, y_cell, sep = "_"))),
    track_id = as.integer(factor(file_name)),
    month_id = as.integer(month),
    year_id = as.integer(year),
    tide_s = as.numeric(scale(tide_height_m)),
    moon_s = as.numeric(scale(moon_illumination))
  )

d_cells_nb <- list(
  N = nrow(gps_cells),

```

```

count = gps_cells$n_fixes_cell,
month = gps_cells$month_id,
year = gps_cells$year_id,
track = gps_cells$track_id,
cell = gps_cells$cell_id,
tide = gps_cells$tide_s,
moon = gps_cells$moon_s,
log_effort = gps_cells$log_effort,
n_month = nlevels(gps_use$month),
n_year = nlevels(gps_use$year),
n_track = length(unique(gps_cells$track_id)),
n_cell = length(unique(gps_cells$cell_id))
)

# Negative binomial (gamma-Poisson) with effort offset and cell effect
m_space_use <- ulam(
  alist(
    count ~ dgamma_pois(lambda, phi),
    log(lambda) <- log_effort +
      a +
      a_month[month] +
      a_year[year] +
      a_track[track] +
      a_cell[cell] +
      b_tide * tide +
      b_moon * moon,

    a ~ dnorm(0, 1),

    a_month[month] ~ dnorm(0, sigma_month),
    a_year[year] ~ dnorm(0, sigma_year),
    a_track[track] ~ dnorm(0, sigma_track),
    a_cell[cell] ~ dnorm(0, sigma_cell),

    b_tide ~ dnorm(0, 1),
    b_moon ~ dnorm(0, 1),

    sigma_month ~ dexp(1),
    sigma_year ~ dexp(1),
    sigma_track ~ dexp(1),
    sigma_cell ~ dexp(1),

```

```

    phi ~ dexp(1)
  ),
  data = d_cells_nb,
  chains = 4, cores = 4, iter = 2000
)

```

Running MCMC with 4 parallel chains, with 1 thread(s) per chain...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)

Chain 2 Iteration: 1 / 2000 [0%] (Warmup)

Chain 3 Iteration: 1 / 2000 [0%] (Warmup)

Chain 4 Iteration: 1 / 2000 [0%] (Warmup)

Chain 2 Iteration: 100 / 2000 [5%] (Warmup)

Chain 1 Iteration: 100 / 2000 [5%] (Warmup)

Chain 4 Iteration: 100 / 2000 [5%] (Warmup)

Chain 3 Iteration: 100 / 2000 [5%] (Warmup)

Chain 1 Iteration: 200 / 2000 [10%] (Warmup)

Chain 2 Iteration: 200 / 2000 [10%] (Warmup)

Chain 3 Iteration: 200 / 2000 [10%] (Warmup)

Chain 4 Iteration: 200 / 2000 [10%] (Warmup)

Chain 1 Iteration: 300 / 2000 [15%] (Warmup)

Chain 2 Iteration: 300 / 2000 [15%] (Warmup)

Chain 3 Iteration: 300 / 2000 [15%] (Warmup)

Chain 1 Iteration: 400 / 2000 [20%] (Warmup)

Chain 4 Iteration: 300 / 2000 [15%] (Warmup)

Chain 2 Iteration: 400 / 2000 [20%] (Warmup)

Chain 3 Iteration: 400 / 2000 [20%] (Warmup)

Chain 1 Iteration: 500 / 2000 [25%] (Warmup)

Chain 2 Iteration: 500 / 2000 [25%] (Warmup)

Chain 3 Iteration: 500 / 2000 [25%] (Warmup)

Chain 4 Iteration: 400 / 2000 [20%] (Warmup)

Chain 1 Iteration: 600 / 2000 [30%] (Warmup)

Chain 2 Iteration: 600 / 2000 [30%] (Warmup)

Chain 1 Iteration: 700 / 2000 [35%] (Warmup)

Chain 2 Iteration: 700 / 2000 [35%] (Warmup)

Chain 4 Iteration: 500 / 2000 [25%] (Warmup)

Chain 3 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
Chain 2 Iteration: 800 / 2000 [40%] (Warmup)
Chain 3 Iteration: 700 / 2000 [35%] (Warmup)
Chain 2 Iteration: 900 / 2000 [45%] (Warmup)
Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 2 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 2 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
Chain 3 Iteration: 800 / 2000 [40%] (Warmup)
Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
Chain 3 Iteration: 900 / 2000 [45%] (Warmup)
Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
Chain 3 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 3 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 2 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 3 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [85%] (Sampling)
Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 3 Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [95%] (Sampling)
Chain 3 Iteration: 1300 / 2000 [65%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [75%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 495.0 seconds.

```

Chain 3 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 611.3 seconds.
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 716.3 seconds.
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 737.0 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 639.9 seconds.
 Total execution time: 737.1 seconds.

```

#### Diagnostics and posterior summaries for track-level models
#| cache: true

## posterior summaries
precis(m_track_y, depth = 2)

```

	mean	sd	5.5%	94.5%	rhat	ess_bulk
a	-0.15441531	0.33581375	-0.72854235	0.36253937	1.003315	960.7755
a_month[1]	0.01263489	0.25481475	-0.38274836	0.42806300	1.001405	1195.0082
a_month[2]	0.20647433	0.27334405	-0.21502070	0.66452041	1.004213	1263.2286
a_month[3]	0.21922199	0.33554981	-0.28624698	0.77555519	1.000855	1778.2623
a_month[4]	0.10161994	0.38795557	-0.49982646	0.72540347	1.001955	1905.6010
a_month[5]	0.37646041	0.33930815	-0.15206888	0.91443003	1.001002	1919.9137
a_month[6]	-0.48968902	0.34626489	-1.04980569	0.05643183	1.000199	1853.1001
a_month[7]	-0.13810860	0.31469868	-0.65374728	0.35046435	1.001140	1606.6194
a_month[8]	-0.66127325	0.33293554	-1.21312143	-0.14839988	1.000983	1636.8963

```

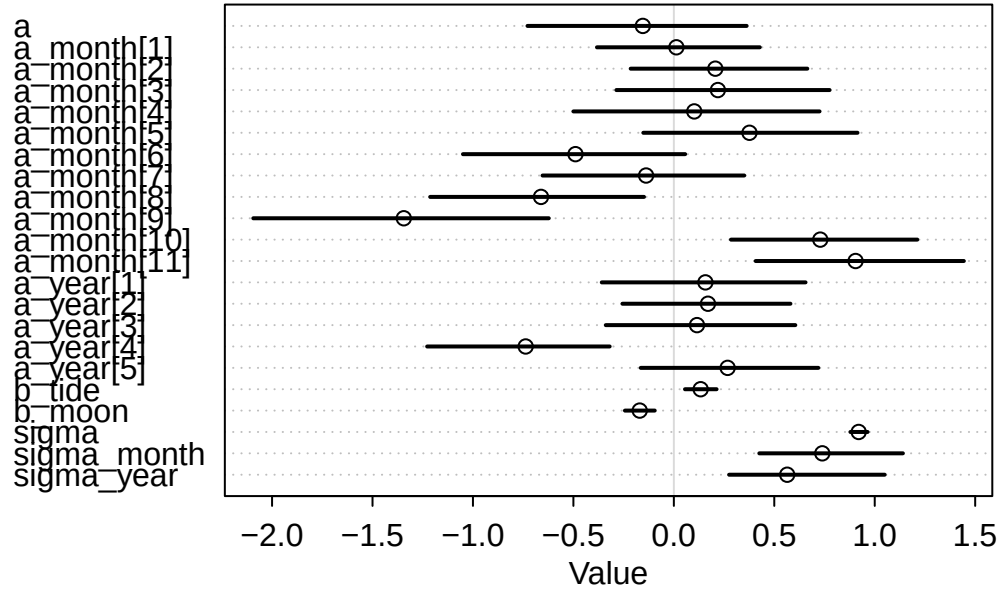
a_month[9] -1.34428672 0.46151363 -2.09355076 -0.62243011 1.000471 1835.5135
a_month[10] 0.72887758 0.29302735 0.28442844 1.21197012 1.005622 1066.5724
a_month[11] 0.90456604 0.32534067 0.40717301 1.44376041 1.005875 1145.4673
a_year[1] 0.15711379 0.32081119 -0.35884192 0.65569461 1.002749 1669.6210
a_year[2] 0.16986053 0.26717999 -0.25607342 0.58050282 1.002918 1301.9252
a_year[3] 0.11524060 0.29400239 -0.33903126 0.60481536 1.002659 1203.8678
a_year[4] -0.73766819 0.28471222 -1.22811978 -0.31937985 1.004821 1273.6141
a_year[5] 0.26771044 0.28349765 -0.16486283 0.71920215 1.002759 1386.8922
b_tide 0.13343244 0.04957119 0.05536572 0.21278279 1.001300 3095.7545
b_moon -0.16988186 0.04709438 -0.24359361 -0.09504275 1.000499 2450.8845
sigma 0.92009966 0.02687643 0.87863372 0.96454553 1.000142 3693.8177
sigma_month 0.73906236 0.23571479 0.42532905 1.14035230 1.001871 1687.0596
sigma_year 0.56431645 0.26968350 0.27489013 1.04944323 0.999615 2015.7107

```

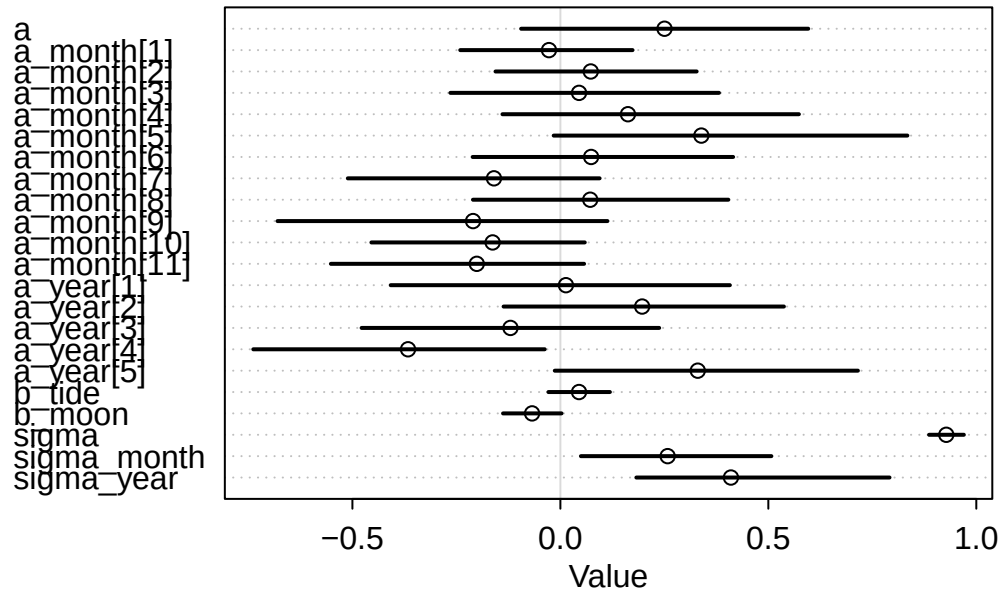
```
precis(m_track_x, depth = 2)
```

	mean	sd	5.5%	94.5%	rhat	ess_bulk
a	0.25036130	0.24168002	-0.09440024	0.596011321	1.0126454	901.3141
a_month[1]	-0.02743895	0.13134258	-0.24080734	0.173549659	1.0022165	2013.7189
a_month[2]	0.07305739	0.15255063	-0.15603355	0.327734043	1.0010822	2111.0641
a_month[3]	0.04492743	0.20167283	-0.26444002	0.381386349	1.0018743	2569.4299
a_month[4]	0.16245684	0.23305317	-0.13928134	0.573481586	1.0003646	2014.6676
a_month[5]	0.33892408	0.27829137	-0.01613279	0.834290801	1.0028510	785.9011
a_month[6]	0.07392015	0.19715979	-0.21097537	0.415405133	1.0004180	2968.4605
a_month[7]	-0.15966882	0.19641373	-0.51108550	0.094426097	1.0033700	1645.3304
a_month[8]	0.07178167	0.19013291	-0.21016929	0.403845087	1.0005041	2680.7286
a_month[9]	-0.20994499	0.25869546	-0.68007816	0.113257807	0.9999161	1356.5148
a_month[10]	-0.16291478	0.16581580	-0.45439442	0.058706095	1.0035533	965.3686
a_month[11]	-0.20096874	0.19669067	-0.55177965	0.057053938	1.0036549	919.8301
a_year[1]	0.01334688	0.27532774	-0.40800887	0.407560156	1.0057824	1265.9709
a_year[2]	0.19657965	0.23477700	-0.13660226	0.537463158	1.0104239	948.1765
a_year[3]	-0.12001017	0.24170658	-0.47794471	0.237609050	1.0097094	964.4482
a_year[4]	-0.36625649	0.24225658	-0.73833578	-0.037544303	1.0076759	926.1197
a_year[5]	0.33031972	0.25022714	-0.01351704	0.715386443	1.0073327	1058.6658
b_tide	0.04486975	0.04627803	-0.02884511	0.119149882	1.0001928	3458.0614
b_moon	-0.06797600	0.04366751	-0.13790192	0.002819479	1.0018976	3793.0322
sigma	0.92779845	0.02659438	0.88638650	0.970113489	1.0016092	3196.9172
sigma_month	0.25785732	0.14633200	0.04935783	0.507365212	1.0059446	409.8665
sigma_year	0.41007515	0.22996563	0.18262287	0.791388246	1.0072107	1176.5078

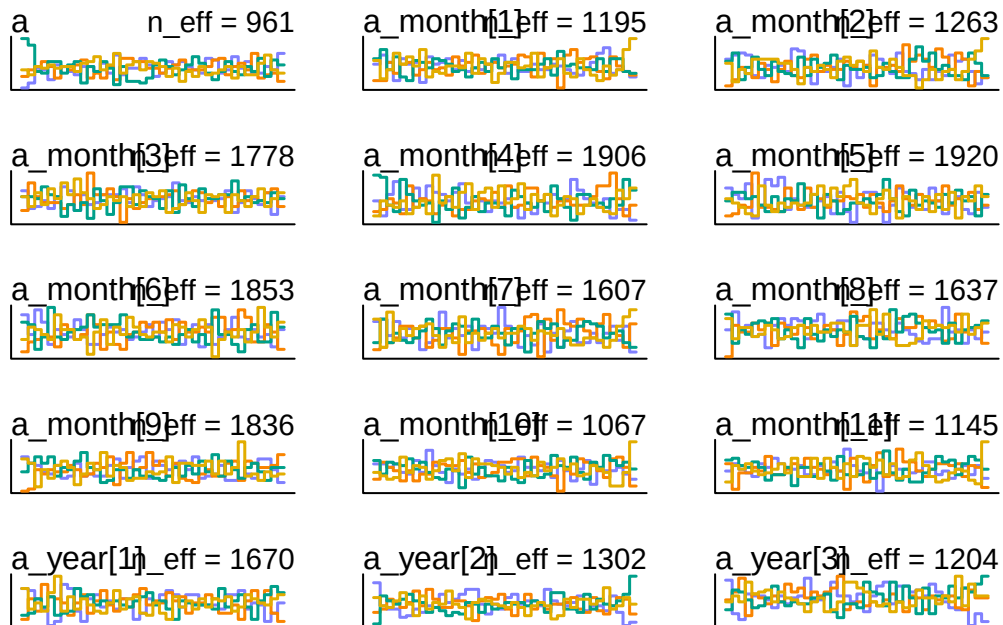
```
plot(precis(m_track_y, depth = 2))
```



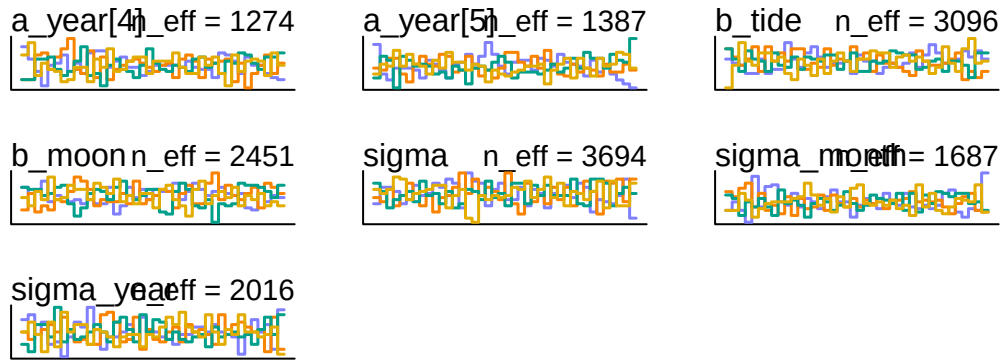
```
plot(precis(m_track_x, depth = 2))
```



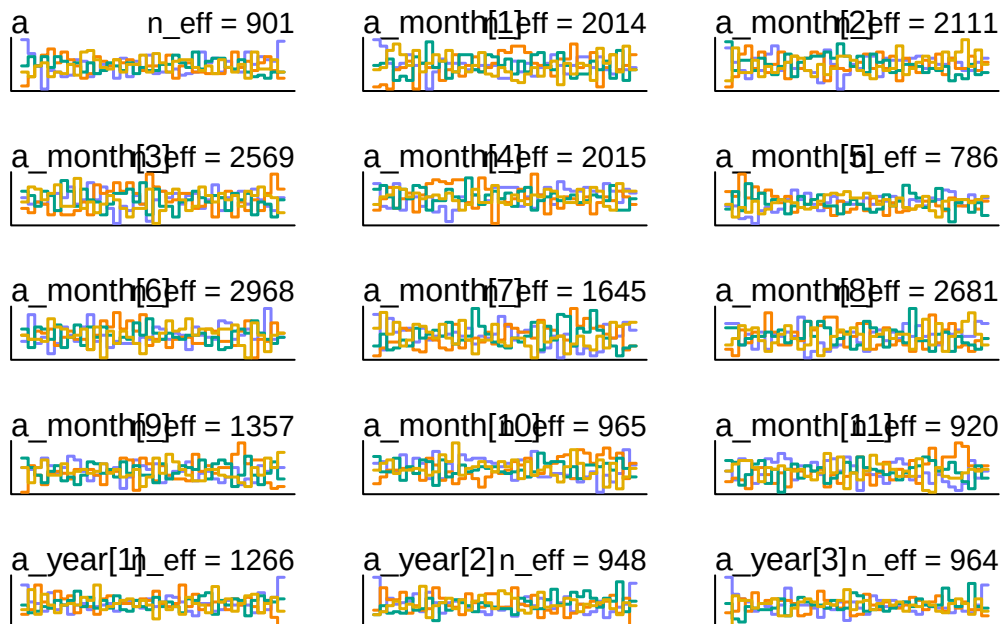
```
## trace/rank plots
# Track Y
par(mfrow = c(2,2))
trankplot(m_track_y, ask = FALSE)
```



```
par(mfrow = c(1,1))
```



```
# Track X
par(mfrow = c(2,2))
trankplot(m_track_x, ask = FALSE)
```



```
par(mfrow = c(1,1))
```

a_year[4] n_eff = 926



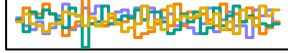
a_year[5] n_eff = 1059



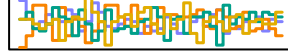
b_tide n_eff = 3458



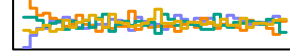
b_moon n_eff = 3793



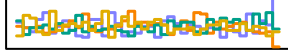
sigma n_eff = 3197



sigma_month n_eff = 410



sigma_year n_eff = 1177



```
## convergence check
summary(m_track_y)
```

	mean	sd	5.5%	94.5%	rhat	ess_bulk
a	-0.15441531	0.33581375	-0.72854235	0.36253937	1.003315	960.7755
a_month[1]	0.01263489	0.25481475	-0.38274836	0.42806300	1.001405	1195.0082
a_month[2]	0.20647433	0.27334405	-0.21502070	0.66452041	1.004213	1263.2286
a_month[3]	0.21922199	0.33554981	-0.28624698	0.77555519	1.000855	1778.2623
a_month[4]	0.10161994	0.38795557	-0.49982646	0.72540347	1.001955	1905.6010
a_month[5]	0.37646041	0.33930815	-0.15206888	0.91443003	1.001002	1919.9137
a_month[6]	-0.48968902	0.34626489	-1.04980569	0.05643183	1.000199	1853.1001
a_month[7]	-0.13810860	0.31469868	-0.65374728	0.35046435	1.001140	1606.6194
a_month[8]	-0.66127325	0.33293554	-1.21312143	-0.14839988	1.000983	1636.8963
a_month[9]	-1.34428672	0.46151363	-2.09355076	-0.62243011	1.000471	1835.5135
a_month[10]	0.72887758	0.29302735	0.28442844	1.21197012	1.005622	1066.5724
a_month[11]	0.90456604	0.32534067	0.40717301	1.44376041	1.005875	1145.4673
a_year[1]	0.15711379	0.32081119	-0.35884192	0.65569461	1.002749	1669.6210
a_year[2]	0.16986053	0.26717999	-0.25607342	0.58050282	1.002918	1301.9252
a_year[3]	0.11524060	0.29400239	-0.33903126	0.60481536	1.002659	1203.8678
a_year[4]	-0.73766819	0.28471222	-1.22811978	-0.31937985	1.004821	1273.6141

```

a_year[5]      0.26771044 0.28349765 -0.16486283  0.71920215 1.002759 1386.8922
b_tide         0.13343244 0.04957119  0.05536572  0.21278279 1.001300 3095.7545
b_moon        -0.16988186 0.04709438 -0.24359361 -0.09504275 1.000499 2450.8845
sigma          0.92009966 0.02687643  0.87863372  0.96454553 1.000142 3693.8177
sigma_month    0.73906236 0.23571479  0.42532905  1.14035230 1.001871 1687.0596
sigma_year     0.56431645 0.26968350  0.27489013  1.04944323 0.999615 2015.7107

```

```
summary(m_track_x)
```

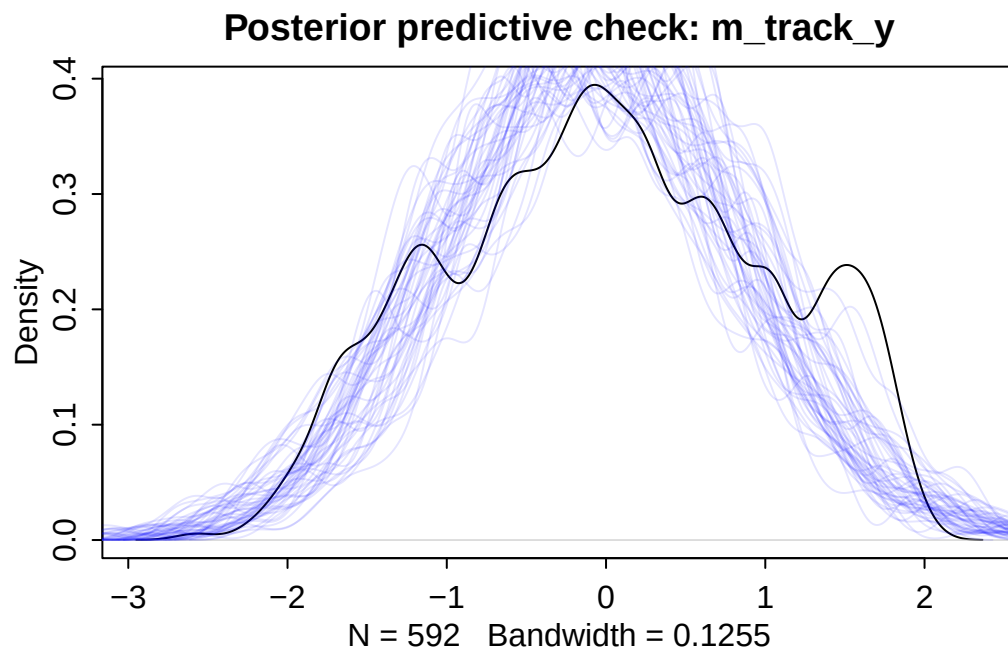
	mean	sd	5.5%	94.5%	rhat	ess_bulk
a	0.25036130	0.24168002	-0.09440024	0.596011321	1.0126454	901.3141
a_month[1]	-0.02743895	0.13134258	-0.24080734	0.173549659	1.0022165	2013.7189
a_month[2]	0.07305739	0.15255063	-0.15603355	0.327734043	1.0010822	2111.0641
a_month[3]	0.04492743	0.20167283	-0.26444002	0.381386349	1.0018743	2569.4299
a_month[4]	0.16245684	0.23305317	-0.13928134	0.573481586	1.0003646	2014.6676
a_month[5]	0.33892408	0.27829137	-0.01613279	0.834290801	1.0028510	785.9011
a_month[6]	0.07392015	0.19715979	-0.21097537	0.415405133	1.0004180	2968.4605
a_month[7]	-0.15966882	0.19641373	-0.51108550	0.094426097	1.0033700	1645.3304
a_month[8]	0.07178167	0.19013291	-0.21016929	0.403845087	1.0005041	2680.7286
a_month[9]	-0.20994499	0.25869546	-0.68007816	0.113257807	0.9999161	1356.5148
a_month[10]	-0.16291478	0.16581580	-0.45439442	0.058706095	1.0035533	965.3686
a_month[11]	-0.20096874	0.19669067	-0.55177965	0.057053938	1.0036549	919.8301
a_year[1]	0.01334688	0.27532774	-0.40800887	0.407560156	1.0057824	1265.9709
a_year[2]	0.19657965	0.23477700	-0.13660226	0.537463158	1.0104239	948.1765
a_year[3]	-0.12001017	0.24170658	-0.47794471	0.237609050	1.0097094	964.4482
a_year[4]	-0.36625649	0.24225658	-0.73833578	-0.037544303	1.0076759	926.1197
a_year[5]	0.33031972	0.25022714	-0.01351704	0.715386443	1.0073327	1058.6658
b_tide	0.04486975	0.04627803	-0.02884511	0.119149882	1.0001928	3458.0614
b_moon	-0.06797600	0.04366751	-0.13790192	0.002819479	1.0018976	3793.0322
sigma	0.92779845	0.02659438	0.88638650	0.970113489	1.0016092	3196.9172
sigma_month	0.25785732	0.14633200	0.04935783	0.507365212	1.0059446	409.8665
sigma_year	0.41007515	0.22996563	0.18262287	0.791388246	1.0072107	1176.5078

```

## posterior predictive check
y_sim <- sim(m_track_y, n = 1000)

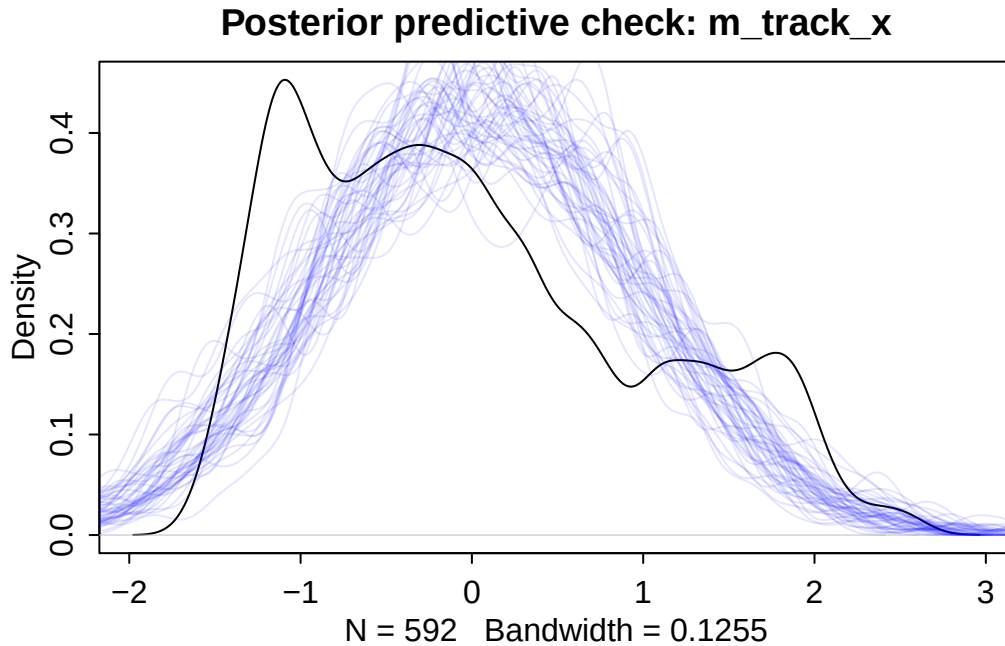
dens(d_track_y_scaled$Y)
for(i in 1:50) dens(y_sim[i,], add = TRUE, col = col.alpha("blue", 0.1))
title("Posterior predictive check: m_track_y")

```



```
x_sim <- sim(m_track_x, n = 1000)

dens(d_track_x_scaled$Y)
for(i in 1:50) dens(x_sim[i,], add = TRUE, col = col.alpha("blue", 0.1))
title("Posterior predictive check: m_track_x")
```



```
#### Diagnostics and posterior summaries for space-use model
```

```
## posterior summaries
```

```
precis(m_space_use, depth = 2)
```

	mean	sd	5.5%	94.5%	rhat
a	-1.443416e+00	0.31772862	-1.9008596000	-0.891525708	1.0033606
a_month[1]	-1.455363e-01	0.14172713	-0.3719307257	0.063107953	1.0012422
a_month[2]	-3.906011e-01	0.15342327	-0.6388620941	-0.163849693	1.0001936
a_month[3]	-5.578440e-01	0.20476477	-0.8956567976	-0.248473121	1.0004658
a_month[4]	-2.794663e-01	0.22348087	-0.6427039755	0.066589664	0.9999326
a_month[5]	1.186992e-01	0.18683196	-0.1703581423	0.421656400	1.0019113
a_month[6]	-3.940112e-02	0.18575301	-0.3370622745	0.254229770	1.0025291
a_month[7]	3.304022e-01	0.17726085	0.0605341534	0.619589952	1.0024881
a_month[8]	3.320704e-01	0.17927608	0.0613548326	0.627316181	1.0012451
a_month[9]	5.232711e-01	0.24320030	0.1480252010	0.922137896	1.0018850
a_month[10]	-6.867962e-02	0.16168614	-0.3303186157	0.177425450	1.0006250
a_month[11]	-5.361176e-02	0.17909464	-0.3498015397	0.213794166	0.9999216
a_year[1]	-6.395059e-02	0.31579706	-0.5910462643	0.395183839	1.0044381
a_year[2]	-5.174193e-01	0.29721446	-1.0327834155	-0.113414082	1.0047807
a_year[3]	-5.167922e-01	0.30894705	-1.0472717600	-0.081741826	1.0062572
a_year[4]	6.146400e-01	0.29593223	0.1200466088	1.037216326	1.0052510

a_year[5]	-9.918494e-02	0.30240587	-0.6281338379	0.323640206	1.0035983
a_track[1]	-3.784405e-01	0.21452266	-0.7231816447	-0.038294257	1.0001347
a_track[2]	-1.482837e-01	0.22581283	-0.5053903228	0.215297050	1.0012433
a_track[3]	-1.563535e-01	0.22308093	-0.5194174827	0.200602638	1.0006139
a_track[4]	-1.573815e-01	0.22635458	-0.5194772813	0.203604623	0.9999474
a_track[5]	2.234266e-01	0.24872150	-0.1719330459	0.621069274	1.0027773
a_track[6]	-7.012662e-03	0.30555474	-0.5036624161	0.489075130	1.0013949
a_track[7]	-3.125709e-02	0.21451768	-0.3739545449	0.320098485	1.0004229
a_track[8]	-3.386844e-02	0.21747863	-0.3742655438	0.319567551	1.0022301
a_track[9]	5.678529e-02	0.22042274	-0.2877946596	0.416850912	1.0003835
a_track[10]	-3.079188e-01	0.20510107	-0.6372329684	0.026906450	1.0005776
a_track[11]	-2.839937e-02	0.22917372	-0.3913975641	0.341777393	1.0011605
a_track[12]	-1.113668e-01	0.21182870	-0.4366556018	0.230956884	1.0005471
a_track[13]	-1.111013e-01	0.21191781	-0.4432752807	0.228759281	1.0000459
a_track[14]	-1.708842e-02	0.21990248	-0.3600331612	0.343031132	1.0017649
a_track[15]	4.176242e-02	0.22159792	-0.3001885236	0.398925688	1.0008064
a_track[16]	1.342036e-01	0.22873126	-0.2327860348	0.495950201	1.0008230
a_track[17]	1.348190e-01	0.23454571	-0.2444349310	0.510619709	1.0002926
a_track[18]	-9.544415e-02	0.21103761	-0.4275860068	0.246948840	1.0012011
a_track[19]	-1.053057e-01	0.21850432	-0.4502818914	0.248760563	1.0014229
a_track[20]	-3.099761e-01	0.19569922	-0.6188518275	0.003499808	1.0016196
a_track[21]	-2.568898e-01	0.20433107	-0.5847490756	0.076854259	1.0014456
a_track[22]	-1.776298e-01	0.20684537	-0.5064415099	0.151473318	1.0003005
a_track[23]	-1.790698e-01	0.21521761	-0.5158987720	0.174599666	1.0007779
a_track[24]	9.548518e-02	0.26334761	-0.3270465522	0.520533724	1.0003558
a_track[25]	-1.361169e-02	0.22054384	-0.3651558580	0.347354927	1.0031565
a_track[26]	-1.568681e-01	0.21502485	-0.4989140294	0.182572568	1.0014458
a_track[27]	3.296035e-01	0.27823577	-0.1143301528	0.765935110	1.0002647
a_track[28]	4.054724e-01	0.32710837	-0.1182906967	0.931412117	1.0017483
a_track[29]	3.820639e-01	0.25344268	-0.0162025946	0.800522802	1.0000987
a_track[30]	1.954581e-01	0.23939242	-0.1860176078	0.577548573	1.0016061
a_track[31]	1.046897e-01	0.22909135	-0.2579022031	0.473377472	1.0029502
a_track[32]	-1.498181e-01	0.20771598	-0.4814543123	0.187182501	1.0008414
a_track[33]	7.941826e-03	0.22202174	-0.3436165110	0.368392430	1.0002209
a_track[34]	7.002660e-03	0.22390716	-0.3454527063	0.373445436	1.0006952
a_track[35]	-7.135861e-02	0.22047581	-0.4148469534	0.288512328	1.0003612
a_track[36]	9.516973e-03	0.21697977	-0.3306881360	0.354549724	1.0014350
a_track[37]	-2.922104e-01	0.20694437	-0.6201478404	0.037766099	1.0011988
a_track[38]	-2.977393e-01	0.20460477	-0.6232242443	0.029205604	1.0031089
a_track[39]	-4.247741e-01	0.20140582	-0.7446917434	-0.099300673	1.0022149
a_track[40]	-5.328440e-01	0.19517177	-0.8426612450	-0.223376835	1.0028886
a_track[41]	-7.986461e-02	0.22394040	-0.4376062964	0.281451856	1.0030721
a_track[42]	4.153948e-01	0.29669231	-0.0585679990	0.881763623	1.0013517

a_track[43]	-2.203684e-01	0.21099698	-0.5488828455	0.127808539	0.9999966
a_track[44]	-1.389338e-01	0.22010035	-0.4815875218	0.216583530	1.0012210
a_track[45]	1.922240e-01	0.24666681	-0.1952252403	0.582754142	1.0024210
a_track[46]	1.808225e-03	0.22158540	-0.3537673071	0.349141053	0.9995564
a_track[47]	7.583716e-02	0.23179382	-0.2886629692	0.444269487	1.0012468
a_track[48]	1.496624e-01	0.23353276	-0.2141836226	0.531100333	1.0032855
a_track[49]	2.358134e-01	0.24261116	-0.1504094854	0.630686253	1.0000690
a_track[50]	2.630612e-02	0.26543611	-0.3810170443	0.450466257	1.0006505
a_track[51]	-4.155599e-01	0.22276648	-0.7661751210	-0.070207594	1.0000882
a_track[52]	1.261101e-01	0.26548247	-0.2929258685	0.554690812	1.0012501
a_track[53]	3.174885e-01	0.29963145	-0.1749256283	0.783424925	0.9998175
a_track[54]	-7.339881e-02	0.25483737	-0.4733725203	0.340427215	1.0007023
a_track[55]	5.721717e-01	0.31491017	0.0771857984	1.075578705	1.0030917
a_track[56]	-4.897689e-01	0.22401312	-0.8383313747	-0.131283400	1.0025707
a_track[57]	1.312533e-01	0.25730332	-0.2710981956	0.536097123	1.0009568
a_track[58]	-2.476931e-01	0.23494003	-0.6332894201	0.133497182	1.0014222
a_track[59]	-2.419256e-01	0.24220263	-0.6243741726	0.146418570	1.0012433
a_track[60]	4.537464e-01	0.29862859	-0.0122285331	0.932320957	1.0005935
a_track[61]	-7.366795e-02	0.24311051	-0.4624487394	0.318072885	0.9999925
a_track[62]	6.467350e-01	0.30734153	0.1551640113	1.139246997	1.0023035
a_track[63]	1.138670e-01	0.25483897	-0.2843086716	0.530638114	1.0009494
a_track[64]	-3.758934e-01	0.22028453	-0.7146119851	-0.019091909	1.0013905
a_track[65]	2.897127e-01	0.37001099	-0.2984271493	0.884209300	1.0042238
a_track[66]	-4.456588e-01	0.22183709	-0.7920875947	-0.089488317	0.9995230
a_track[67]	2.580289e-01	0.29352669	-0.2098510285	0.734077442	1.0030054
a_track[68]	-6.526174e-02	0.23247724	-0.4298827025	0.306755801	1.0006402
a_track[69]	-1.453950e-01	0.22972358	-0.4999732589	0.226449442	1.0012921
a_track[70]	-1.437195e-01	0.23601298	-0.5204868226	0.234956784	0.9995598
a_track[71]	-1.436210e-01	0.23257666	-0.5080296228	0.229393077	1.0005091
a_track[72]	-1.792853e-02	0.26788620	-0.4469160926	0.415308955	1.0002625
a_track[73]	2.272802e-01	0.27891284	-0.2169489707	0.675642098	1.0012897
a_track[74]	-2.983020e-01	0.23652057	-0.6773751441	0.068336474	1.0027106
a_track[75]	2.319852e-01	0.28850078	-0.2337828190	0.689184296	1.0031341
a_track[76]	-1.982466e-01	0.24300629	-0.5871202663	0.190952361	1.0003508
a_track[77]	-1.986248e-01	0.24995828	-0.5857026169	0.215074963	1.0004128
a_track[78]	1.458704e-01	0.27021285	-0.2785197329	0.585095863	1.0032943
a_track[79]	-1.983942e-01	0.24849270	-0.5927504169	0.205176623	0.9998448
a_track[80]	-3.039392e-01	0.23554596	-0.6782500752	0.069132160	1.0003310
a_track[81]	-2.174157e-03	0.25678803	-0.3978751408	0.407586206	1.0007898
a_track[82]	1.205457e-01	0.27012659	-0.3053902972	0.559489273	1.0002166
a_track[83]	-1.109623e-01	0.27519759	-0.5457482657	0.325954440	1.0002371
a_track[84]	5.233104e-01	0.29564490	0.0547320775	1.000966473	1.0000208
a_track[85]	-2.198621e-01	0.24064305	-0.6097989287	0.176054539	1.0002397

a_track[86]	-2.084978e-01	0.25095778	-0.6110342182	0.192025247	1.0008121
a_track[87]	-2.185008e-01	0.24762051	-0.6084913928	0.177296917	1.0004613
a_track[88]	3.465887e-01	0.30654838	-0.1444635945	0.835301216	1.0008219
a_track[89]	4.762922e-01	0.32760851	-0.0417392171	1.010608634	1.0008649
a_track[90]	2.350749e-01	0.29271164	-0.2296079699	0.710886053	1.0012104
a_track[91]	-1.285265e-01	0.25196334	-0.5176750733	0.272955331	1.0011321
a_track[92]	-4.254287e-01	0.23269330	-0.7914021410	-0.052098082	1.0011895
a_track[93]	-1.190687e-01	0.25693946	-0.5237470928	0.294001877	1.0006064
a_track[94]	3.383544e-01	0.30202789	-0.1404542299	0.822386920	1.0023518
a_track[95]	1.260987e-01	0.27596134	-0.3073013174	0.587145885	1.0003449
a_track[96]	-9.919157e-02	0.25851988	-0.5101053533	0.321163072	1.0018824
a_track[97]	-3.908786e-01	0.23831782	-0.7699473251	-0.005231445	0.9999301
a_track[98]	3.437083e-01	0.31226917	-0.1449253320	0.846803424	1.0011298
a_track[99]	2.141909e-02	0.26258702	-0.3859581447	0.447192367	1.0012314
a_track[100]	3.814308e-01	0.29251121	-0.0738098624	0.840199062	1.0026674
a_track[101]	2.422341e-01	0.28947800	-0.2156776649	0.712195170	0.9995927
a_track[102]	2.420697e-01	0.35954918	-0.3520981960	0.801384684	1.0028973
a_track[103]	1.294755e-01	0.27263433	-0.3005629249	0.566061646	0.9992386
a_track[104]	-9.126023e-02	0.24975401	-0.4779704123	0.312519168	1.0003287
a_track[105]	-2.013458e-01	0.24088179	-0.5893917653	0.179384014	1.0040470
a_track[106]	-2.059370e-01	0.24602166	-0.5880952932	0.194976526	1.0010239
a_track[107]	2.654836e-01	0.28218787	-0.1800812419	0.722718702	1.0012959
a_track[108]	-3.717478e-01	0.19904871	-0.6753291621	-0.043396787	1.0023755
a_track[109]	-1.026875e-01	0.21554629	-0.4471620637	0.246123679	1.0014622
a_track[110]	-2.523349e-01	0.20966780	-0.5879924438	0.077198291	1.0025327
a_track[111]	5.091776e-01	0.34176475	-0.0270018169	1.061708001	1.0005118
a_track[112]	-3.717497e-01	0.19881067	-0.6805639650	-0.048309458	1.0009013
a_track[113]	3.410658e-01	0.35581280	-0.2231385019	0.908689908	1.0014987
a_track[114]	2.793468e-01	0.33324943	-0.2546488595	0.812587210	1.0010859
a_track[115]	3.360232e-01	0.31764301	-0.1688744192	0.834511062	1.0008265
a_track[116]	3.041925e-01	0.32684666	-0.2196830380	0.825615978	1.0038552
a_track[117]	3.061547e-01	0.27847441	-0.1264410553	0.761571836	1.0004114
a_track[118]	1.976986e-01	0.26801689	-0.2231172678	0.637047392	1.0001091
a_track[119]	3.924083e-01	0.34223350	-0.1569796135	0.950452519	1.0034495
a_track[120]	1.518860e-01	0.26990925	-0.2790902469	0.584456675	1.0007368
a_track[121]	3.820094e-01	0.31229884	-0.1085116841	0.891449263	1.0022239
a_track[122]	2.567955e-01	0.29158435	-0.2099083138	0.718948841	1.0006869
a_track[123]	-1.868311e-01	0.24213060	-0.5676212600	0.206942724	0.9998290
a_track[124]	3.152255e-01	0.32341319	-0.2120790142	0.819656725	1.0007265
a_track[125]	2.644384e-01	0.36250763	-0.3101348526	0.849302003	1.0029139
a_track[126]	-4.522903e-01	0.22600861	-0.8072555141	-0.078430572	1.0020340
a_track[127]	-1.088750e-01	0.25105347	-0.5126367748	0.297493051	1.0008981
a_track[128]	-4.535367e-01	0.21940423	-0.7962492382	-0.096618541	1.0001346

a_track[129]	-3.445741e-01	0.23227925	-0.7143001344	0.029365531	1.0015922
a_track[130]	3.004230e-01	0.32418562	-0.2214837758	0.819687267	0.9999141
a_track[131]	1.549176e-01	0.27801364	-0.2910856136	0.605614903	1.0014077
a_track[132]	-6.115313e-01	0.21490473	-0.9570664879	-0.265783907	1.0011625
a_track[133]	1.190780e-01	0.29061636	-0.3457657451	0.592478428	1.0004677
a_track[134]	-5.405939e-01	0.21633885	-0.8842705044	-0.202791456	1.0006619
a_track[135]	3.202783e-01	0.29990656	-0.1618757708	0.801770327	1.0035347
a_track[136]	4.097504e-01	0.34219527	-0.1252703665	0.963634531	1.0027045
a_track[137]	3.361373e-02	0.25884119	-0.3802491990	0.453919348	1.0014215
a_track[138]	-6.149473e-01	0.21195408	-0.9497400296	-0.274659960	1.0041357
a_track[139]	1.574939e-01	0.26779200	-0.2705269646	0.582082929	1.0001839
a_track[140]	2.386535e-01	0.29306646	-0.2228332428	0.708406056	1.0002893
a_track[141]	4.492204e-01	0.32374906	-0.0718534034	0.965722830	0.9996506
a_track[142]	6.937882e-02	0.24826420	-0.3146323863	0.480832185	1.0008382
a_track[143]	-2.904456e-01	0.21898253	-0.6345534779	0.062138005	1.0026567
a_track[144]	1.757639e-01	0.25985123	-0.2412583776	0.597660662	0.9996998
a_track[145]	2.887379e-01	0.26364127	-0.1351235287	0.713731889	1.0025018
a_track[146]	-5.047672e-01	0.20606999	-0.8231710165	-0.166527811	1.0032341
a_track[147]	-1.193851e-01	0.22163874	-0.4729161170	0.238675867	1.0008534
a_track[148]	1.848576e-01	0.24900477	-0.2105405009	0.584457297	1.0018122
a_track[149]	8.168991e-02	0.23645353	-0.2909502398	0.467841372	1.0038074
a_track[150]	-1.626466e-02	0.23583550	-0.3870689893	0.367828022	1.0010623
a_track[151]	2.949416e-01	0.26078645	-0.1144879628	0.723307333	1.0009282
a_track[152]	5.334628e-01	0.28047802	0.0878446961	0.983954477	1.0029714
a_track[153]	-7.201685e-01	0.18667155	-1.0181163740	-0.425441451	1.0003428
a_track[154]	-4.923785e-01	0.20678560	-0.8193839732	-0.162325929	1.0012906
a_track[155]	-4.173815e-01	0.20837759	-0.7491360080	-0.081871333	1.0017051
a_track[156]	-2.715406e-01	0.22229887	-0.6216432470	0.089872427	1.0001510
a_track[157]	8.572790e-02	0.23968483	-0.2911684540	0.467905482	1.0015062
a_track[158]	9.116333e-02	0.25029172	-0.3008827748	0.503442337	1.0014681
a_track[159]	-2.029673e-01	0.22413006	-0.5546755372	0.150955653	1.0002767
a_track[160]	2.940336e-02	0.24029567	-0.3484018010	0.427669600	1.0017770
a_track[161]	-4.896559e-02	0.23835074	-0.4247213037	0.333116169	1.0026401
a_track[162]	-3.373651e-01	0.21237052	-0.6793397849	-0.005519788	1.0018146
a_track[163]	2.107907e-01	0.25232583	-0.1946496667	0.620968712	1.0003716
a_track[164]	-5.109433e-01	0.21206800	-0.8403829429	-0.168929516	1.0012384
a_track[165]	1.932627e-01	0.25917276	-0.2182384110	0.609864564	1.0003566
a_track[166]	-7.474072e-01	0.19738189	-1.0607919380	-0.423344041	1.0004053
a_track[167]	-4.351519e-01	0.20915840	-0.7627281876	-0.102135047	1.0004271
a_track[168]	3.901026e-01	0.28045503	-0.0584173379	0.844163578	1.0007053
a_track[169]	1.604418e-01	0.26142229	-0.2534828500	0.576835389	1.0013011
a_track[170]	1.207448e-02	0.25542401	-0.3911101262	0.421041665	1.0006183
a_track[171]	5.346326e-01	0.30910116	0.0558546443	1.036909917	1.0015350

a_track[172]	-8.647154e-02	0.25535180	-0.4875568182	0.325669225	1.0034054
a_track[173]	2.168788e-01	0.29796974	-0.2503634450	0.694072470	1.0009821
a_track[174]	-8.717586e-02	0.25489359	-0.4871651429	0.333623706	1.0007452
a_track[175]	1.313811e-01	0.27871837	-0.3127303791	0.579456732	1.0003911
a_track[176]	2.727639e-01	0.32586671	-0.2434254325	0.785301434	0.9998361
a_track[177]	5.539272e-01	0.30275594	0.0765404645	1.054890425	1.0000824
a_track[178]	2.981223e-02	0.25339448	-0.3677770861	0.437049515	1.0002791
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a_track[180]	-9.238075e-02	0.24662107	-0.4900803163	0.304324726	0.9999133
a_track[181]	-5.244940e-01	0.22108442	-0.8731306148	-0.167839522	1.0005616
a_track[182]	-8.890871e-02	0.29814981	-0.5614301162	0.387939844	1.0011574
a_track[183]	-1.107717e+00	0.22821740	-1.4643214755	-0.731296367	1.0004709
a_track[184]	-7.496938e-01	0.24717491	-1.1435359760	-0.350800849	1.0018828
a_track[185]	-3.793607e-01	0.27904752	-0.8148121194	0.075959374	1.0022049
a_track[186]	-6.297288e-01	0.25945055	-1.0352615800	-0.212556880	1.0013137
a_track[187]	-2.238769e-01	0.27598027	-0.6681478272	0.222020237	1.0008055
a_track[188]	-3.464981e-01	0.28259340	-0.7852120609	0.109892232	1.0008495
a_track[189]	-7.001247e-02	0.32549013	-0.5905905597	0.450580983	1.0009954
a_track[190]	2.177827e-01	0.33709111	-0.3146028178	0.767768648	1.0020653
a_track[191]	-6.246043e-01	0.25695405	-1.0288367050	-0.214053305	1.0013366
a_track[192]	7.130586e-02	0.30992618	-0.4127554184	0.568710226	1.0007166
a_track[193]	5.323575e-02	0.33447604	-0.4733740809	0.588320169	1.0024414
a_track[194]	2.629550e-01	0.32131418	-0.2397936092	0.770818660	1.0021653
a_track[195]	2.795163e-01	0.33158971	-0.2520686456	0.810602877	1.0033994
a_track[196]	2.155362e-01	0.34573705	-0.3294158510	0.767477538	1.0027928
a_track[197]	2.848883e-01	0.34281994	-0.2548235558	0.843267077	1.0016962
a_track[198]	-6.043283e-01	0.25219007	-1.0020959435	-0.193072460	1.0014170
a_track[199]	2.645223e-01	0.32611429	-0.2626001820	0.795852984	0.9997331
a_track[200]	-7.378729e-01	0.24015212	-1.1147905270	-0.344689105	1.0038149
a_track[201]	2.566405e-01	0.30720835	-0.2347269202	0.749025394	1.0011275
a_track[202]	-1.297816e+00	0.22780733	-1.6609627130	-0.931983195	1.0014988
a_track[203]	5.094572e-02	0.31490496	-0.4476864753	0.559066635	1.0008827
a_track[204]	8.709961e-02	0.30728229	-0.3963703860	0.584809008	1.0007377
a_track[205]	-2.242448e-01	0.27879012	-0.6586455933	0.229410344	1.0039429
a_track[206]	-6.608986e-02	0.29922172	-0.5241196586	0.414318498	1.0000463
a_track[207]	1.638554e-01	0.36319089	-0.4256969505	0.728977686	1.0023238
a_track[208]	9.402189e-02	0.32363817	-0.4178868157	0.607428594	1.0026739
a_track[209]	-4.212259e-02	0.28917968	-0.4980107559	0.426937674	1.0011837
a_track[210]	-3.341382e-01	0.27187681	-0.7632753696	0.106522838	1.0027514
a_track[211]	2.782514e-01	0.32437976	-0.2140669966	0.807631417	1.0011024
a_track[212]	-3.209988e-01	0.26711161	-0.7449173813	0.107079194	0.9999316
a_track[213]	1.781577e-01	0.33264538	-0.3522441128	0.716829904	0.9993847
a_track[214]	-4.191357e-02	0.30827628	-0.5332215592	0.454443801	1.0026834

a_track[215]	-3.224949e-02	0.30475193	-0.5045985263	0.451536052	1.0023658
a_track[216]	1.050408e-01	0.32323452	-0.4095384988	0.626421736	1.0007755
a_track[217]	8.185925e-02	0.32213770	-0.4338818222	0.606250408	1.0003901
a_track[218]	-3.967578e-02	0.29438344	-0.5095301753	0.442665039	1.0015316
a_track[219]	1.549178e-01	0.35285203	-0.4128286024	0.714261954	1.0009129
a_track[220]	-3.593443e-01	0.27359696	-0.7947510322	0.085092636	1.0006895
a_track[221]	-3.932787e-01	0.27707761	-0.8323829259	0.049282373	0.9997385
a_track[222]	8.391469e-02	0.34101044	-0.4640450472	0.635091460	1.0005875
a_track[223]	1.529773e-01	0.36623796	-0.4252821106	0.746948707	1.0014242
a_track[224]	1.035681e-01	0.31167601	-0.3834428979	0.609693195	1.0023380
a_track[225]	9.241876e-02	0.33213975	-0.4334359602	0.617645874	0.9997191
a_track[226]	9.682109e-02	0.31340123	-0.4011508814	0.604235936	1.0011596
a_track[227]	1.185595e-01	0.29557246	-0.3453803422	0.603677456	1.0016541
a_track[228]	2.293159e-01	0.33170479	-0.2978902587	0.763790769	1.0004390
a_track[229]	-5.137888e-02	0.32117809	-0.5674040795	0.451120301	1.0006640
a_track[230]	2.242822e-01	0.33994728	-0.3085531184	0.769579591	1.0038001
a_track[231]	2.716413e-01	0.33401950	-0.2561099031	0.809190901	1.0015548
a_track[232]	2.373472e-01	0.33360550	-0.2942834355	0.774073582	1.0007360
a_track[233]	-4.846153e-02	0.29943806	-0.5191760115	0.423582776	1.0019933
a_track[234]	9.278006e-02	0.31748331	-0.4147736711	0.597878221	0.9997044
a_track[235]	-1.945565e-01	0.29275989	-0.6557114071	0.276084307	0.9999971
a_track[236]	1.112318e-01	0.31108759	-0.3777735946	0.623546091	1.0037167
a_track[237]	2.261637e-01	0.33483432	-0.2972951126	0.770222253	1.0009497
a_track[238]	-1.133906e+00	0.23744487	-1.5083829475	-0.758480140	1.0009102
a_track[239]	-4.375881e-01	0.29359319	-0.9095428574	0.028461301	0.9997704
a_track[240]	-5.647219e-01	0.28386774	-1.0089973735	-0.117087146	1.0009840
a_track[241]	-7.445615e-02	0.30159154	-0.5483100917	0.408744197	1.0005952
a_track[242]	-5.704873e-01	0.27253492	-0.9953451900	-0.129320872	1.0002352
a_track[243]	-2.101859e-01	0.28990584	-0.6603244039	0.256811571	1.0025855
a_track[244]	-6.034800e-02	0.30736290	-0.5457765343	0.437978867	1.0024247
a_track[245]	7.163426e-02	0.31648012	-0.4317195641	0.582773484	1.0008368
a_track[246]	2.281322e-01	0.32962722	-0.3015910341	0.767876622	1.0023611
a_track[247]	-9.212899e-02	0.30703069	-0.5786518161	0.410384136	0.9998344
a_track[248]	2.296238e-01	0.33146321	-0.2923804209	0.764109982	1.0007018
a_track[249]	1.759773e-01	0.34619846	-0.3714578557	0.735806580	1.0026217
a_track[250]	2.117822e-01	0.32346763	-0.3074180634	0.732294424	1.0010589
a_track[251]	1.735392e-01	0.36433325	-0.4106715146	0.766810796	1.0001542
a_track[252]	-9.464596e-02	0.31016601	-0.5838240134	0.398835486	1.0036455
a_track[253]	6.096202e-02	0.30792797	-0.4343636402	0.542176557	1.0004412
a_track[254]	5.965948e-02	0.31681723	-0.4429166331	0.562614513	1.0024388
a_track[255]	-2.285494e-01	0.28540668	-0.6730912810	0.228586679	0.9998302
a_track[256]	-8.188763e-02	0.29780239	-0.5517955091	0.402307807	1.0023611
a_track[257]	2.132309e-01	0.33914190	-0.3218376178	0.763255553	1.0008206

a_track[258]	1.975660e-01	0.35461524	-0.3794933585	0.759701081	1.0014339
a_track[259]	2.111778e-01	0.34374655	-0.3369735287	0.761000795	1.0005807
a_track[260]	-7.960204e-02	0.30199266	-0.5666199817	0.403374177	1.0005359
a_track[261]	7.309940e-02	0.30520985	-0.4030055887	0.568348126	0.9996776
a_track[262]	9.193263e-02	0.30982374	-0.3976870886	0.589790105	1.0009095
a_track[263]	7.684890e-02	0.31178367	-0.4181987337	0.577716074	0.9999340
a_track[264]	7.015804e-02	0.32266568	-0.4326461783	0.584960685	1.0010446
a_track[265]	2.129643e-01	0.34925758	-0.3393947273	0.786486065	1.0028669
a_track[266]	7.099086e-02	0.30379969	-0.4120446956	0.554753278	1.0003805
a_track[267]	-7.894282e-02	0.30835470	-0.5667182483	0.419332165	1.0033263
a_track[268]	5.981519e-02	0.31768265	-0.4188098103	0.566471183	1.0003740
a_track[269]	-7.236945e-02	0.31133967	-0.5555186291	0.429128242	1.0015563
a_track[270]	2.261932e-01	0.33255159	-0.2940908124	0.758729359	1.0017587
a_track[271]	-8.291263e-02	0.30790234	-0.5819453156	0.416939692	0.9997892
a_track[272]	7.258097e-02	0.31570116	-0.4194275656	0.575059034	1.0036054
a_track[273]	-8.398596e-02	0.29948075	-0.5600900354	0.392406123	1.0009987
a_track[274]	1.770889e-01	0.34284966	-0.3704302538	0.725987800	1.0005838
a_track[275]	-8.043447e-02	0.30804119	-0.5697817813	0.409930181	1.0011570
a_track[276]	7.174570e-02	0.31766890	-0.4339962771	0.590380594	1.0016652
a_track[277]	7.101203e-02	0.32531930	-0.4420050156	0.591998813	1.0035521
a_track[278]	-8.272463e-02	0.29002044	-0.5366335992	0.381590099	1.0000851
a_track[279]	4.494128e-02	0.32694231	-0.4631658950	0.566816100	1.0018963
a_track[280]	1.757519e-01	0.34934229	-0.3854781594	0.749383916	1.0008527
a_track[281]	1.313986e-01	0.35196089	-0.4317956468	0.688949571	1.0008642
a_track[282]	2.094642e-01	0.33704052	-0.3214678853	0.749145702	0.9999720
a_track[283]	-2.166643e-01	0.29357622	-0.6832571777	0.249182956	1.0022737
a_track[284]	2.564045e-01	0.32444657	-0.2496374939	0.769305940	1.0001301
a_track[285]	2.436588e-01	0.33394489	-0.2873365603	0.789038641	1.0000736
a_track[286]	8.483463e-02	0.32410231	-0.4331926529	0.605701151	1.0013828
a_track[287]	1.970346e-01	0.34300805	-0.3299240495	0.756072156	1.0007809
a_track[288]	1.659830e-01	0.36271228	-0.4123485000	0.743354315	1.0006360
a_track[289]	1.721639e-01	0.33902048	-0.3688504684	0.711383692	1.0013601
a_track[290]	-2.015282e-01	0.29537660	-0.6762001597	0.257080194	1.0009518
a_track[291]	4.168600e-02	0.33920916	-0.5086474448	0.583758540	1.0049665
a_track[292]	-6.175913e-01	0.25452522	-1.0158869835	-0.207354631	1.0016681
a_track[293]	6.213930e-02	0.32195381	-0.4442489923	0.580873628	1.0014333
a_track[294]	-2.017186e-01	0.31780304	-0.7113779798	0.304261171	1.0039664
a_track[295]	-7.631888e-02	0.30081827	-0.5577838741	0.405936089	1.0006846
a_track[296]	6.459980e-02	0.33041015	-0.4629332671	0.596799839	0.9997217
a_track[297]	1.930644e-01	0.34824196	-0.3495096765	0.763881138	1.0003812
a_track[298]	2.581450e-01	0.32033016	-0.2428855642	0.773170408	1.0008378
a_track[299]	8.262123e-02	0.30567592	-0.4097109076	0.572399984	1.0028504
a_track[300]	2.359096e-01	0.33426919	-0.2842744697	0.760039446	1.0008209

a_track[301]	2.406112e-01	0.32171274	-0.2648563463	0.750330212	1.0021721
a_track[302]	8.393917e-02	0.31429529	-0.4193617553	0.589355439	1.0017433
a_track[303]	-7.022229e-02	0.30587500	-0.5581326339	0.415205614	1.0019138
a_track[304]	7.272404e-02	0.30516430	-0.4089552277	0.557796379	1.0012357
a_track[305]	-6.369544e-02	0.30239443	-0.5500909937	0.421140650	1.0033071
a_track[306]	1.764937e-01	0.35590170	-0.3981389917	0.748866740	1.0019050
a_track[307]	2.392666e-01	0.33171165	-0.2968573092	0.784049833	1.0022648
a_track[308]	8.038986e-02	0.31643352	-0.4119452116	0.611194251	1.0011404
a_track[309]	5.735749e-02	0.33212711	-0.4585485576	0.587001769	1.0021597
a_track[310]	8.568888e-02	0.31332383	-0.4187029165	0.591740125	1.0034035
a_track[311]	1.939422e-01	0.35105204	-0.3636031153	0.752689567	1.0008121
a_track[312]	-2.069908e-01	0.29302769	-0.6731067402	0.277842008	1.0009053
a_track[313]	-6.799141e-02	0.29833415	-0.5493325839	0.404182995	1.0004640
a_track[314]	8.310836e-02	0.30809272	-0.4000056901	0.582364463	1.0020218
a_track[315]	-2.145272e-01	0.28064228	-0.6515894258	0.239944055	1.0012746
a_track[316]	-6.892909e-02	0.28225265	-0.5195161058	0.392744508	1.0002642
a_track[317]	1.047643e-01	0.30353804	-0.3701987816	0.602274397	1.0020251
a_track[318]	7.613844e-02	0.31981571	-0.4310250156	0.578846578	1.0020425
a_track[319]	-1.727955e-01	0.30813768	-0.6764846871	0.319962246	1.0030568
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a_track[323]	2.394696e-01	0.33467215	-0.2844913775	0.770223959	1.0027589
a_track[324]	7.915291e-02	0.32359683	-0.4378078168	0.598597424	1.0023816
a_track[325]	1.552129e-01	0.36146065	-0.4154462890	0.733501172	1.0014283
a_track[326]	-5.914168e-02	0.28651872	-0.5133253035	0.403365594	1.0000188
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a_track[330]	8.718207e-02	0.31893048	-0.4184819906	0.603943261	1.0016533
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a_track[332]	7.693845e-02	0.31516774	-0.4217540578	0.594525590	1.0021088
a_track[333]	-1.998268e-01	0.28037295	-0.6507077706	0.248296151	1.0022535
a_track[334]	9.797643e-02	0.30780675	-0.3954217949	0.599113828	1.0000367
a_track[335]	8.757234e-02	0.30903158	-0.3838820712	0.577957129	1.0015573
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a_track[339]	-6.679975e-02	0.30184962	-0.5474609420	0.417305365	1.0000746
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a_track[341]	-6.668675e-02	0.28670138	-0.5102000968	0.401580042	0.9993738
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a_track[343]	7.511840e-02	0.32062216	-0.4376647254	0.576283039	1.0015026

a_track[344]	-2.304908e-01	0.28511106	-0.6824463757	0.223249547	1.0005954
a_track[345]	2.382638e-01	0.32889585	-0.2850262484	0.756621162	0.9997650
a_track[346]	2.342943e-01	0.32131169	-0.2754987791	0.763702144	1.0015555
a_track[347]	-6.719158e-02	0.29805235	-0.5398399118	0.406139846	1.0009977
a_track[348]	7.490891e-02	0.31604624	-0.4227794824	0.585410890	1.0029605
a_track[349]	4.546449e-02	0.32407463	-0.4813997938	0.566756027	1.0009550
a_track[350]	2.104084e-01	0.33484895	-0.3196433864	0.753667894	0.9995552
a_track[351]	-2.335851e-01	0.29787811	-0.7153319097	0.248705988	1.0004143
a_track[352]	1.800597e-01	0.34670477	-0.3805679744	0.731030310	1.0014531
a_track[353]	7.262129e-02	0.30241856	-0.4076376835	0.558147945	1.0000598
a_track[354]	6.379909e-02	0.32401355	-0.4634318956	0.578122765	1.0027075
a_track[355]	-8.711878e-02	0.29251003	-0.5449810398	0.383067348	1.0000257
a_track[356]	-2.293457e-01	0.30098620	-0.7191325658	0.253339061	1.0001751
a_track[357]	4.520503e-02	0.33509556	-0.4853415416	0.588102677	1.0014832
a_track[358]	1.345414e-01	0.35225064	-0.4296029242	0.696790044	1.0005219
a_track[359]	4.858801e-02	0.34229745	-0.5022495492	0.589731730	1.0012707
a_track[360]	7.237005e-02	0.30918114	-0.4200981883	0.571302611	1.0009917
a_track[361]	1.939650e-01	0.34898242	-0.3537971705	0.742801452	1.0026484
a_track[362]	5.974256e-02	0.32773646	-0.4607111304	0.585009126	1.0002116
a_track[363]	-8.157076e-02	0.30404600	-0.5663990373	0.410956411	1.0012545
a_track[364]	-8.467977e-02	0.30575956	-0.5798695276	0.404629273	1.0013000
a_track[365]	-7.845569e-02	0.29624128	-0.5591028423	0.411829799	1.0009823
a_track[366]	4.535863e-02	0.33663032	-0.4872103377	0.579163267	1.0009370
a_track[367]	5.195050e-02	0.32848498	-0.4605460008	0.582040044	1.0018130
a_track[368]	1.621553e-01	0.35457114	-0.3987300668	0.731903010	1.0025508
a_track[369]	4.989199e-02	0.32025637	-0.4625629630	0.561322114	1.0051685
a_track[370]	5.648545e-02	0.32361945	-0.4535183286	0.573941713	1.0009407
a_track[371]	-5.028954e-01	0.27468888	-0.9390209038	-0.062217148	1.0010261
a_track[372]	-7.183681e-02	0.31035368	-0.5598237347	0.420833756	1.0003286
a_track[373]	-2.260371e-01	0.28578735	-0.6745838835	0.233244231	1.0011728
a_track[374]	4.613842e-02	0.32878172	-0.4805056394	0.560504295	1.0010805
a_track[375]	1.328152e-01	0.35061393	-0.4266128596	0.678825674	1.0005436
a_track[376]	-7.486384e-02	0.32524497	-0.5947923279	0.437030788	1.0011049
a_track[377]	2.078916e-01	0.33894866	-0.3191930284	0.754979044	1.0045135
a_track[378]	2.284477e-01	0.32943592	-0.2944281224	0.767522936	1.0018190
a_track[379]	1.860583e-01	0.33668187	-0.3517874906	0.710239932	1.0000789
a_track[380]	7.597075e-02	0.32870970	-0.4588228591	0.586675264	1.0014128
a_track[381]	8.402592e-02	0.32788633	-0.4321321296	0.612716975	0.9995604
a_track[382]	6.812119e-02	0.31820765	-0.4396137192	0.571122491	1.0005965
a_track[383]	2.787175e-01	0.32227163	-0.2200594250	0.800816513	0.9997160
a_track[384]	9.314736e-02	0.31021015	-0.3919049200	0.595754320	1.0019401
a_track[385]	-4.651200e-02	0.28980966	-0.5016352282	0.425774685	1.0007877
a_track[386]	-5.860473e-02	0.31261524	-0.5521244726	0.437848189	1.0006663

a_track[387]	2.298567e-01	0.33587361	-0.2956869047	0.782019688	1.0019941
a_track[388]	-5.568951e-02	0.28670022	-0.5178965852	0.410723452	1.0008241
a_track[389]	9.951359e-02	0.30540946	-0.3773375997	0.593670994	1.0022710
a_track[390]	-5.340214e-02	0.29503914	-0.5240779328	0.417471512	1.0002827
a_track[391]	-6.858561e-02	0.28688592	-0.5174344274	0.390423715	1.0014804
a_track[392]	-5.841916e-02	0.29207225	-0.5162508062	0.402465105	1.0019374
a_track[393]	2.090931e-01	0.34048390	-0.3394843912	0.759151938	1.0035972
a_track[394]	-2.167067e-01	0.28832019	-0.6794263518	0.240620571	1.0012093
a_track[395]	-6.197664e-02	0.29985689	-0.5343818840	0.419770160	1.0005561
a_track[396]	-6.768296e-02	0.29051269	-0.5262244256	0.405965714	1.0013700
a_track[397]	7.958904e-02	0.32540361	-0.4355993955	0.609108478	1.0020313
a_track[398]	8.686755e-02	0.31177738	-0.4085799950	0.593265599	1.0008878
a_track[399]	7.966220e-02	0.32661396	-0.4493389626	0.609819280	1.0005316
a_track[400]	8.610011e-02	0.29539237	-0.3741262829	0.559101498	0.9999022
a_track[401]	-6.161162e-02	0.30003682	-0.5363911597	0.418849970	1.0012810
a_track[402]	8.709949e-02	0.31322676	-0.4095214807	0.580186271	1.0001417
a_track[403]	-5.845685e-02	0.31035957	-0.5539338856	0.437032228	1.0025615
a_track[404]	-6.180695e-02	0.30496790	-0.5433686644	0.427883134	0.9997303
a_track[405]	7.561278e-02	0.32931471	-0.4496183016	0.610586076	1.0013374
a_track[406]	1.688500e-01	0.34396661	-0.3667021331	0.708989561	1.0006047
a_track[407]	-5.987183e-02	0.30337758	-0.5378748643	0.409477573	0.9996103
a_track[408]	-5.817899e-02	0.30959051	-0.5510906836	0.426628302	1.0034739
a_track[409]	-4.669874e-02	0.30944135	-0.5420561700	0.440913950	1.0005536
a_track[410]	-3.467938e-01	0.26743065	-0.7700818854	0.091926203	1.0007811
a_track[411]	4.249712e-02	0.33740664	-0.4917991922	0.571153646	1.0017717
a_track[412]	4.056505e-02	0.35046656	-0.5060779660	0.577070238	1.0016130
a_track[413]	-3.321513e-01	0.28216278	-0.7778564706	0.105482686	1.0010628
a_track[414]	1.910551e-01	0.34345404	-0.3485602304	0.737828087	0.9998066
a_track[415]	-5.333245e-02	0.28488951	-0.5057264623	0.402315266	1.0008554
a_track[416]	7.518558e-02	0.31760462	-0.4400158604	0.584522445	1.0004936
a_track[417]	2.230775e-01	0.33553270	-0.3163697142	0.767746837	1.0020821
a_track[418]	-6.223958e-02	0.29557483	-0.5342580462	0.410641280	1.0012253
a_track[419]	-6.299253e-02	0.29836960	-0.5320410323	0.412285727	1.0015615
a_track[420]	-6.021738e-02	0.30440748	-0.5397658390	0.420031914	1.0025854
a_track[421]	-6.316498e-02	0.29258080	-0.5231551337	0.414718799	0.9997107
a_track[422]	1.673157e-01	0.35122382	-0.3990074426	0.737862888	0.9997885
a_track[423]	8.903486e-02	0.31905510	-0.4150735923	0.601474343	1.0009961
a_track[424]	1.872642e-01	0.34776116	-0.3795400681	0.737250226	1.0026920
a_track[425]	-6.319126e-02	0.30931787	-0.5692850299	0.416585161	0.9996524
a_track[426]	1.030377e-01	0.30976740	-0.3932061769	0.603912934	1.0020645
a_track[427]	-2.201624e-01	0.28967550	-0.6784872984	0.233304539	1.0009045
a_track[428]	7.631996e-02	0.31457892	-0.4163546282	0.580268357	1.0000892
a_track[429]	-7.130103e-02	0.30923302	-0.5650086876	0.417664819	1.0023696

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a_track[432]	2.143325e-01	0.33338631	-0.3120103900	0.744464452	1.0002749
a_track[433]	7.097340e-02	0.32365625	-0.4493000503	0.603484956	1.0007842
a_track[434]	2.306408e-01	0.32498896	-0.3023451810	0.757525128	1.0023288
a_track[435]	-3.577952e-01	0.27319752	-0.7877662514	0.079750464	1.0013652
a_track[436]	1.869666e-01	0.34178567	-0.3599232934	0.734359387	1.0024786
a_track[437]	6.572705e-02	0.31692181	-0.4358529699	0.579461144	1.0002833
a_track[438]	-6.698623e-02	0.29343134	-0.5345712793	0.407645077	0.9999705
a_track[439]	8.373097e-02	0.31342886	-0.4177778763	0.583656772	1.0022775
a_track[440]	-6.479477e-02	0.31314058	-0.5684930755	0.424633295	1.0013053
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a_track[462]	8.439890e-02	0.31725937	-0.4164883027	0.609664580	1.0012500
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a_track[467]	-5.153956e-02	0.30300066	-0.5334334180	0.435494854	1.0008417
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a_track[496]	-3.253458e-02	0.21588068	-0.3767332531	0.310340409	1.0015326
a_track[497]	-3.319846e-02	0.22835360	-0.3961836338	0.349470822	0.9999875
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a_cell[8]	5.283670e-03	0.06738530	-0.1034247881	0.113270360	1.0858759
a_cell[9]	2.238846e-02	0.07028744	-0.0766860819	0.142465830	1.0826179

a_cell[10]	-1.129470e-03	0.06878033	-0.1023316333	0.105380265	1.0931661
a_cell[11]	1.785528e-03	0.06993104	-0.1077815529	0.114142687	1.0936197
a_cell[12]	-3.216621e-03	0.07133947	-0.1179166717	0.107630732	1.1152801
a_cell[13]	-1.834570e-02	0.07033972	-0.1389267055	0.083484706	1.1022965
a_cell[14]	-1.335512e-02	0.07326436	-0.1353549326	0.091935003	1.0732921
a_cell[15]	3.498195e-02	0.07508010	-0.0616812135	0.177468037	1.0714659
a_cell[16]	-3.784342e-03	0.06787911	-0.1146415693	0.104369165	1.1081542
a_cell[17]	-4.076221e-03	0.07266521	-0.1211268648	0.105224028	1.1134398
a_cell[18]	2.621917e-02	0.07081639	-0.0665694921	0.153603632	1.0846531
a_cell[19]	1.028314e-02	0.06915544	-0.0919046280	0.123702307	1.1030995
a_cell[20]	-4.045970e-03	0.06818145	-0.1139822238	0.103482995	1.0906143
a_cell[21]	3.200361e-03	0.06687594	-0.1041167619	0.111865052	1.1089343
a_cell[22]	-1.616368e-02	0.06818721	-0.1308034006	0.082526082	1.0793002
a_cell[23]	-1.812203e-02	0.07049419	-0.1414631909	0.084490585	1.1062224
a_cell[24]	2.265367e-02	0.06758690	-0.0701176425	0.138980690	1.0778663
a_cell[25]	-1.189482e-02	0.06634630	-0.1225287239	0.087160414	1.0822672
a_cell[26]	1.089523e-02	0.06489293	-0.0875081569	0.122050530	1.0893492
a_cell[27]	1.144042e-03	0.06927254	-0.1057011157	0.110622574	1.1010941
a_cell[28]	2.071304e-03	0.06977858	-0.1083271790	0.110528476	1.0931955
a_cell[29]	4.220115e-04	0.07055582	-0.1131001960	0.111348003	1.0922773
a_cell[30]	-8.321524e-03	0.07006244	-0.1212337207	0.096395988	1.0947443
a_cell[31]	2.320060e-03	0.07078353	-0.1076378414	0.116501366	1.0972865
a_cell[32]	2.019138e-03	0.06889232	-0.1050448613	0.108283833	1.1059651
a_cell[33]	-8.113242e-03	0.07216780	-0.1248125880	0.100222896	1.0991107
a_cell[34]	-3.593687e-03	0.06829563	-0.1138935780	0.100277033	1.0989412
a_cell[35]	7.652269e-03	0.06934097	-0.0977304390	0.122152077	1.0988556
a_cell[36]	2.382154e-03	0.06936957	-0.1032849152	0.112762005	1.0957608
a_cell[37]	-1.248090e-03	0.06987349	-0.1135665345	0.107460122	1.1151013
a_cell[38]	-1.817294e-02	0.06738875	-0.1328633947	0.078302883	1.0894560
a_cell[39]	1.133642e-02	0.06533925	-0.0858469084	0.122419187	1.0912194
a_cell[40]	9.269018e-03	0.06383949	-0.0900286793	0.113054842	1.0992074
a_cell[41]	8.656246e-03	0.06217947	-0.0821319230	0.113650822	1.0945984
a_cell[42]	-3.374369e-02	0.06786218	-0.1573299905	0.054220491	1.0711794
a_cell[43]	1.471943e-02	0.06743279	-0.0831883455	0.132848544	1.0990383
a_cell[44]	-1.542407e-02	0.06608626	-0.1324335903	0.082553055	1.1050310
a_cell[45]	-4.582506e-03	0.06194703	-0.1063398037	0.092527460	1.0788705
a_cell[46]	-2.048353e-03	0.06662547	-0.1085653158	0.107375950	1.0946930
a_cell[47]	-2.933332e-04	0.06718590	-0.1096627969	0.106569301	1.0952156
a_cell[48]	1.479015e-02	0.07081954	-0.0902335511	0.136286699	1.0844399
a_cell[49]	-1.094216e-02	0.06677436	-0.1227821591	0.090418960	1.0825974
a_cell[50]	1.002833e-02	0.06484832	-0.0884369836	0.118929579	1.0992860
a_cell[51]	1.263614e-02	0.06976747	-0.0921517407	0.126764545	1.0857562
a_cell[52]	4.832534e-03	0.06688191	-0.0962921604	0.110503755	1.1015079

a_cell[53]	-3.392510e-03	0.06995013	-0.1144591403	0.104320522	1.0938221
a_cell[54]	3.578041e-03	0.06967483	-0.1041118176	0.118087071	1.0981669
a_cell[55]	-4.900822e-03	0.06665144	-0.1119238919	0.097156265	1.0852972
a_cell[56]	-1.124815e-02	0.06542707	-0.1235955255	0.086930625	1.0943266
a_cell[57]	-4.381069e-03	0.06607154	-0.1094608791	0.094502996	1.0898024
a_cell[58]	1.456563e-02	0.06661440	-0.0824647577	0.128722356	1.0893673
a_cell[59]	-3.099446e-02	0.06910482	-0.1551098214	0.062605070	1.0603960
a_cell[60]	2.268666e-02	0.06710699	-0.0723165021	0.138437664	1.0928817
a_cell[61]	-2.570799e-02	0.06914689	-0.1477538739	0.067352550	1.0926342
a_cell[62]	5.577116e-05	0.06283741	-0.1004828241	0.099674943	1.0856774
a_cell[63]	2.146926e-02	0.06653431	-0.0721495733	0.139415992	1.0940994
a_cell[64]	-3.477805e-02	0.07080295	-0.1655793191	0.057798825	1.0621792
a_cell[65]	-1.839522e-03	0.06748799	-0.1126045951	0.100546940	1.1106773
a_cell[66]	-1.030484e-02	0.06696178	-0.1238453354	0.088940118	1.0829486
a_cell[67]	3.034527e-03	0.06668722	-0.0995874519	0.109344801	1.0942683
a_cell[68]	3.891178e-03	0.06536906	-0.1004882163	0.112074413	1.0880960
a_cell[69]	-1.371931e-02	0.06713074	-0.1257418839	0.086128113	1.0815295
a_cell[70]	-9.767838e-04	0.06972121	-0.1164306959	0.111802466	1.1054500
a_cell[71]	7.873299e-03	0.06970899	-0.0975650668	0.127335163	1.1199142
a_cell[72]	-8.047829e-03	0.06827826	-0.1224745995	0.096975236	1.0874804
a_cell[73]	-4.869920e-03	0.07144829	-0.1182918192	0.103926154	1.1012225
a_cell[74]	-1.292622e-02	0.06838408	-0.1292548569	0.086854195	1.0855313
a_cell[75]	1.426035e-02	0.07046962	-0.0871958035	0.134228979	1.0897917
a_cell[76]	-8.842148e-03	0.06807996	-0.1206697121	0.097449418	1.1055799
a_cell[77]	-4.514557e-03	0.06780764	-0.1101392910	0.097805319	1.0859038
a_cell[78]	2.400249e-02	0.07026563	-0.0742059506	0.149877757	1.0818030
a_cell[79]	1.345846e-02	0.06726848	-0.0840735531	0.128505180	1.0859788
a_cell[80]	-1.825924e-02	0.06876204	-0.1375206703	0.081458315	1.0882890
a_cell[81]	-2.292516e-02	0.07083946	-0.1470367708	0.073722439	1.1060996
a_cell[82]	2.901439e-03	0.06655603	-0.0996369863	0.110580930	1.0928791
a_cell[83]	7.798380e-03	0.06385306	-0.0910157604	0.115543189	1.0951839
a_cell[84]	-3.980718e-02	0.07219633	-0.1737001145	0.052187331	1.0662149
a_cell[85]	-1.233976e-02	0.06593089	-0.1236038833	0.086305922	1.0894143
a_cell[86]	-6.511730e-03	0.06504660	-0.1155893318	0.097812232	1.0852510
a_cell[87]	2.135831e-02	0.06949358	-0.0796324966	0.143041807	1.0913485
a_cell[88]	-1.011687e-02	0.06753334	-0.1219801308	0.089943862	1.1006358
a_cell[89]	2.423853e-02	0.06996675	-0.0706457716	0.148425694	1.0783291
a_cell[90]	-1.847171e-02	0.06978126	-0.1342797260	0.081572265	1.1001873
a_cell[91]	-8.200057e-03	0.07007196	-0.1207438822	0.096843585	1.0949645
a_cell[92]	-6.777408e-03	0.06745249	-0.1137718861	0.096281484	1.1016531
a_cell[93]	6.799590e-03	0.06818556	-0.0971113361	0.122348410	1.0861902
a_cell[94]	2.532087e-04	0.07206233	-0.1148221736	0.116770132	1.1025386
a_cell[95]	-1.040280e-02	0.07385075	-0.1309492779	0.097672114	1.1187292

a_cell[96]	6.170429e-03	0.06502998	-0.0898026965	0.114100322	1.0935706
a_cell[97]	-2.651834e-03	0.06654008	-0.1111989543	0.101567985	1.0811497
a_cell[98]	-1.456039e-02	0.06987521	-0.1315717550	0.086299987	1.0931841
a_cell[99]	4.287374e-02	0.07651101	-0.0538969337	0.184933063	1.0616734
a_cell[100]	-2.560728e-03	0.06816771	-0.1088909074	0.102507389	1.0939781
a_cell[101]	-1.038542e-02	0.06980232	-0.1284642098	0.094768167	1.0986237
a_cell[102]	1.137185e-02	0.06876483	-0.0879433232	0.127707221	1.0940581
a_cell[103]	-2.785974e-03	0.06884014	-0.1165239329	0.100261443	1.1013456
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a_cell[106]	-9.990885e-03	0.06892798	-0.1291016663	0.094762410	1.0930107
a_cell[107]	-2.691944e-02	0.06923446	-0.1479304244	0.066103067	1.0770226
a_cell[108]	-3.594708e-03	0.06612829	-0.1137467388	0.098138060	1.0950990
a_cell[109]	-1.628254e-03	0.06775795	-0.1101433015	0.105682392	1.1079057
a_cell[110]	-1.896587e-02	0.06748021	-0.1372993143	0.078055952	1.0733092
a_cell[111]	-2.715810e-02	0.06789449	-0.1467548962	0.064787867	1.0676200
a_cell[112]	-1.082750e-02	0.06752343	-0.1193187665	0.088391453	1.0805453
a_cell[113]	-1.370261e-02	0.06842397	-0.1318533763	0.086916105	1.0809168
a_cell[114]	1.503729e-03	0.06640814	-0.1045460144	0.109917745	1.0997911
a_cell[115]	-4.060688e-03	0.07118808	-0.1218663014	0.107989532	1.0972626
a_cell[116]	-1.390175e-02	0.06965734	-0.1293829307	0.088701953	1.1017972
a_cell[117]	-3.889852e-03	0.07250272	-0.1182791282	0.103470328	1.0954495
a_cell[118]	-4.958540e-03	0.06725648	-0.1135213875	0.099234994	1.0822433
a_cell[119]	3.937503e-02	0.07530248	-0.0552361895	0.172286539	1.0756334
a_cell[120]	1.744303e-03	0.06605528	-0.1043709367	0.105325923	1.1117303
a_cell[121]	7.462748e-03	0.06887442	-0.0934543398	0.121682596	1.0967216
a_cell[122]	2.081003e-02	0.06822479	-0.0751901855	0.141777104	1.0875178
a_cell[123]	-8.337231e-03	0.07003244	-0.1282837124	0.101007453	1.0982586
a_cell[124]	-9.368393e-03	0.06656399	-0.1236213519	0.091646020	1.0801595
a_cell[125]	-1.608618e-02	0.06832148	-0.1357292446	0.085944959	1.0897004
a_cell[126]	1.377754e-02	0.06740614	-0.0884122715	0.126862943	1.0853534
a_cell[127]	6.409765e-03	0.06629138	-0.0960640211	0.116754672	1.0994834
a_cell[128]	-1.559724e-02	0.06627972	-0.1289545671	0.083875772	1.0918130
a_cell[129]	1.850023e-02	0.06676416	-0.0764023519	0.133356920	1.0800190
a_cell[130]	9.454129e-03	0.06660933	-0.0902203996	0.122516505	1.0905111
a_cell[131]	-2.978390e-02	0.07163886	-0.1580264385	0.068700630	1.0830831
a_cell[132]	8.019923e-03	0.06468065	-0.0887628685	0.116894442	1.0834855
a_cell[133]	-6.529350e-03	0.06964432	-0.1226513827	0.099865539	1.1022625
a_cell[134]	-8.712161e-03	0.06515160	-0.1161841712	0.089106061	1.0967621
a_cell[135]	-2.436766e-02	0.06890715	-0.1463170911	0.070037723	1.0775635
a_cell[136]	1.878190e-02	0.06622613	-0.0751053717	0.133090572	1.0991602
a_cell[137]	1.094890e-02	0.06682019	-0.0870005230	0.121186744	1.0759836
a_cell[138]	2.836539e-03	0.06726570	-0.1016063453	0.114810749	1.0928636

a_cell[139]	-9.746989e-03	0.06649808	-0.1223349276	0.089032522	1.0830071
a_cell[140]	1.182359e-02	0.06863410	-0.0926159863	0.125724046	1.0956976
a_cell[141]	-1.290718e-02	0.07021902	-0.1335130655	0.090329679	1.0845716
a_cell[142]	-2.556098e-03	0.06952472	-0.1187394550	0.105147646	1.1109770
a_cell[143]	1.404551e-03	0.07091584	-0.1084345735	0.110575487	1.0897515
a_cell[144]	2.091404e-03	0.06661737	-0.1061657869	0.110748413	1.0882268
a_cell[145]	-1.351581e-02	0.06901076	-0.1283244334	0.088851266	1.0919058
a_cell[146]	5.989504e-03	0.06557098	-0.0948383448	0.111449763	1.0893776
a_cell[147]	1.074779e-03	0.07068424	-0.1066421637	0.113398517	1.1085472
a_cell[148]	-3.486526e-03	0.06805567	-0.1158650000	0.099108184	1.0930724
a_cell[149]	-9.166957e-03	0.07351939	-0.1287292791	0.097828874	1.1125650
a_cell[150]	1.439523e-02	0.06800297	-0.0853748172	0.131287400	1.0916819
a_cell[151]	-2.849497e-03	0.06731023	-0.1101403267	0.102128108	1.0947502
a_cell[152]	1.392321e-02	0.06632562	-0.0843832127	0.129011459	1.0793809
a_cell[153]	1.660009e-04	0.06667590	-0.1050452959	0.105570750	1.0956617
a_cell[154]	2.184697e-02	0.07243172	-0.0794996525	0.144098772	1.0864936
a_cell[155]	7.569545e-03	0.06467483	-0.0894558556	0.120552335	1.0785445
a_cell[156]	-1.972906e-02	0.06720282	-0.1328326780	0.076203024	1.0828992
a_cell[157]	-7.448714e-03	0.06393727	-0.1193687131	0.090836439	1.0825500
a_cell[158]	8.247663e-03	0.06597473	-0.0910455143	0.115160910	1.0940945
a_cell[159]	4.139974e-02	0.07183560	-0.0514023470	0.173719138	1.0690123
a_cell[160]	-2.633933e-03	0.06646575	-0.1133613138	0.103000863	1.1055708
a_cell[161]	-1.148312e-02	0.07119898	-0.1279909597	0.094879654	1.1055807
a_cell[162]	-3.893582e-03	0.06517339	-0.1081327822	0.097864016	1.0843679
a_cell[163]	-3.934661e-03	0.06623851	-0.1098294261	0.094509687	1.0872055
a_cell[164]	-5.978633e-03	0.06582821	-0.1113107595	0.092386194	1.0732796
a_cell[165]	-9.844400e-03	0.07169326	-0.1273242848	0.095912760	1.0920203
a_cell[166]	-1.544446e-03	0.06882011	-0.1110750049	0.109332746	1.0917318
a_cell[167]	-3.461860e-03	0.07189390	-0.1197680984	0.102327278	1.1076737
a_cell[168]	-1.648707e-02	0.07408447	-0.1423227097	0.091382840	1.0940847
a_cell[169]	5.182073e-02	0.07897650	-0.0450358745	0.196374907	1.0751911
a_cell[170]	6.104897e-03	0.06888083	-0.0977223941	0.117709116	1.0961645
a_cell[171]	8.426400e-03	0.06678913	-0.0937568851	0.121526399	1.0977138
a_cell[172]	-1.527008e-03	0.07049996	-0.1153863597	0.106893341	1.0963161
a_cell[173]	1.162195e-02	0.06773181	-0.0878439251	0.126096316	1.0929378
a_cell[174]	-1.489248e-03	0.07016434	-0.1123616559	0.108333290	1.1109392
a_cell[175]	-2.321530e-03	0.06939019	-0.1091688286	0.107747385	1.1056119
a_cell[176]	1.058503e-03	0.06888453	-0.1068048396	0.110440447	1.0953409
a_cell[177]	-6.775828e-03	0.06637857	-0.1178419360	0.096541187	1.0757031
a_cell[178]	-1.345647e-02	0.06585764	-0.1260298097	0.084152366	1.1077786
a_cell[179]	-6.847646e-03	0.06737236	-0.1124098177	0.091624230	1.1059627
a_cell[180]	1.481736e-02	0.06563491	-0.0796824709	0.127598233	1.0895007
a_cell[181]	1.186922e-02	0.06531363	-0.0866184547	0.120264325	1.0945615

a_cell[182]	-1.717390e-02	0.06974745	-0.1365583229	0.082151103	1.1047197
a_cell[183]	-1.371361e-02	0.06920748	-0.1336172161	0.089139410	1.0962157
a_cell[184]	7.373622e-03	0.07027767	-0.1000340979	0.125085414	1.0997775
a_cell[185]	6.503856e-03	0.06834088	-0.0992813755	0.117612552	1.0971899
a_cell[186]	-7.505248e-03	0.06384752	-0.1131142710	0.090434523	1.0806676
a_cell[187]	-3.592976e-03	0.06685415	-0.1092666111	0.099588607	1.1005146
a_cell[188]	-1.757414e-03	0.06968197	-0.1100489985	0.106516904	1.0963080
a_cell[189]	9.098665e-04	0.07107101	-0.1132121840	0.113485593	1.1008725
a_cell[190]	-2.706095e-03	0.07326799	-0.1198067540	0.109403198	1.1090423
a_cell[191]	2.453007e-03	0.07106374	-0.1090509931	0.120766839	1.0978300
a_cell[192]	1.115258e-02	0.06856162	-0.0911555426	0.125341913	1.0781327
a_cell[193]	-3.718315e-03	0.06852941	-0.1150028321	0.100414966	1.1102859
a_cell[194]	7.064946e-03	0.06942047	-0.0990110855	0.122731365	1.0961006
a_cell[195]	3.212323e-02	0.07056065	-0.0623549332	0.155486654	1.0682424
a_cell[196]	-1.016290e-02	0.07001839	-0.1301640226	0.098515255	1.0932909
a_cell[197]	2.970793e-02	0.07185810	-0.0672869405	0.156049329	1.0928859
a_cell[198]	2.044252e-03	0.06955403	-0.1043280126	0.109701605	1.0959951
a_cell[199]	7.920271e-05	0.06979816	-0.1093285727	0.105727744	1.0982657
a_cell[200]	-4.664073e-03	0.07270282	-0.1188078763	0.103772294	1.1120286
a_cell[201]	-5.882106e-03	0.06671292	-0.1179017926	0.095851639	1.0787763
a_cell[202]	1.630748e-02	0.06863466	-0.0806106560	0.134197322	1.0958136
a_cell[203]	1.560112e-02	0.06823457	-0.0866377585	0.131957873	1.1008819
a_cell[204]	5.590168e-02	0.07920367	-0.0383540722	0.205388058	1.0826162
a_cell[205]	2.859078e-02	0.06993993	-0.0688778342	0.150018728	1.0838526
a_cell[206]	-8.459864e-03	0.06862382	-0.1179604525	0.096545073	1.0774504
a_cell[207]	-1.084593e-02	0.06477541	-0.1188924272	0.088604366	1.0914012
a_cell[208]	-2.900720e-02	0.07155385	-0.1552409543	0.071327900	1.0738702
a_cell[209]	-5.429843e-03	0.06724928	-0.1173628294	0.101766731	1.0898159
a_cell[210]	7.106108e-03	0.06743489	-0.0938133334	0.116923112	1.0946908
a_cell[211]	5.741858e-02	0.08106303	-0.0374713763	0.212034343	1.0872080
a_cell[212]	-1.938442e-02	0.06905877	-0.1417799332	0.078955702	1.0807747
a_cell[213]	-5.222300e-03	0.06677047	-0.1143548735	0.099447657	1.1041859
a_cell[214]	2.126798e-02	0.07008865	-0.0732661040	0.145791611	1.0913342
a_cell[215]	-4.676900e-03	0.07302733	-0.1239106966	0.108117682	1.1113688
a_cell[216]	-4.318243e-03	0.06944592	-0.1177007979	0.104431124	1.0969000
a_cell[217]	-4.004380e-03	0.07117965	-0.1140764788	0.105598467	1.1011581
a_cell[218]	1.615353e-02	0.07050744	-0.0853228502	0.136988139	1.0959872
a_cell[219]	7.228363e-03	0.07135735	-0.0982884281	0.121877919	1.0914505
a_cell[220]	-6.179381e-03	0.07161093	-0.1265971040	0.103910620	1.1192588
a_cell[221]	8.679546e-03	0.06972023	-0.0969572951	0.123485074	1.0853587
a_cell[222]	1.476842e-03	0.06853119	-0.1030856365	0.110735164	1.0849472
a_cell[223]	1.858558e-03	0.06593607	-0.1007596858	0.110743445	1.0912904
a_cell[224]	7.585033e-04	0.06897512	-0.1065944693	0.111546688	1.1084177

a_cell[225]	1.509415e-02	0.07104214	-0.0894683585	0.135876231	1.0945656
a_cell[226]	5.176249e-05	0.07198221	-0.1120245278	0.116215185	1.1074477
a_cell[227]	1.430146e-02	0.06782158	-0.0846167241	0.124509463	1.0814605
a_cell[228]	-4.612494e-03	0.06937698	-0.1222454132	0.098768781	1.0906897
a_cell[229]	-1.017235e-02	0.06808184	-0.1231322926	0.089664470	1.0947535
a_cell[230]	-3.035015e-02	0.07313041	-0.1600086648	0.064749446	1.0770847
a_cell[231]	2.759883e-04	0.06789939	-0.1090373643	0.104599649	1.0899134
a_cell[232]	-2.743581e-02	0.07351799	-0.1580208136	0.072815038	1.0832127
a_cell[233]	-3.349297e-03	0.06619228	-0.1114127531	0.101452160	1.1037978
a_cell[234]	6.734632e-03	0.06654616	-0.0919067776	0.112459396	1.0803500
a_cell[235]	1.193081e-03	0.07231434	-0.1064040262	0.113070868	1.0969411
a_cell[236]	1.270549e-02	0.06774666	-0.0882731381	0.128373678	1.0719585
a_cell[237]	-3.741091e-03	0.06528254	-0.1110772338	0.099512376	1.0881004
a_cell[238]	-8.903947e-03	0.06758949	-0.1223490591	0.090753776	1.0966068
a_cell[239]	-7.202354e-03	0.07093355	-0.1229112149	0.094704777	1.1013991
a_cell[240]	-3.082504e-03	0.07092830	-0.1192782214	0.108642706	1.0927225
a_cell[241]	3.004419e-02	0.07376192	-0.0677333583	0.162094628	1.0778009
a_cell[242]	-8.958647e-03	0.06867913	-0.1245799171	0.091435890	1.0999239
a_cell[243]	-8.257053e-03	0.07320904	-0.1305219410	0.106488034	1.0952295
a_cell[244]	-7.458035e-03	0.07156207	-0.1272825545	0.099922653	1.0951309
a_cell[245]	-5.182350e-03	0.06808319	-0.1189959079	0.099787843	1.1063989
a_cell[246]	2.282601e-02	0.07288587	-0.0757863419	0.151713420	1.0853879
a_cell[247]	-9.213347e-03	0.07209844	-0.1328798044	0.095679875	1.0881641
a_cell[248]	-3.875740e-03	0.07138520	-0.1178750207	0.107580730	1.0926414
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a_cell[250]	2.491702e-03	0.06931410	-0.1104988771	0.115340100	1.1027221
a_cell[251]	-1.959238e-02	0.07281608	-0.1490536342	0.080033693	1.1035061
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a_cell[253]	-1.839247e-02	0.07162849	-0.1409002895	0.088043190	1.1013541
a_cell[254]	-4.316985e-03	0.06990037	-0.1150514592	0.106365688	1.0939699
a_cell[255]	1.362058e-03	0.07025865	-0.1090533780	0.117078987	1.0928779
a_cell[256]	7.238886e-04	0.06754951	-0.1042915033	0.106654890	1.0929910
a_cell[257]	-1.349593e-02	0.07075333	-0.1306361220	0.090542706	1.0924138
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a_cell[260]	-1.271667e-02	0.07010734	-0.1254652095	0.089492003	1.0889668
a_cell[261]	7.635381e-03	0.06192955	-0.0864122216	0.106160939	1.0784120
a_cell[262]	5.860550e-03	0.06477207	-0.0948421765	0.110787409	1.0920390
a_cell[263]	-1.566146e-02	0.06721146	-0.1338185739	0.081818910	1.0769879
a_cell[264]	1.410571e-02	0.07027317	-0.0885431802	0.132943435	1.0931969
a_cell[265]	-4.132421e-03	0.07179368	-0.1198460830	0.104820167	1.1111113
a_cell[266]	8.657810e-03	0.07095946	-0.0983441506	0.127821189	1.0798562
a_cell[267]	-1.411339e-02	0.07073212	-0.1341143129	0.089591103	1.0904997

a_cell[268]	1.560052e-03	0.07173676	-0.1114928266	0.116544873	1.0985282
a_cell[269]	-7.121048e-03	0.07452715	-0.1311643717	0.101935340	1.1027013
a_cell[270]	-1.685319e-02	0.07199575	-0.1408321556	0.089576134	1.0974298
a_cell[271]	-1.370908e-02	0.06997304	-0.1371501470	0.088098654	1.1039951
a_cell[272]	-1.191984e-02	0.07217499	-0.1341133572	0.096086038	1.0915284
a_cell[273]	3.641408e-03	0.06696141	-0.0986967637	0.110182151	1.1077646
a_cell[274]	-4.626263e-03	0.07117606	-0.1187807427	0.106048909	1.0874115
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a_cell[276]	1.428465e-02	0.07078049	-0.0905595030	0.136527423	1.1094989
a_cell[277]	-1.653481e-02	0.07020459	-0.1362736214	0.084042703	1.0801573
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a_cell[280]	-9.613736e-04	0.07079799	-0.1103182405	0.112608215	1.1050362
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a_cell[289]	5.253034e-04	0.06668034	-0.1049624770	0.110645103	1.1047918
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a_cell[291]	-6.422835e-03	0.07088919	-0.1228824859	0.103051424	1.1141028
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a_cell[293]	1.185041e-02	0.06854743	-0.0850506705	0.126082112	1.0952287
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a_cell[295]	-8.103497e-03	0.06953987	-0.1260208873	0.099280213	1.0911326
a_cell[296]	-5.565360e-03	0.06889298	-0.1168374437	0.101141971	1.0898063
a_cell[297]	-1.363421e-02	0.07457077	-0.1419320400	0.094026764	1.1088551
a_cell[298]	-1.023952e-02	0.06899034	-0.1251841348	0.096410393	1.1057912
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a_cell[301]	1.375080e-02	0.06813150	-0.0859549223	0.125886718	1.1069821
a_cell[302]	8.193849e-04	0.06805400	-0.1071597805	0.107424677	1.1069244
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a_cell[304]	-1.506204e-02	0.07162912	-0.1323732970	0.090049235	1.0938021
a_cell[305]	-1.475555e-02	0.06900796	-0.1303865390	0.087654458	1.0738817
a_cell[306]	-2.115386e-02	0.06971901	-0.1466614927	0.075488926	1.0931444
a_cell[307]	-5.840516e-03	0.07078057	-0.1191192716	0.104286792	1.0894849
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a_cell[310]	-1.154690e-02	0.06934609	-0.1228432714	0.088218835	1.1171979

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a_cell[312]	-2.777065e-02	0.07419875	-0.1551054668	0.067617153	1.0859958
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a_cell[317]	-2.264618e-02	0.07179109	-0.1461015251	0.075354086	1.0812198
a_cell[318]	2.215400e-03	0.06919318	-0.1013636095	0.109850501	1.0856357
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a_cell[320]	-1.562631e-03	0.06937127	-0.1115302980	0.109696919	1.1016565
a_cell[321]	1.144097e-02	0.06949215	-0.0882761443	0.128695185	1.0992799
a_cell[322]	2.256211e-03	0.07179124	-0.1088469025	0.116054633	1.0943121
a_cell[323]	2.682609e-03	0.07009592	-0.1039962915	0.115509410	1.1094800
a_cell[324]	4.930452e-03	0.06860691	-0.1031164981	0.113539069	1.1194908
a_cell[325]	2.484904e-02	0.07251456	-0.0747509540	0.153782925	1.0683980
a_cell[326]	-7.291258e-04	0.07079484	-0.1130811656	0.109624659	1.1005713
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a_cell[333]	-1.284774e-02	0.07078185	-0.1372967349	0.090753362	1.0958386
a_cell[334]	3.204272e-02	0.07149846	-0.0610010802	0.158508346	1.0698640
a_cell[335]	2.279362e-02	0.06985585	-0.0721268660	0.147669006	1.0937935
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a_cell[344]	-1.757817e-02	0.06674399	-0.1309513189	0.081414301	1.0821948
a_cell[345]	-4.797759e-03	0.06678716	-0.1133061451	0.100522088	1.1061329
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a_cell[350]	-7.709849e-05	0.06873418	-0.1059670912	0.106427088	1.0887532
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a_cell[356]	-2.244409e-03	0.07313763	-0.1147113539	0.109173097	1.0943254
a_cell[357]	-5.180797e-03	0.06963839	-0.1180816092	0.100453492	1.0942423
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a_cell[359]	-1.257304e-02	0.07265298	-0.1368977162	0.094897008	1.1098376
a_cell[360]	-2.445964e-04	0.07031801	-0.1139160432	0.109657756	1.1123997
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a_cell[363]	-1.971141e-02	0.06908520	-0.1397667212	0.078983697	1.1075590
a_cell[364]	-1.333041e-02	0.06886234	-0.1322896212	0.086362723	1.0908624
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a_cell[366]	-4.652807e-03	0.06957653	-0.1209908491	0.101826625	1.0933487
a_cell[367]	-1.624148e-02	0.07056300	-0.1326592096	0.086998893	1.0758173
a_cell[368]	-1.865685e-02	0.06812996	-0.1374868847	0.079901901	1.0947956
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a_cell[263]	1336.692473
a_cell[264]	3360.088862
a_cell[265]	4772.354561
a_cell[266]	4726.190753
a_cell[267]	2336.630683
a_cell[268]	5358.110068
a_cell[269]	5807.586512
a_cell[270]	1506.716426
a_cell[271]	3466.146772

a_cell[272]	4397.220408
a_cell[273]	5468.359956
a_cell[274]	5638.807091
a_cell[275]	6069.773400
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a_cell[277]	2350.416168
a_cell[278]	5454.468273
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a_cell[280]	5519.013094
a_cell[281]	3928.694107
a_cell[282]	4947.512999
a_cell[283]	5378.938156
a_cell[284]	246.688610
a_cell[285]	5142.097909
a_cell[286]	174.055665
a_cell[287]	4248.646320
a_cell[288]	891.827969
a_cell[289]	6214.514918
a_cell[290]	4789.950395
a_cell[291]	6157.246036
a_cell[292]	5787.658623
a_cell[293]	5660.591510
a_cell[294]	4519.103460
a_cell[295]	5544.399511
a_cell[296]	4727.479621
a_cell[297]	2244.909869
a_cell[298]	4745.257448
a_cell[299]	5283.676314
a_cell[300]	5675.746770
a_cell[301]	4286.913434
a_cell[302]	5515.786037
a_cell[303]	3708.867653
a_cell[304]	4021.487844
a_cell[305]	3527.107734
a_cell[306]	732.463901
a_cell[307]	4267.996250
a_cell[308]	4204.257215
a_cell[309]	4578.018196
a_cell[310]	5390.878271
a_cell[311]	4936.322065
a_cell[312]	286.658783
a_cell[313]	3839.939839
a_cell[314]	5848.730317

a_cell[315]	158.285407
a_cell[316]	113.191233
a_cell[317]	304.187895
a_cell[318]	5777.043474
a_cell[319]	4988.454195
a_cell[320]	5599.059368
a_cell[321]	3955.564327
a_cell[322]	5860.136011
a_cell[323]	5870.615477
a_cell[324]	5873.754597
a_cell[325]	228.365339
a_cell[326]	7028.440462
a_cell[327]	4961.744442
a_cell[328]	5018.819700
a_cell[329]	5168.898289
a_cell[330]	5709.154908
a_cell[331]	5963.155581
a_cell[332]	5616.994969
a_cell[333]	5468.304401
a_cell[334]	98.118024
a_cell[335]	464.789981
a_cell[336]	5404.382005
a_cell[337]	4168.052879
a_cell[338]	5277.145921
a_cell[339]	4677.719996
a_cell[340]	5626.165882
a_cell[341]	4370.101116
a_cell[342]	5669.949865
a_cell[343]	4252.154963
a_cell[344]	1848.309441
a_cell[345]	6016.998361
a_cell[346]	4555.404398
a_cell[347]	4421.247769
a_cell[348]	4946.709497
a_cell[349]	5264.487108
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a_cell[351]	3557.093952
a_cell[352]	1848.939878
a_cell[353]	5516.284133
a_cell[354]	5216.180888
a_cell[355]	5450.431263
a_cell[356]	5406.330573
a_cell[357]	4831.834073

a_cell[358]	3985.160317
a_cell[359]	3307.567562
a_cell[360]	6253.235980
a_cell[361]	4973.004771
a_cell[362]	5013.216974
a_cell[363]	735.101266
a_cell[364]	4702.765212
a_cell[365]	5266.168391
a_cell[366]	4857.088073
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a_cell[374]	1009.426462
a_cell[375]	5849.163655
a_cell[376]	5113.164378
a_cell[377]	5477.766393
a_cell[378]	1048.335258
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a_cell[380]	5480.148503
a_cell[381]	5135.639990
a_cell[382]	115.792080
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a_cell[384]	4526.283485
a_cell[385]	1021.098806
a_cell[386]	4281.629475
a_cell[387]	4891.466883
a_cell[388]	2996.437459
a_cell[389]	5145.142034
a_cell[390]	1452.399389
a_cell[391]	5652.536198
a_cell[392]	5295.623371
a_cell[393]	5443.399470
a_cell[394]	4234.813421
a_cell[395]	1785.671859
a_cell[396]	3428.947168
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a_cell[398]	5192.809273
a_cell[399]	5638.824165
a_cell[400]	5597.061106

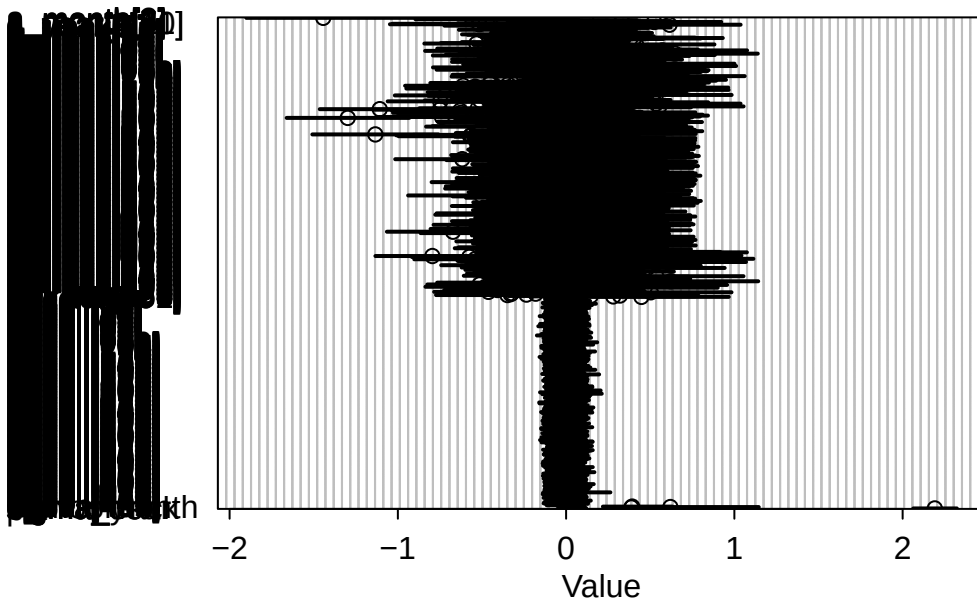
a_cell[401]	4869.551163
a_cell[402]	5151.157044
a_cell[403]	5759.894139
a_cell[404]	4873.578137
a_cell[405]	5622.366922
a_cell[406]	4316.118520
a_cell[407]	4636.450768
a_cell[408]	93.701719
a_cell[409]	5385.446017
a_cell[410]	4603.459067
a_cell[411]	5040.220597
a_cell[412]	4112.578297
a_cell[413]	290.249702
a_cell[414]	4895.268343
a_cell[415]	4118.620452
a_cell[416]	2860.375615
a_cell[417]	4283.134383
a_cell[418]	3552.351748
a_cell[419]	4774.281839
a_cell[420]	5516.298243
a_cell[421]	5239.557259
a_cell[422]	5862.942874
a_cell[423]	5387.823487
a_cell[424]	5754.246852
a_cell[425]	4384.450185
a_cell[426]	19.812132
a_cell[427]	2964.587070
a_cell[428]	6072.386365
a_cell[429]	5111.984325
a_cell[430]	3368.392042
a_cell[431]	4609.311900
a_cell[432]	4689.748452
a_cell[433]	4596.217801
a_cell[434]	4445.881647
a_cell[435]	4360.842460
a_cell[436]	5733.620809
a_cell[437]	5562.269903
a_cell[438]	4121.811013
a_cell[439]	4364.768914
a_cell[440]	6617.039688
a_cell[441]	5621.877922
a_cell[442]	5223.700887
a_cell[443]	4283.120379

```

a_cell[444]    941.904677
a_cell[445]    5332.987652
a_cell[446]    6418.107029
a_cell[447]    4969.273141
a_cell[448]    4766.690084
a_cell[449]    3750.192436
a_cell[450]    2359.989651
a_cell[451]    3823.098379
a_cell[452]    4991.383879
a_cell[453]    6080.655377
a_cell[454]    5674.484258
b_tide         2699.834886
b_moon         2149.208096
sigma_month    1939.591836
sigma_year     2365.847965
sigma_track    1033.536912
sigma_cell      7.709065
phi            1344.589308

```

```
plot(precis(m_space_use, depth = 2))
```



```
par(mfrow = c(2,2))
trankplot(m_space_use, ask = FALSE)
```



a_track[14]eff = 5062



a_track[15]eff = 5743



a_track[16]eff = 6362



a_track[17]eff = 5610



a_track[18]eff = 5386



a_track[19]eff = 5896



a_track[20]eff = 5065



a_track[21]eff = 5379



a_track[22]eff = 5295



a_track[23]eff = 4968



a_track[24]eff = 5911



a_track[25]eff = 4981



a_track[26]eff = 4951



a_track[27]eff = 5017



a_track[28]eff = 5527



a_track[29]eff = 6072



a_track[30]eff = 5996



a_track[31]eff = 5770



a_track[32]eff = 5986



a_track[33]eff = 5915



a_track[34]eff = 6192



a_track[35]eff = 4852



a_track[36]eff = 5764



a_track[37]eff = 5236



a_track[38]eff = 5865



a_track[39]eff = 5623



a_track[40]eff = 4897



a_track[41]eff = 5777



a_track[42]eff = 5547



a_track[43]eff = 5069





a_track[74]eff = 4991



a_track[75]eff = 6206



a_track[76]eff = 5663



a_track[77]eff = 5534



a_track[78]eff = 6401



a_track[79]eff = 5610



a_track[80]eff = 4686



a_track[81]eff = 5556



a_track[82]eff = 5671



a_track[83]eff = 5988



a_track[84]eff = 6151



a_track[85]eff = 6961



a_track[86]eff = 5630



a_track[87]eff = 5409



a_track[88]eff = 6254



a_track[89]eff = 5424



a_track[90]eff = 7296



a_track[91]eff = 6035



a_track[92]eff = 5407



a_track[93]eff = 5297



a_track[94]eff = 6224



a_track[95]eff = 6569



a_track[96]eff = 5756



a_track[97]eff = 4953



a_track[98]eff = 6409



a_track[99]eff = 5029



a_track[100]eff = 5573



a_track[101]eff = 5902



a_track[102]eff = 6970



a_track[103]eff = 5539

















a_track[314]f = 6712



a_track[315]f = 5797



a_track[316]f = 6363



a_track[317]f = 6119



a_track[318]f = 6505



a_track[319]f = 5778



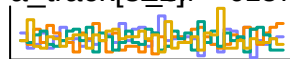
a_track[320]f = 6133



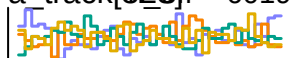
a_track[321]f = 6732



a_track[322]f = 6137



a_track[323]f = 6019



a_track[324]f = 6631



a_track[325]f = 6488



a_track[326]f = 6085



a_track[327]f = 5814



a_track[328]f = 6522



a_track[329]f = 5670



a_track[330]f = 6475



a_track[331]f = 7043



a_track[332]f = 6063



a_track[333]f = 6160



a_track[334]f = 6163



a_track[335]f = 5616



a_track[336]f = 6516



a_track[337]f = 5564



a_track[338]f = 6793



a_track[339]f = 5963



a_track[340]f = 6324



a_track[341]f = 6832



a_track[342]f = 6197



a_track[343]f = 6654













a_track[494]f = 5623



a_track[495]f = 6172



a_track[496]f = 5269



a_track[497]f = 5830



a_track[498]f = 5079



a_track[499]f = 6195



a_track[500]f = 6360



a_track[501]f = 5293



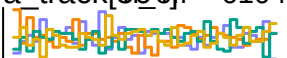
a_track[502]f = 5704



a_track[503]f = 4378



a_track[504]f = 6104



a_track[505]f = 4431



a_track[506]f = 4639



a_track[507]f = 6109



a_track[508]f = 4223



a_track[509]f = 5308



a_track[510]f = 3234



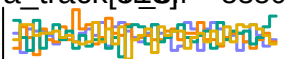
a_track[511]f = 4770



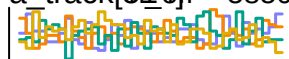
a_track[512]f = 5507



a_track[513]f = 5350



a_track[514]f = 5850



a_track[515]f = 5058



a_track[516]f = 6400



a_track[517]f = 5624



a_track[518]f = 5699



a_track[519]f = 4934



a_track[520]f = 6029



a_track[521]f = 5373



a_track[522]f = 6285



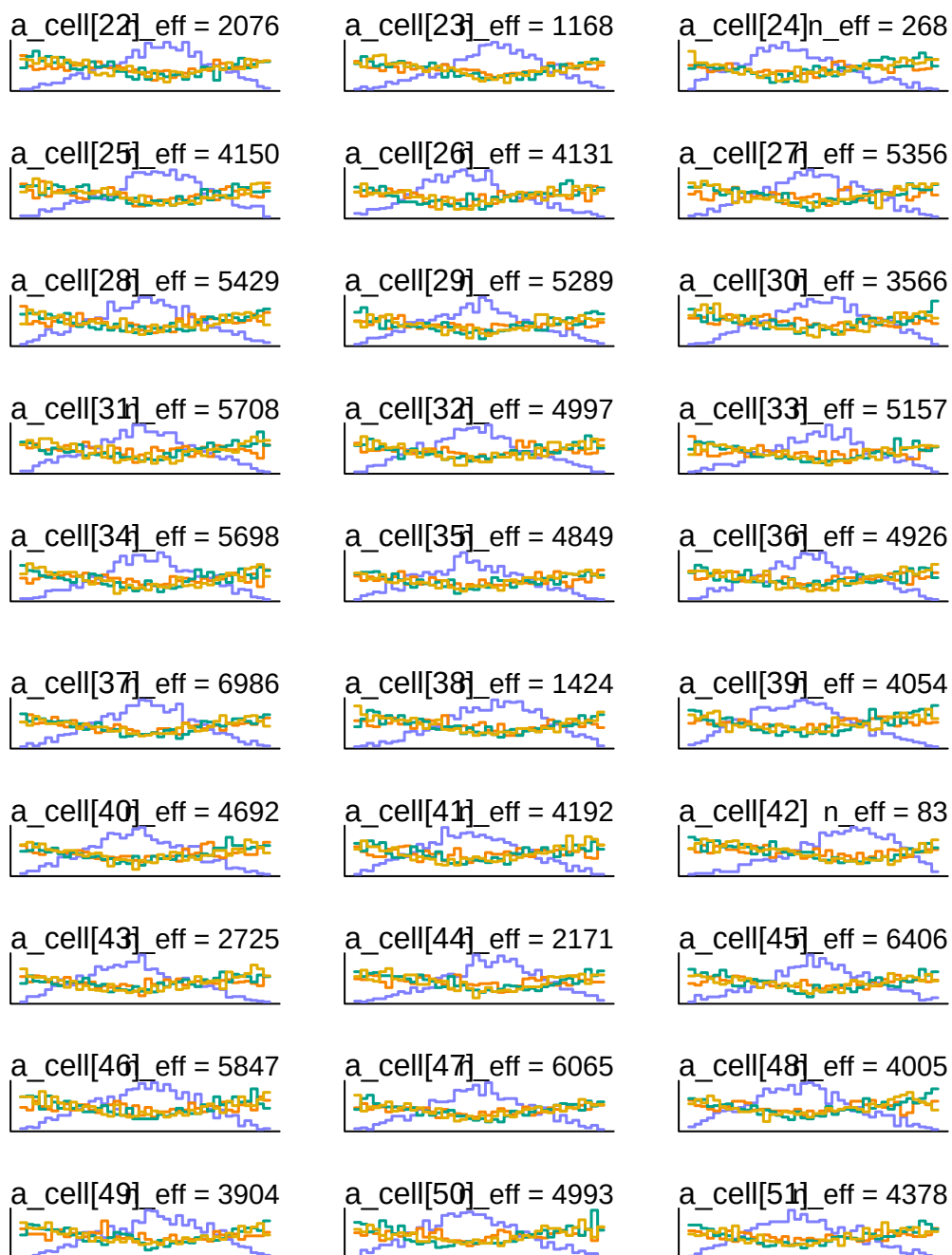
a_track[523]f = 5020

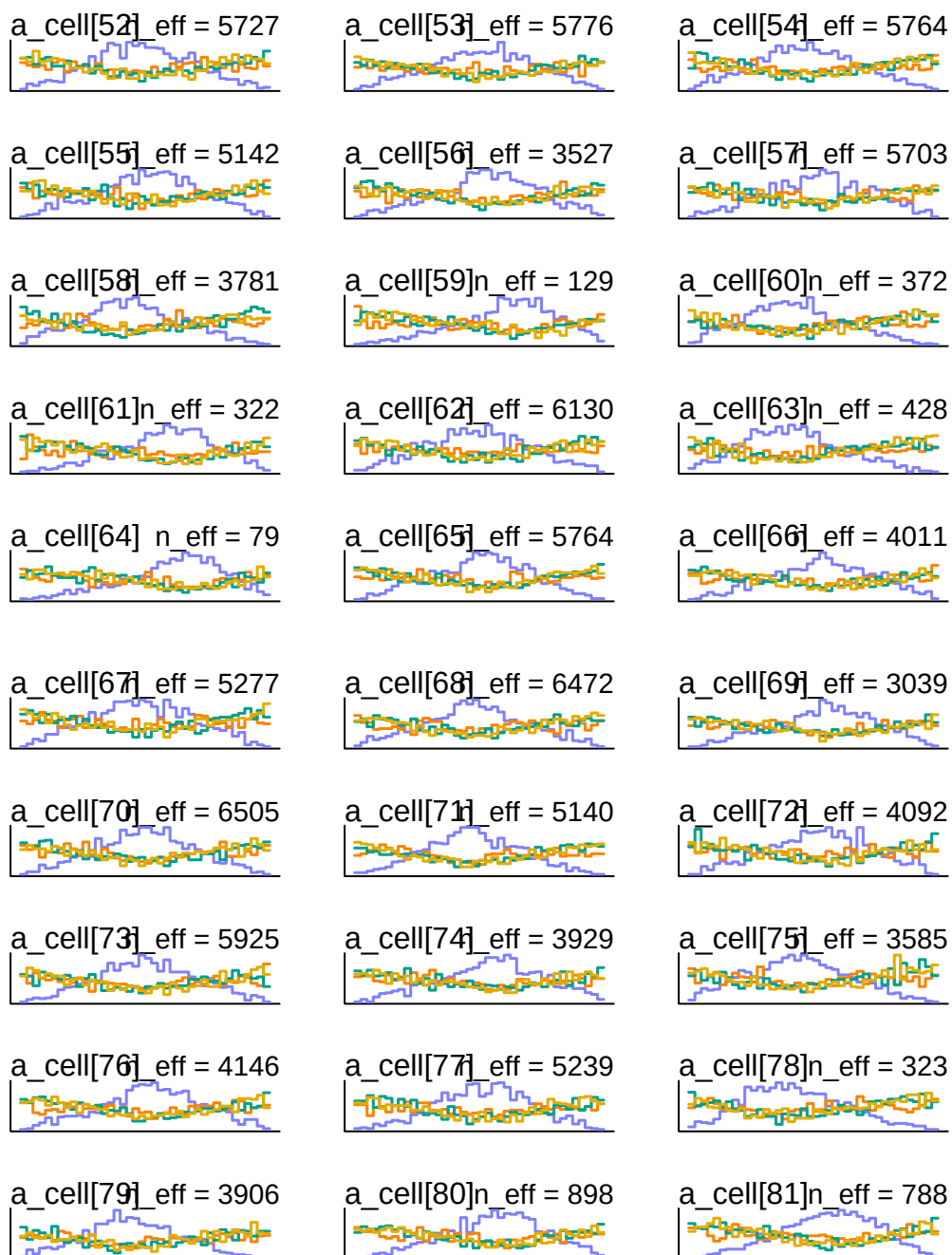


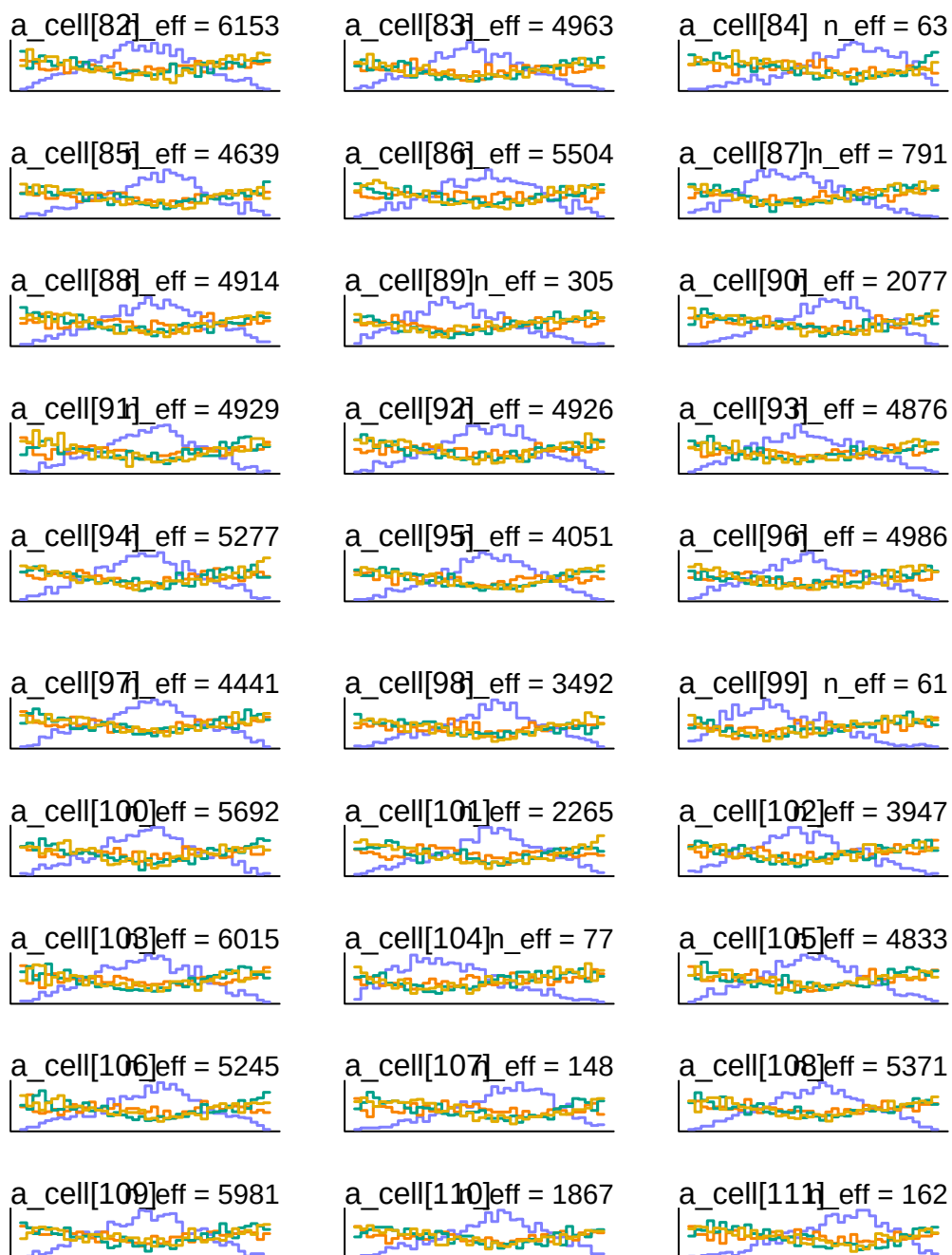


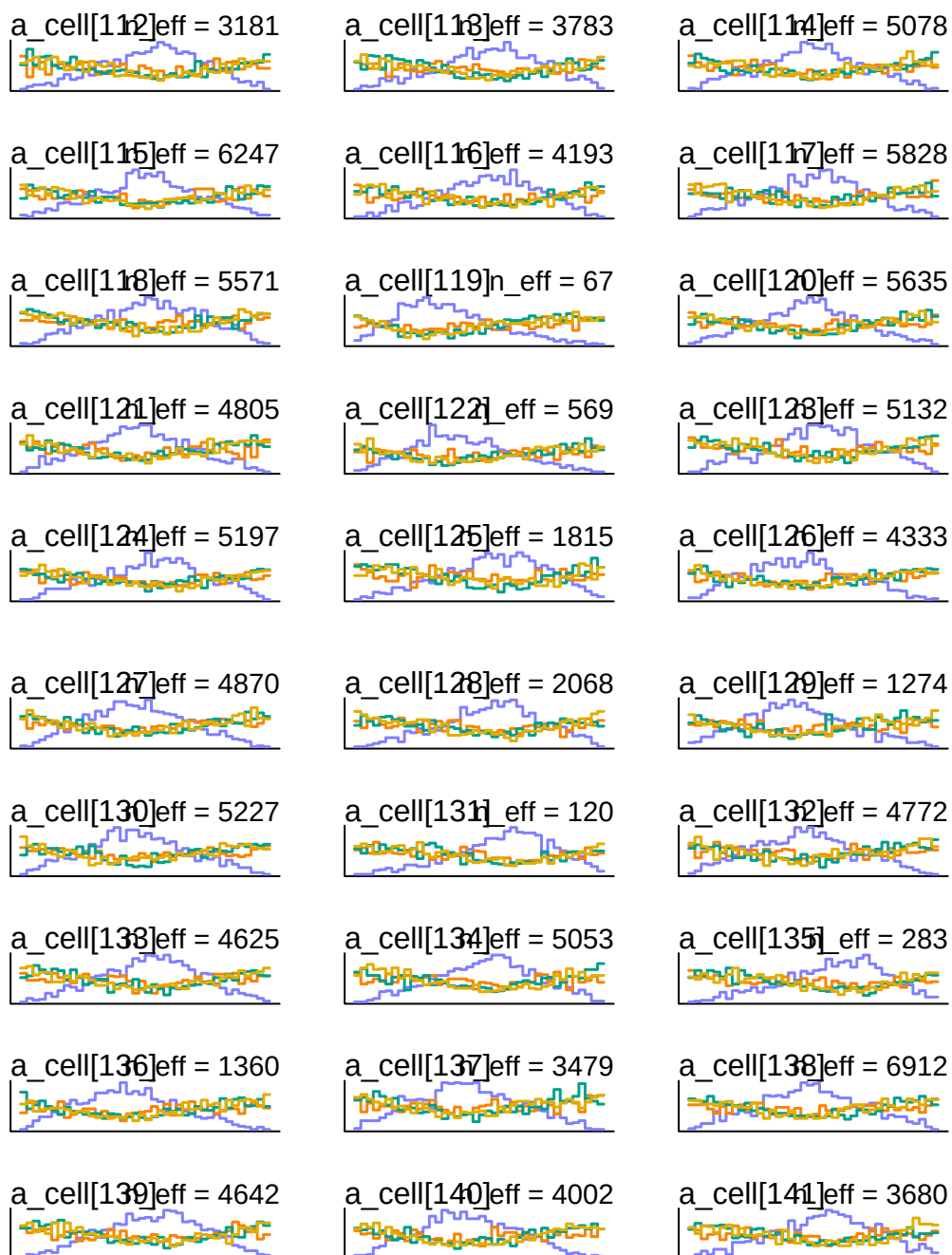










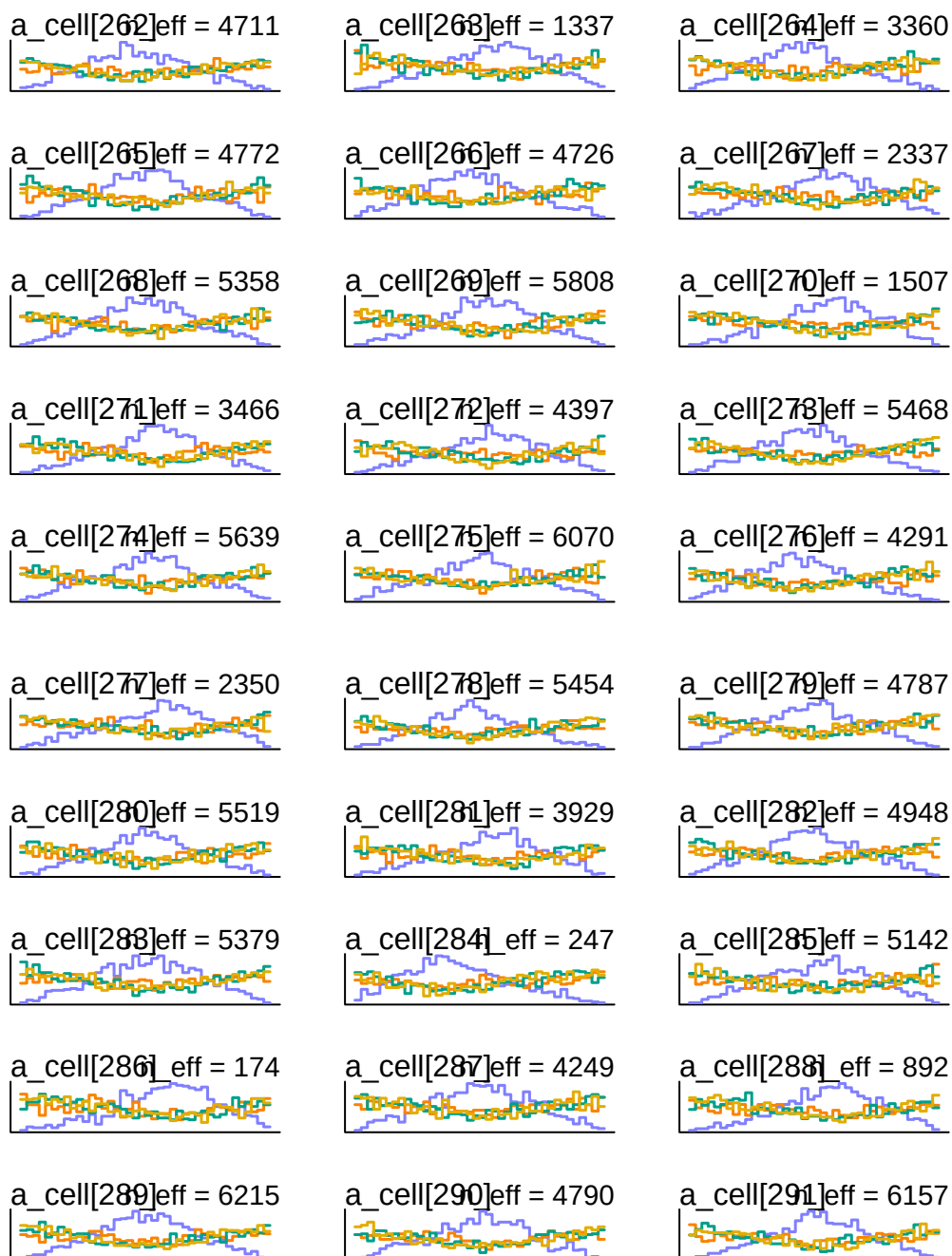








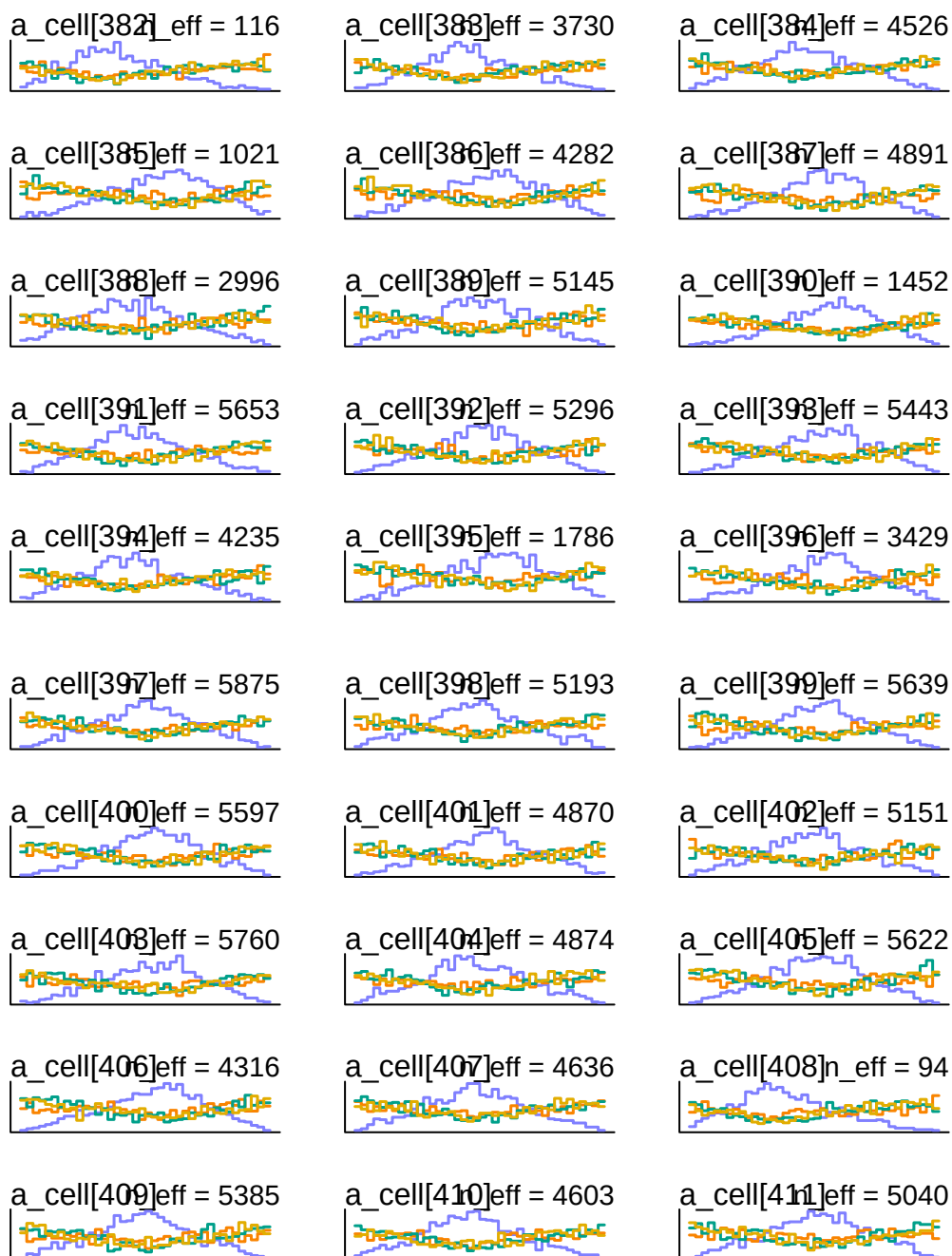




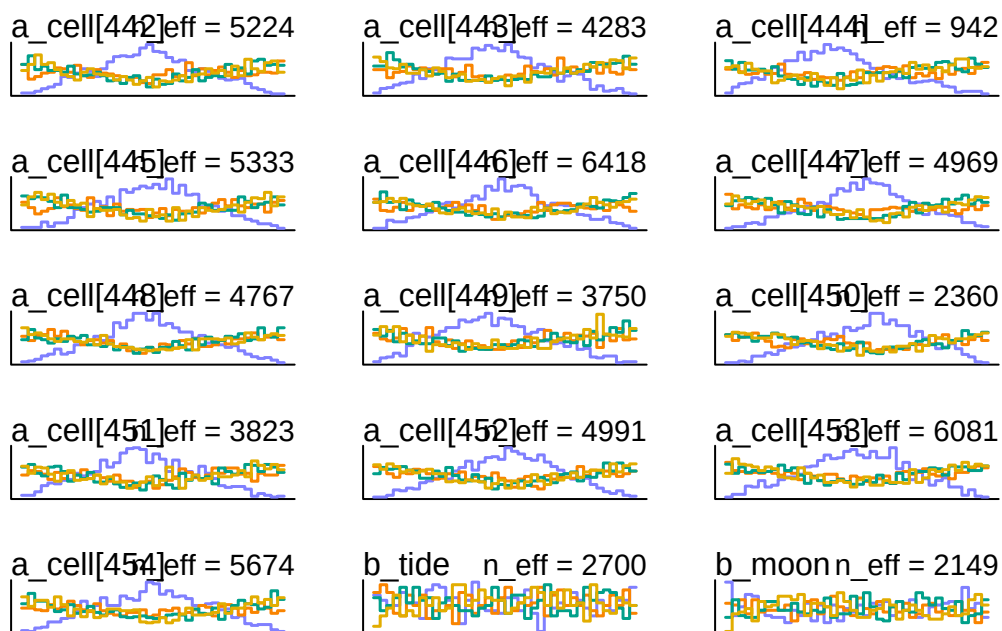




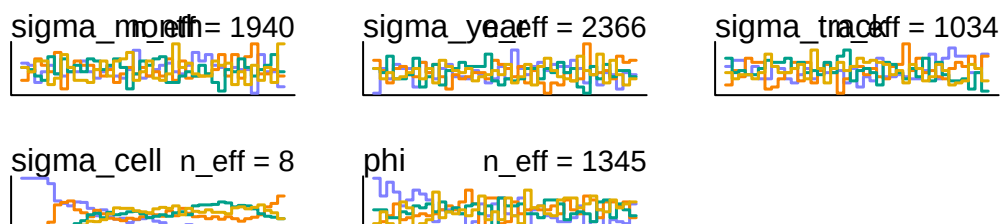








```
par(mfrow = c(1,1))
```



```
## convergence check
summary(m_space_use)
```

	mean	sd	5.5%	94.5%	rhat
a	-1.443416e+00	0.31772862	-1.9008596000	-0.891525708	1.0033606
a_month[1]	-1.455363e-01	0.14172713	-0.3719307257	0.063107953	1.0012422
a_month[2]	-3.906011e-01	0.15342327	-0.6388620941	-0.163849693	1.0001936
a_month[3]	-5.578440e-01	0.20476477	-0.8956567976	-0.248473121	1.0004658
a_month[4]	-2.794663e-01	0.22348087	-0.6427039755	0.066589664	0.9999326
a_month[5]	1.186992e-01	0.18683196	-0.1703581423	0.421656400	1.0019113
a_month[6]	-3.940112e-02	0.18575301	-0.3370622745	0.254229770	1.0025291
a_month[7]	3.304022e-01	0.17726085	0.0605341534	0.619589952	1.0024881
a_month[8]	3.320704e-01	0.17927608	0.0613548326	0.627316181	1.0012451
a_month[9]	5.232711e-01	0.24320030	0.1480252010	0.922137896	1.0018850
a_month[10]	-6.867962e-02	0.16168614	-0.3303186157	0.177425450	1.0006250
a_month[11]	-5.361176e-02	0.17909464	-0.3498015397	0.213794166	0.9999216
a_year[1]	-6.395059e-02	0.31579706	-0.5910462643	0.395183839	1.0044381
a_year[2]	-5.174193e-01	0.29721446	-1.0327834155	-0.113414082	1.0047807
a_year[3]	-5.167922e-01	0.30894705	-1.0472717600	-0.081741826	1.0062572
a_year[4]	6.146400e-01	0.29593223	0.1200466088	1.037216326	1.0052510
a_year[5]	-9.918494e-02	0.30240587	-0.6281338379	0.323640206	1.0035983
a_track[1]	-3.784405e-01	0.21452266	-0.7231816447	-0.038294257	1.0001347
a_track[2]	-1.482837e-01	0.22581283	-0.5053903228	0.215297050	1.0012433
a_track[3]	-1.563535e-01	0.22308093	-0.5194174827	0.200602638	1.0006139
a_track[4]	-1.573815e-01	0.22635458	-0.5194772813	0.203604623	0.9999474
a_track[5]	2.234266e-01	0.24872150	-0.1719330459	0.621069274	1.0027773
a_track[6]	-7.012662e-03	0.30555474	-0.5036624161	0.489075130	1.0013949
a_track[7]	-3.125709e-02	0.21451768	-0.3739545449	0.320098485	1.0004229
a_track[8]	-3.386844e-02	0.21747863	-0.3742655438	0.319567551	1.0022301
a_track[9]	5.678529e-02	0.22042274	-0.2877946596	0.416850912	1.0003835
a_track[10]	-3.079188e-01	0.20510107	-0.6372329684	0.026906450	1.0005776
a_track[11]	-2.839937e-02	0.22917372	-0.3913975641	0.341777393	1.0011605
a_track[12]	-1.113668e-01	0.21182870	-0.4366556018	0.230956884	1.0005471
a_track[13]	-1.111013e-01	0.21191781	-0.4432752807	0.228759281	1.0000459
a_track[14]	-1.708842e-02	0.21990248	-0.3600331612	0.343031132	1.0017649
a_track[15]	4.176242e-02	0.22159792	-0.3001885236	0.398925688	1.0008064
a_track[16]	1.342036e-01	0.22873126	-0.2327860348	0.495950201	1.0008230
a_track[17]	1.348190e-01	0.23454571	-0.2444349310	0.510619709	1.0002926
a_track[18]	-9.544415e-02	0.21103761	-0.4275860068	0.246948840	1.0012011
a_track[19]	-1.053057e-01	0.21850432	-0.4502818914	0.248760563	1.0014229
a_track[20]	-3.099761e-01	0.19569922	-0.6188518275	0.003499808	1.0016196
a_track[21]	-2.568898e-01	0.20433107	-0.5847490756	0.076854259	1.0014456

a_track[22]	-1.776298e-01	0.20684537	-0.5064415099	0.151473318	1.0003005
a_track[23]	-1.790698e-01	0.21521761	-0.5158987720	0.174599666	1.0007779
a_track[24]	9.548518e-02	0.26334761	-0.3270465522	0.520533724	1.0003558
a_track[25]	-1.361169e-02	0.22054384	-0.3651558580	0.347354927	1.0031565
a_track[26]	-1.568681e-01	0.21502485	-0.4989140294	0.182572568	1.0014458
a_track[27]	3.296035e-01	0.27823577	-0.1143301528	0.765935110	1.0002647
a_track[28]	4.054724e-01	0.32710837	-0.1182906967	0.931412117	1.0017483
a_track[29]	3.820639e-01	0.25344268	-0.0162025946	0.800522802	1.0000987
a_track[30]	1.954581e-01	0.23939242	-0.1860176078	0.577548573	1.0016061
a_track[31]	1.046897e-01	0.22909135	-0.2579022031	0.473377472	1.0029502
a_track[32]	-1.498181e-01	0.20771598	-0.4814543123	0.187182501	1.0008414
a_track[33]	7.941826e-03	0.22202174	-0.3436165110	0.368392430	1.0002209
a_track[34]	7.002660e-03	0.22390716	-0.3454527063	0.373445436	1.0006952
a_track[35]	-7.135861e-02	0.22047581	-0.4148469534	0.288512328	1.0003612
a_track[36]	9.516973e-03	0.21697977	-0.3306881360	0.354549724	1.0014350
a_track[37]	-2.922104e-01	0.20694437	-0.6201478404	0.037766099	1.0011988
a_track[38]	-2.977393e-01	0.20460477	-0.6232242443	0.029205604	1.0031089
a_track[39]	-4.247741e-01	0.20140582	-0.7446917434	-0.099300673	1.0022149
a_track[40]	-5.328440e-01	0.19517177	-0.8426612450	-0.223376835	1.0028886
a_track[41]	-7.986461e-02	0.22394040	-0.4376062964	0.281451856	1.0030721
a_track[42]	4.153948e-01	0.29669231	-0.0585679990	0.881763623	1.0013517
a_track[43]	-2.203684e-01	0.21099698	-0.5488828455	0.127808539	0.9999966
a_track[44]	-1.389338e-01	0.22010035	-0.4815875218	0.216583530	1.0012210
a_track[45]	1.922240e-01	0.24666681	-0.1952252403	0.582754142	1.0024210
a_track[46]	1.808225e-03	0.22158540	-0.3537673071	0.349141053	0.9995564
a_track[47]	7.583716e-02	0.23179382	-0.2886629692	0.444269487	1.0012468
a_track[48]	1.496624e-01	0.23353276	-0.2141836226	0.531100333	1.0032855
a_track[49]	2.358134e-01	0.24261116	-0.1504094854	0.630686253	1.0000690
a_track[50]	2.630612e-02	0.26543611	-0.3810170443	0.450466257	1.0006505
a_track[51]	-4.155599e-01	0.22276648	-0.7661751210	-0.070207594	1.0000882
a_track[52]	1.261101e-01	0.26548247	-0.2929258685	0.554690812	1.0012501
a_track[53]	3.174885e-01	0.29963145	-0.1749256283	0.783424925	0.9998175
a_track[54]	-7.339881e-02	0.25483737	-0.4733725203	0.340427215	1.0007023
a_track[55]	5.721717e-01	0.31491017	0.0771857984	1.075578705	1.0030917
a_track[56]	-4.897689e-01	0.22401312	-0.8383313747	-0.131283400	1.0025707
a_track[57]	1.312533e-01	0.25730332	-0.2710981956	0.536097123	1.0009568
a_track[58]	-2.476931e-01	0.23494003	-0.6332894201	0.133497182	1.0014222
a_track[59]	-2.419256e-01	0.24220263	-0.6243741726	0.146418570	1.0012433
a_track[60]	4.537464e-01	0.29862859	-0.0122285331	0.932320957	1.0005935
a_track[61]	-7.366795e-02	0.24311051	-0.4624487394	0.318072885	0.9999925
a_track[62]	6.467350e-01	0.30734153	0.1551640113	1.139246997	1.0023035
a_track[63]	1.138670e-01	0.25483897	-0.2843086716	0.530638114	1.0009494
a_track[64]	-3.758934e-01	0.22028453	-0.7146119851	-0.019091909	1.0013905

a_track[65]	2.897127e-01	0.37001099	-0.2984271493	0.884209300	1.0042238
a_track[66]	-4.456588e-01	0.22183709	-0.7920875947	-0.089488317	0.9995230
a_track[67]	2.580289e-01	0.29352669	-0.2098510285	0.734077442	1.0030054
a_track[68]	-6.526174e-02	0.23247724	-0.4298827025	0.306755801	1.0006402
a_track[69]	-1.453950e-01	0.22972358	-0.4999732589	0.226449442	1.0012921
a_track[70]	-1.437195e-01	0.23601298	-0.5204868226	0.234956784	0.9995598
a_track[71]	-1.436210e-01	0.23257666	-0.5080296228	0.229393077	1.0005091
a_track[72]	-1.792853e-02	0.26788620	-0.4469160926	0.415308955	1.0002625
a_track[73]	2.272802e-01	0.27891284	-0.2169489707	0.675642098	1.0012897
a_track[74]	-2.983020e-01	0.23652057	-0.6773751441	0.068336474	1.0027106
a_track[75]	2.319852e-01	0.28850078	-0.2337828190	0.689184296	1.0031341
a_track[76]	-1.982466e-01	0.24300629	-0.5871202663	0.190952361	1.0003508
a_track[77]	-1.986248e-01	0.24995828	-0.5857026169	0.215074963	1.0004128
a_track[78]	1.458704e-01	0.27021285	-0.2785197329	0.585095863	1.0032943
a_track[79]	-1.983942e-01	0.24849270	-0.5927504169	0.205176623	0.9998448
a_track[80]	-3.039392e-01	0.23554596	-0.6782500752	0.069132160	1.0003310
a_track[81]	-2.174157e-03	0.25678803	-0.3978751408	0.407586206	1.0007898
a_track[82]	1.205457e-01	0.27012659	-0.3053902972	0.559489273	1.0002166
a_track[83]	-1.109623e-01	0.27519759	-0.5457482657	0.325954440	1.0002371
a_track[84]	5.233104e-01	0.29564490	0.0547320775	1.000966473	1.0000208
a_track[85]	-2.198621e-01	0.24064305	-0.6097989287	0.176054539	1.0002397
a_track[86]	-2.084978e-01	0.25095778	-0.6110342182	0.192025247	1.0008121
a_track[87]	-2.185008e-01	0.24762051	-0.6084913928	0.177296917	1.0004613
a_track[88]	3.465887e-01	0.30654838	-0.1444635945	0.835301216	1.0008219
a_track[89]	4.762922e-01	0.32760851	-0.0417392171	1.010608634	1.0008649
a_track[90]	2.350749e-01	0.29271164	-0.2296079699	0.710886053	1.0012104
a_track[91]	-1.285265e-01	0.25196334	-0.5176750733	0.272955331	1.0011321
a_track[92]	-4.254287e-01	0.23269330	-0.7914021410	-0.052098082	1.0011895
a_track[93]	-1.190687e-01	0.25693946	-0.5237470928	0.294001877	1.0006064
a_track[94]	3.383544e-01	0.30202789	-0.1404542299	0.822386920	1.0023518
a_track[95]	1.260987e-01	0.27596134	-0.3073013174	0.587145885	1.0003449
a_track[96]	-9.919157e-02	0.25851988	-0.5101053533	0.321163072	1.0018824
a_track[97]	-3.908786e-01	0.23831782	-0.7699473251	-0.005231445	0.9999301
a_track[98]	3.437083e-01	0.31226917	-0.1449253320	0.846803424	1.0011298
a_track[99]	2.141909e-02	0.26258702	-0.3859581447	0.447192367	1.0012314
a_track[100]	3.814308e-01	0.29251121	-0.0738098624	0.840199062	1.0026674
a_track[101]	2.422341e-01	0.28947800	-0.2156776649	0.712195170	0.9995927
a_track[102]	2.420697e-01	0.35954918	-0.3520981960	0.801384684	1.0028973
a_track[103]	1.294755e-01	0.27263433	-0.3005629249	0.566061646	0.9992386
a_track[104]	-9.126023e-02	0.24975401	-0.4779704123	0.312519168	1.0003287
a_track[105]	-2.013458e-01	0.24088179	-0.5893917653	0.179384014	1.0040470
a_track[106]	-2.059370e-01	0.24602166	-0.5880952932	0.194976526	1.0010239
a_track[107]	2.654836e-01	0.28218787	-0.1800812419	0.722718702	1.0012959

a_track[108]	-3.717478e-01	0.19904871	-0.6753291621	-0.043396787	1.0023755
a_track[109]	-1.026875e-01	0.21554629	-0.4471620637	0.246123679	1.0014622
a_track[110]	-2.523349e-01	0.20966780	-0.5879924438	0.077198291	1.0025327
a_track[111]	5.091776e-01	0.34176475	-0.0270018169	1.061708001	1.0005118
a_track[112]	-3.717497e-01	0.19881067	-0.6805639650	-0.048309458	1.0009013
a_track[113]	3.410658e-01	0.35581280	-0.2231385019	0.908689908	1.0014987
a_track[114]	2.793468e-01	0.33324943	-0.2546488595	0.812587210	1.0010859
a_track[115]	3.360232e-01	0.31764301	-0.1688744192	0.834511062	1.0008265
a_track[116]	3.041925e-01	0.32684666	-0.2196830380	0.825615978	1.0038552
a_track[117]	3.061547e-01	0.27847441	-0.1264410553	0.761571836	1.0004114
a_track[118]	1.976986e-01	0.26801689	-0.2231172678	0.637047392	1.0001091
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a_track[120]	1.518860e-01	0.26990925	-0.2790902469	0.584456675	1.0007368
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a_track[122]	2.567955e-01	0.29158435	-0.2099083138	0.718948841	1.0006869
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a_track[125]	2.644384e-01	0.36250763	-0.3101348526	0.849302003	1.0029139
a_track[126]	-4.522903e-01	0.22600861	-0.8072555141	-0.078430572	1.0020340
a_track[127]	-1.088750e-01	0.25105347	-0.5126367748	0.297493051	1.0008981
a_track[128]	-4.535367e-01	0.21940423	-0.7962492382	-0.096618541	1.0001346
a_track[129]	-3.445741e-01	0.23227925	-0.7143001344	0.029365531	1.0015922
a_track[130]	3.004230e-01	0.32418562	-0.2214837758	0.819687267	0.9999141
a_track[131]	1.549176e-01	0.27801364	-0.2910856136	0.605614903	1.0014077
a_track[132]	-6.115313e-01	0.21490473	-0.9570664879	-0.265783907	1.0011625
a_track[133]	1.190780e-01	0.29061636	-0.3457657451	0.592478428	1.0004677
a_track[134]	-5.405939e-01	0.21633885	-0.8842705044	-0.202791456	1.0006619
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a_track[137]	3.361373e-02	0.25884119	-0.3802491990	0.453919348	1.0014215
a_track[138]	-6.149473e-01	0.21195408	-0.9497400296	-0.274659960	1.0041357
a_track[139]	1.574939e-01	0.26779200	-0.2705269646	0.582082929	1.0001839
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a_track[141]	4.492204e-01	0.32374906	-0.0718534034	0.965722830	0.9996506
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a_track[143]	-2.904456e-01	0.21898253	-0.6345534779	0.062138005	1.0026567
a_track[144]	1.757639e-01	0.25985123	-0.2412583776	0.597660662	0.9996998
a_track[145]	2.887379e-01	0.26364127	-0.1351235287	0.713731889	1.0025018
a_track[146]	-5.047672e-01	0.20606999	-0.8231710165	-0.166527811	1.0032341
a_track[147]	-1.193851e-01	0.22163874	-0.4729161170	0.238675867	1.0008534
a_track[148]	1.848576e-01	0.24900477	-0.2105405009	0.584457297	1.0018122
a_track[149]	8.168991e-02	0.23645353	-0.2909502398	0.467841372	1.0038074
a_track[150]	-1.626466e-02	0.23583550	-0.3870689893	0.367828022	1.0010623

a_track[151]	2.949416e-01	0.26078645	-0.1144879628	0.723307333	1.0009282
a_track[152]	5.334628e-01	0.28047802	0.0878446961	0.983954477	1.0029714
a_track[153]	-7.201685e-01	0.18667155	-1.0181163740	-0.425441451	1.0003428
a_track[154]	-4.923785e-01	0.20678560	-0.8193839732	-0.162325929	1.0012906
a_track[155]	-4.173815e-01	0.20837759	-0.7491360080	-0.081871333	1.0017051
a_track[156]	-2.715406e-01	0.22229887	-0.6216432470	0.089872427	1.0001510
a_track[157]	8.572790e-02	0.23968483	-0.2911684540	0.467905482	1.0015062
a_track[158]	9.116333e-02	0.25029172	-0.3008827748	0.503442337	1.0014681
a_track[159]	-2.029673e-01	0.22413006	-0.5546755372	0.150955653	1.0002767
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a_track[161]	-4.896559e-02	0.23835074	-0.4247213037	0.333116169	1.0026401
a_track[162]	-3.373651e-01	0.21237052	-0.6793397849	-0.005519788	1.0018146
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a_track[164]	-5.109433e-01	0.21206800	-0.8403829429	-0.168929516	1.0012384
a_track[165]	1.932627e-01	0.25917276	-0.2182384110	0.609864564	1.0003566
a_track[166]	-7.474072e-01	0.19738189	-1.0607919380	-0.423344041	1.0004053
a_track[167]	-4.351519e-01	0.20915840	-0.7627281876	-0.102135047	1.0004271
a_track[168]	3.901026e-01	0.28045503	-0.0584173379	0.844163578	1.0007053
a_track[169]	1.604418e-01	0.26142229	-0.2534828500	0.576835389	1.0013011
a_track[170]	1.207448e-02	0.25542401	-0.3911101262	0.421041665	1.0006183
a_track[171]	5.346326e-01	0.30910116	0.0558546443	1.036909917	1.0015350
a_track[172]	-8.647154e-02	0.25535180	-0.4875568182	0.325669225	1.0034054
a_track[173]	2.168788e-01	0.29796974	-0.2503634450	0.694072470	1.0009821
a_track[174]	-8.717586e-02	0.25489359	-0.4871651429	0.333623706	1.0007452
a_track[175]	1.313811e-01	0.27871837	-0.3127303791	0.579456732	1.0003911
a_track[176]	2.727639e-01	0.32586671	-0.2434254325	0.785301434	0.9998361
a_track[177]	5.539272e-01	0.30275594	0.0765404645	1.054890425	1.0000824
a_track[178]	2.981223e-02	0.25339448	-0.3677770861	0.437049515	1.0002791
a_track[179]	2.971885e-01	0.27371765	-0.1264900099	0.742724334	1.0006685
a_track[180]	-9.238075e-02	0.24662107	-0.4900803163	0.304324726	0.9999133
a_track[181]	-5.244940e-01	0.22108442	-0.8731306148	-0.167839522	1.0005616
a_track[182]	-8.890871e-02	0.29814981	-0.5614301162	0.387939844	1.0011574
a_track[183]	-1.107717e+00	0.22821740	-1.4643214755	-0.731296367	1.0004709
a_track[184]	-7.496938e-01	0.24717491	-1.1435359760	-0.350800849	1.0018828
a_track[185]	-3.793607e-01	0.27904752	-0.8148121194	0.075959374	1.0022049
a_track[186]	-6.297288e-01	0.25945055	-1.0352615800	-0.212556880	1.0013137
a_track[187]	-2.238769e-01	0.27598027	-0.6681478272	0.222020237	1.0008055
a_track[188]	-3.464981e-01	0.28259340	-0.7852120609	0.109892232	1.0008495
a_track[189]	-7.001247e-02	0.32549013	-0.5905905597	0.450580983	1.0009954
a_track[190]	2.177827e-01	0.33709111	-0.3146028178	0.767768648	1.0020653
a_track[191]	-6.246043e-01	0.25695405	-1.0288367050	-0.214053305	1.0013366
a_track[192]	7.130586e-02	0.30992618	-0.4127554184	0.568710226	1.0007166
a_track[193]	5.323575e-02	0.33447604	-0.4733740809	0.588320169	1.0024414

a_track[194]	2.629550e-01	0.32131418	-0.2397936092	0.770818660	1.0021653
a_track[195]	2.795163e-01	0.33158971	-0.2520686456	0.810602877	1.0033994
a_track[196]	2.155362e-01	0.34573705	-0.3294158510	0.767477538	1.0027928
a_track[197]	2.848883e-01	0.34281994	-0.2548235558	0.843267077	1.0016962
a_track[198]	-6.043283e-01	0.25219007	-1.0020959435	-0.193072460	1.0014170
a_track[199]	2.645223e-01	0.32611429	-0.2626001820	0.795852984	0.9997331
a_track[200]	-7.378729e-01	0.24015212	-1.1147905270	-0.344689105	1.0038149
a_track[201]	2.566405e-01	0.30720835	-0.2347269202	0.749025394	1.0011275
a_track[202]	-1.297816e+00	0.22780733	-1.6609627130	-0.931983195	1.0014988
a_track[203]	5.094572e-02	0.31490496	-0.4476864753	0.559066635	1.0008827
a_track[204]	8.709961e-02	0.30728229	-0.3963703860	0.584809008	1.0007377
a_track[205]	-2.242448e-01	0.27879012	-0.6586455933	0.229410344	1.0039429
a_track[206]	-6.608986e-02	0.29922172	-0.5241196586	0.414318498	1.0000463
a_track[207]	1.638554e-01	0.36319089	-0.4256969505	0.728977686	1.0023238
a_track[208]	9.402189e-02	0.32363817	-0.4178868157	0.607428594	1.0026739
a_track[209]	-4.212259e-02	0.28917968	-0.4980107559	0.426937674	1.0011837
a_track[210]	-3.341382e-01	0.27187681	-0.7632753696	0.106522838	1.0027514
a_track[211]	2.782514e-01	0.32437976	-0.2140669966	0.807631417	1.0011024
a_track[212]	-3.209988e-01	0.26711161	-0.7449173813	0.107079194	0.9999316
a_track[213]	1.781577e-01	0.33264538	-0.3522441128	0.716829904	0.9993847
a_track[214]	-4.191357e-02	0.30827628	-0.5332215592	0.454443801	1.0026834
a_track[215]	-3.224949e-02	0.30475193	-0.5045985263	0.451536052	1.0023658
a_track[216]	1.050408e-01	0.32323452	-0.4095384988	0.626421736	1.0007755
a_track[217]	8.185925e-02	0.32213770	-0.4338818222	0.606250408	1.0003901
a_track[218]	-3.967578e-02	0.29438344	-0.5095301753	0.442665039	1.0015316
a_track[219]	1.549178e-01	0.35285203	-0.4128286024	0.714261954	1.0009129
a_track[220]	-3.593443e-01	0.27359696	-0.7947510322	0.085092636	1.0006895
a_track[221]	-3.932787e-01	0.27707761	-0.8323829259	0.049282373	0.9997385
a_track[222]	8.391469e-02	0.34101044	-0.4640450472	0.635091460	1.0005875
a_track[223]	1.529773e-01	0.36623796	-0.4252821106	0.746948707	1.0014242
a_track[224]	1.035681e-01	0.31167601	-0.3834428979	0.609693195	1.0023380
a_track[225]	9.241876e-02	0.33213975	-0.4334359602	0.617645874	0.9997191
a_track[226]	9.682109e-02	0.31340123	-0.4011508814	0.604235936	1.0011596
a_track[227]	1.185595e-01	0.29557246	-0.3453803422	0.603677456	1.0016541
a_track[228]	2.293159e-01	0.33170479	-0.2978902587	0.763790769	1.0004390
a_track[229]	-5.137888e-02	0.32117809	-0.5674040795	0.451120301	1.0006640
a_track[230]	2.242822e-01	0.33994728	-0.3085531184	0.769579591	1.0038001
a_track[231]	2.716413e-01	0.33401950	-0.2561099031	0.809190901	1.0015548
a_track[232]	2.373472e-01	0.33360550	-0.2942834355	0.774073582	1.0007360
a_track[233]	-4.846153e-02	0.29943806	-0.5191760115	0.423582776	1.0019933
a_track[234]	9.278006e-02	0.31748331	-0.4147736711	0.597878221	0.9997044
a_track[235]	-1.945565e-01	0.29275989	-0.6557114071	0.276084307	0.9999971
a_track[236]	1.112318e-01	0.31108759	-0.3777735946	0.623546091	1.0037167

a_track[237]	2.261637e-01	0.33483432	-0.2972951126	0.770222253	1.0009497
a_track[238]	-1.133906e+00	0.23744487	-1.5083829475	-0.758480140	1.0009102
a_track[239]	-4.375881e-01	0.29359319	-0.9095428574	0.028461301	0.9997704
a_track[240]	-5.647219e-01	0.28386774	-1.0089973735	-0.117087146	1.0009840
a_track[241]	-7.445615e-02	0.30159154	-0.5483100917	0.408744197	1.0005952
a_track[242]	-5.704873e-01	0.27253492	-0.9953451900	-0.129320872	1.0002352
a_track[243]	-2.101859e-01	0.28990584	-0.6603244039	0.256811571	1.0025855
a_track[244]	-6.034800e-02	0.30736290	-0.5457765343	0.437978867	1.0024247
a_track[245]	7.163426e-02	0.31648012	-0.4317195641	0.582773484	1.0008368
a_track[246]	2.281322e-01	0.32962722	-0.3015910341	0.767876622	1.0023611
a_track[247]	-9.212899e-02	0.30703069	-0.5786518161	0.410384136	0.9998344
a_track[248]	2.296238e-01	0.33146321	-0.2923804209	0.764109982	1.0007018
a_track[249]	1.759773e-01	0.34619846	-0.3714578557	0.735806580	1.0026217
a_track[250]	2.117822e-01	0.32346763	-0.3074180634	0.732294424	1.0010589
a_track[251]	1.735392e-01	0.36433325	-0.4106715146	0.766810796	1.0001542
a_track[252]	-9.464596e-02	0.31016601	-0.5838240134	0.398835486	1.0036455
a_track[253]	6.096202e-02	0.30792797	-0.4343636402	0.542176557	1.0004412
a_track[254]	5.965948e-02	0.31681723	-0.4429166331	0.562614513	1.0024388
a_track[255]	-2.285494e-01	0.28540668	-0.6730912810	0.228586679	0.9998302
a_track[256]	-8.188763e-02	0.29780239	-0.5517955091	0.402307807	1.0023611
a_track[257]	2.132309e-01	0.33914190	-0.3218376178	0.763255553	1.0008206
a_track[258]	1.975660e-01	0.35461524	-0.3794933585	0.759701081	1.0014339
a_track[259]	2.111778e-01	0.34374655	-0.3369735287	0.761000795	1.0005807
a_track[260]	-7.960204e-02	0.30199266	-0.5666199817	0.403374177	1.0005359
a_track[261]	7.309940e-02	0.30520985	-0.4030055887	0.568348126	0.9996776
a_track[262]	9.193263e-02	0.30982374	-0.3976870886	0.589790105	1.0009095
a_track[263]	7.684890e-02	0.31178367	-0.4181987337	0.577716074	0.9999340
a_track[264]	7.015804e-02	0.32266568	-0.4326461783	0.584960685	1.0010446
a_track[265]	2.129643e-01	0.34925758	-0.3393947273	0.786486065	1.0028669
a_track[266]	7.099086e-02	0.30379969	-0.4120446956	0.554753278	1.0003805
a_track[267]	-7.894282e-02	0.30835470	-0.5667182483	0.419332165	1.0033263
a_track[268]	5.981519e-02	0.31768265	-0.4188098103	0.566471183	1.0003740
a_track[269]	-7.236945e-02	0.31133967	-0.5555186291	0.429128242	1.0015563
a_track[270]	2.261932e-01	0.33255159	-0.2940908124	0.758729359	1.0017587
a_track[271]	-8.291263e-02	0.30790234	-0.5819453156	0.416939692	0.9997892
a_track[272]	7.258097e-02	0.31570116	-0.4194275656	0.575059034	1.0036054
a_track[273]	-8.398596e-02	0.29948075	-0.5600900354	0.392406123	1.0009987
a_track[274]	1.770889e-01	0.34284966	-0.3704302538	0.725987800	1.0005838
a_track[275]	-8.043447e-02	0.30804119	-0.5697817813	0.409930181	1.0011570
a_track[276]	7.174570e-02	0.31766890	-0.4339962771	0.590380594	1.0016652
a_track[277]	7.101203e-02	0.32531930	-0.4420050156	0.591998813	1.0035521
a_track[278]	-8.272463e-02	0.29002044	-0.5366335992	0.381590099	1.0000851
a_track[279]	4.494128e-02	0.32694231	-0.4631658950	0.566816100	1.0018963

a_track[280]	1.757519e-01	0.34934229	-0.3854781594	0.749383916	1.0008527
a_track[281]	1.313986e-01	0.35196089	-0.4317956468	0.688949571	1.0008642
a_track[282]	2.094642e-01	0.33704052	-0.3214678853	0.749145702	0.9999720
a_track[283]	-2.166643e-01	0.29357622	-0.6832571777	0.249182956	1.0022737
a_track[284]	2.564045e-01	0.32444657	-0.2496374939	0.769305940	1.0001301
a_track[285]	2.436588e-01	0.33394489	-0.2873365603	0.789038641	1.0000736
a_track[286]	8.483463e-02	0.32410231	-0.4331926529	0.605701151	1.0013828
a_track[287]	1.970346e-01	0.34300805	-0.3299240495	0.756072156	1.0007809
a_track[288]	1.659830e-01	0.36271228	-0.4123485000	0.743354315	1.0006360
a_track[289]	1.721639e-01	0.33902048	-0.3688504684	0.711383692	1.0013601
a_track[290]	-2.015282e-01	0.29537660	-0.6762001597	0.257080194	1.0009518
a_track[291]	4.168600e-02	0.33920916	-0.5086474448	0.583758540	1.0049665
a_track[292]	-6.175913e-01	0.25452522	-1.0158869835	-0.207354631	1.0016681
a_track[293]	6.213930e-02	0.32195381	-0.4442489923	0.580873628	1.0014333
a_track[294]	-2.017186e-01	0.31780304	-0.7113779798	0.304261171	1.0039664
a_track[295]	-7.631888e-02	0.30081827	-0.5577838741	0.405936089	1.0006846
a_track[296]	6.459980e-02	0.33041015	-0.4629332671	0.596799839	0.9997217
a_track[297]	1.930644e-01	0.34824196	-0.3495096765	0.763881138	1.0003812
a_track[298]	2.581450e-01	0.32033016	-0.2428855642	0.773170408	1.0008378
a_track[299]	8.262123e-02	0.30567592	-0.4097109076	0.572399984	1.0028504
a_track[300]	2.359096e-01	0.33426919	-0.2842744697	0.760039446	1.0008209
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a_track[302]	8.393917e-02	0.31429529	-0.4193617553	0.589355439	1.0017433
a_track[303]	-7.022229e-02	0.30587500	-0.5581326339	0.415205614	1.0019138
a_track[304]	7.272404e-02	0.30516430	-0.4089552277	0.557796379	1.0012357
a_track[305]	-6.369544e-02	0.30239443	-0.5500909937	0.421140650	1.0033071
a_track[306]	1.764937e-01	0.35590170	-0.3981389917	0.748866740	1.0019050
a_track[307]	2.392666e-01	0.33171165	-0.2968573092	0.784049833	1.0022648
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a_track[311]	1.939422e-01	0.35105204	-0.3636031153	0.752689567	1.0008121
a_track[312]	-2.069908e-01	0.29302769	-0.6731067402	0.277842008	1.0009053
a_track[313]	-6.799141e-02	0.29833415	-0.5493325839	0.404182995	1.0004640
a_track[314]	8.310836e-02	0.30809272	-0.4000056901	0.582364463	1.0020218
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a_track[317]	1.047643e-01	0.30353804	-0.3701987816	0.602274397	1.0020251
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a_track[319]	-1.727955e-01	0.30813768	-0.6764846871	0.319962246	1.0030568
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a_track[322]	-2.022541e-01	0.28059151	-0.6370338745	0.243932044	1.0002162

a_track[323]	2.394696e-01	0.33467215	-0.2844913775	0.770223959	1.0027589
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a_track[325]	1.552129e-01	0.36146065	-0.4154462890	0.733501172	1.0014283
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a_track[329]	2.483534e-01	0.32244252	-0.2600361207	0.761691079	1.0024240
a_track[330]	8.718207e-02	0.31893048	-0.4184819906	0.603943261	1.0016533
a_track[331]	2.422824e-01	0.32607676	-0.2703600895	0.757903920	1.0010932
a_track[332]	7.693845e-02	0.31516774	-0.4217540578	0.594525590	1.0021088
a_track[333]	-1.998268e-01	0.28037295	-0.6507077706	0.248296151	1.0022535
a_track[334]	9.797643e-02	0.30780675	-0.3954217949	0.599113828	1.0000367
a_track[335]	8.757234e-02	0.30903158	-0.3838820712	0.577957129	1.0015573
a_track[336]	7.905804e-02	0.32741575	-0.4413617431	0.603538443	1.0005357
a_track[337]	2.027988e-01	0.35088077	-0.3607495400	0.770395791	0.9999657
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a_track[339]	-6.679975e-02	0.30184962	-0.5474609420	0.417305365	1.0000746
a_track[340]	-2.124224e-01	0.28334565	-0.6547451406	0.237786621	1.0017509
a_track[341]	-6.668675e-02	0.28670138	-0.5102000968	0.401580042	0.9993738
a_track[342]	-3.623354e-01	0.27570719	-0.7995410920	0.087538936	1.0002824
a_track[343]	7.511840e-02	0.32062216	-0.4376647254	0.576283039	1.0015026
a_track[344]	-2.304908e-01	0.28511106	-0.6824463757	0.223249547	1.0005954
a_track[345]	2.382638e-01	0.32889585	-0.2850262484	0.756621162	0.9997650
a_track[346]	2.342943e-01	0.32131169	-0.2754987791	0.763702144	1.0015555
a_track[347]	-6.719158e-02	0.29805235	-0.5398399118	0.406139846	1.0009977
a_track[348]	7.490891e-02	0.31604624	-0.4227794824	0.585410890	1.0029605
a_track[349]	4.546449e-02	0.32407463	-0.4813997938	0.566756027	1.0009550
a_track[350]	2.104084e-01	0.33484895	-0.3196433864	0.753667894	0.9995552
a_track[351]	-2.335851e-01	0.29787811	-0.7153319097	0.248705988	1.0004143
a_track[352]	1.800597e-01	0.34670477	-0.3805679744	0.731030310	1.0014531
a_track[353]	7.262129e-02	0.30241856	-0.4076376835	0.558147945	1.0000598
a_track[354]	6.379909e-02	0.32401355	-0.4634318956	0.578122765	1.0027075
a_track[355]	-8.711878e-02	0.29251003	-0.5449810398	0.383067348	1.0000257
a_track[356]	-2.293457e-01	0.30098620	-0.7191325658	0.253339061	1.0001751
a_track[357]	4.520503e-02	0.33509556	-0.4853415416	0.588102677	1.0014832
a_track[358]	1.345414e-01	0.35225064	-0.4296029242	0.696790044	1.0005219
a_track[359]	4.858801e-02	0.34229745	-0.5022495492	0.589731730	1.0012707
a_track[360]	7.237005e-02	0.30918114	-0.4200981883	0.571302611	1.0009917
a_track[361]	1.939650e-01	0.34898242	-0.3537971705	0.742801452	1.0026484
a_track[362]	5.974256e-02	0.32773646	-0.4607111304	0.585009126	1.0002116
a_track[363]	-8.157076e-02	0.30404600	-0.5663990373	0.410956411	1.0012545
a_track[364]	-8.467977e-02	0.30575956	-0.5798695276	0.404629273	1.0013000
a_track[365]	-7.845569e-02	0.29624128	-0.5591028423	0.411829799	1.0009823

a_track[366]	4.535863e-02	0.33663032	-0.4872103377	0.579163267	1.0009370
a_track[367]	5.195050e-02	0.32848498	-0.4605460008	0.582040044	1.0018130
a_track[368]	1.621553e-01	0.35457114	-0.3987300668	0.731903010	1.0025508
a_track[369]	4.989199e-02	0.32025637	-0.4625629630	0.561322114	1.0051685
a_track[370]	5.648545e-02	0.32361945	-0.4535183286	0.573941713	1.0009407
a_track[371]	-5.028954e-01	0.27468888	-0.9390209038	-0.062217148	1.0010261
a_track[372]	-7.183681e-02	0.31035368	-0.5598237347	0.420833756	1.0003286
a_track[373]	-2.260371e-01	0.28578735	-0.6745838835	0.233244231	1.0011728
a_track[374]	4.613842e-02	0.32878172	-0.4805056394	0.560504295	1.0010805
a_track[375]	1.328152e-01	0.35061393	-0.4266128596	0.678825674	1.0005436
a_track[376]	-7.486384e-02	0.32524497	-0.5947923279	0.437030788	1.0011049
a_track[377]	2.078916e-01	0.33894866	-0.3191930284	0.754979044	1.0045135
a_track[378]	2.284477e-01	0.32943592	-0.2944281224	0.767522936	1.0018190
a_track[379]	1.860583e-01	0.33668187	-0.3517874906	0.710239932	1.0000789
a_track[380]	7.597075e-02	0.32870970	-0.4588228591	0.586675264	1.0014128
a_track[381]	8.402592e-02	0.32788633	-0.4321321296	0.612716975	0.9995604
a_track[382]	6.812119e-02	0.31820765	-0.4396137192	0.571122491	1.0005965
a_track[383]	2.787175e-01	0.32227163	-0.2200594250	0.800816513	0.9997160
a_track[384]	9.314736e-02	0.31021015	-0.3919049200	0.595754320	1.0019401
a_track[385]	-4.651200e-02	0.28980966	-0.5016352282	0.425774685	1.0007877
a_track[386]	-5.860473e-02	0.31261524	-0.5521244726	0.437848189	1.0006663
a_track[387]	2.298567e-01	0.33587361	-0.2956869047	0.782019688	1.0019941
a_track[388]	-5.568951e-02	0.28670022	-0.5178965852	0.410723452	1.0008241
a_track[389]	9.951359e-02	0.30540946	-0.3773375997	0.593670994	1.0022710
a_track[390]	-5.340214e-02	0.29503914	-0.5240779328	0.417471512	1.0002827
a_track[391]	-6.858561e-02	0.28688592	-0.5174344274	0.390423715	1.0014804
a_track[392]	-5.841916e-02	0.29207225	-0.5162508062	0.402465105	1.0019374
a_track[393]	2.090931e-01	0.34048390	-0.3394843912	0.759151938	1.0035972
a_track[394]	-2.167067e-01	0.28832019	-0.6794263518	0.240620571	1.0012093
a_track[395]	-6.197664e-02	0.29985689	-0.5343818840	0.419770160	1.0005561
a_track[396]	-6.768296e-02	0.29051269	-0.5262244256	0.405965714	1.0013700
a_track[397]	7.958904e-02	0.32540361	-0.4355993955	0.609108478	1.0020313
a_track[398]	8.686755e-02	0.31177738	-0.4085799950	0.593265599	1.0008878
a_track[399]	7.966220e-02	0.32661396	-0.4493389626	0.609819280	1.0005316
a_track[400]	8.610011e-02	0.29539237	-0.3741262829	0.559101498	0.9999022
a_track[401]	-6.161162e-02	0.30003682	-0.5363911597	0.418849970	1.0012810
a_track[402]	8.709949e-02	0.31322676	-0.4095214807	0.580186271	1.0001417
a_track[403]	-5.845685e-02	0.31035957	-0.5539338856	0.437032228	1.0025615
a_track[404]	-6.180695e-02	0.30496790	-0.5433686644	0.427883134	0.9997303
a_track[405]	7.561278e-02	0.32931471	-0.4496183016	0.610586076	1.0013374
a_track[406]	1.688500e-01	0.34396661	-0.3667021331	0.708989561	1.0006047
a_track[407]	-5.987183e-02	0.30337758	-0.5378748643	0.409477573	0.9996103
a_track[408]	-5.817899e-02	0.30959051	-0.5510906836	0.426628302	1.0034739

a_track[409]	-4.669874e-02	0.30944135	-0.5420561700	0.440913950	1.0005536
a_track[410]	-3.467938e-01	0.26743065	-0.7700818854	0.091926203	1.0007811
a_track[411]	4.249712e-02	0.33740664	-0.4917991922	0.571153646	1.0017717
a_track[412]	4.056505e-02	0.35046656	-0.5060779660	0.577070238	1.0016130
a_track[413]	-3.321513e-01	0.28216278	-0.7778564706	0.105482686	1.0010628
a_track[414]	1.910551e-01	0.34345404	-0.3485602304	0.737828087	0.9998066
a_track[415]	-5.333245e-02	0.28488951	-0.5057264623	0.402315266	1.0008554
a_track[416]	7.518558e-02	0.31760462	-0.4400158604	0.584522445	1.0004936
a_track[417]	2.230775e-01	0.33553270	-0.3163697142	0.767746837	1.0020821
a_track[418]	-6.223958e-02	0.29557483	-0.5342580462	0.410641280	1.0012253
a_track[419]	-6.299253e-02	0.29836960	-0.5320410323	0.412285727	1.0015615
a_track[420]	-6.021738e-02	0.30440748	-0.5397658390	0.420031914	1.0025854
a_track[421]	-6.316498e-02	0.29258080	-0.5231551337	0.414718799	0.9997107
a_track[422]	1.673157e-01	0.35122382	-0.3990074426	0.737862888	0.9997885
a_track[423]	8.903486e-02	0.31905510	-0.4150735923	0.601474343	1.0009961
a_track[424]	1.872642e-01	0.34776116	-0.3795400681	0.737250226	1.0026920
a_track[425]	-6.319126e-02	0.30931787	-0.5692850299	0.416585161	0.9996524
a_track[426]	1.030377e-01	0.30976740	-0.3932061769	0.603912934	1.0020645
a_track[427]	-2.201624e-01	0.28967550	-0.6784872984	0.233304539	1.0009045
a_track[428]	7.631996e-02	0.31457892	-0.4163546282	0.580268357	1.0000892
a_track[429]	-7.130103e-02	0.30923302	-0.5650086876	0.417664819	1.0023696
a_track[430]	-6.776796e-02	0.31538942	-0.5743174823	0.436212130	1.0033159
a_track[431]	6.897854e-02	0.31510460	-0.4237936288	0.582285627	1.0003948
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a_track[433]	7.097340e-02	0.32365625	-0.4493000503	0.603484956	1.0007842
a_track[434]	2.306408e-01	0.32498896	-0.3023451810	0.757525128	1.0023288
a_track[435]	-3.577952e-01	0.27319752	-0.7877662514	0.079750464	1.0013652
a_track[436]	1.869666e-01	0.34178567	-0.3599232934	0.734359387	1.0024786
a_track[437]	6.572705e-02	0.31692181	-0.4358529699	0.579461144	1.0002833
a_track[438]	-6.698623e-02	0.29343134	-0.5345712793	0.407645077	0.9999705
a_track[439]	8.373097e-02	0.31342886	-0.4177778763	0.583656772	1.0022775
a_track[440]	-6.479477e-02	0.31314058	-0.5684930755	0.424633295	1.0013053
a_track[441]	-7.382694e-02	0.30114598	-0.5496687532	0.412106194	1.0015905
a_track[442]	7.730829e-02	0.34210650	-0.4618200317	0.611745972	1.0011975
a_track[443]	-2.233212e-01	0.28647494	-0.6647595703	0.235850725	1.0017391
a_track[444]	-3.641686e-01	0.27810986	-0.8058562832	0.084979360	1.0027350
a_track[445]	-6.302739e-02	0.31483252	-0.5713873903	0.446168383	0.9996112
a_track[446]	6.836435e-02	0.33051247	-0.4557072870	0.590759819	1.0004424
a_track[447]	1.869422e-01	0.34669672	-0.3689353471	0.740074737	1.0006678
a_track[448]	6.923629e-02	0.31228694	-0.4260667291	0.578337119	1.0001797
a_track[449]	9.510813e-02	0.30963058	-0.3907509451	0.606294158	1.0007827
a_track[450]	-6.723632e-01	0.24917092	-1.0651906790	-0.267621841	1.0011697
a_track[451]	4.938850e-02	0.34673648	-0.4963323881	0.601388825	1.0006896

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a_track[453]	-4.429674e-01	0.26980656	-0.8651660238	-0.013883384	1.0003017
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a_track[455]	7.014585e-02	0.32179694	-0.4334129886	0.594126406	1.0015965
a_track[456]	9.748456e-02	0.31646953	-0.4040002800	0.609654925	1.0000092
a_track[457]	7.451999e-02	0.31148678	-0.4301483428	0.578030811	1.0001413
a_track[458]	-6.651364e-02	0.31004239	-0.5602834852	0.428479876	1.0001262
a_track[459]	8.951757e-02	0.31985477	-0.4188643725	0.608132165	1.0047904
a_track[460]	9.570207e-02	0.32103665	-0.4079540788	0.616516583	1.0004186
a_track[461]	1.969097e-01	0.34686685	-0.3586101512	0.757815628	1.0005598
a_track[462]	8.439890e-02	0.31725937	-0.4164883027	0.609664580	1.0012500
a_track[463]	-6.413578e-02	0.29759834	-0.5308707210	0.417332077	1.0009543
a_track[464]	-6.097163e-02	0.30155612	-0.5426089148	0.420143642	1.0001056
a_track[465]	6.697566e-02	0.32224685	-0.4419219094	0.591941400	1.0022233
a_track[466]	1.757125e-01	0.35518628	-0.3792183733	0.753087921	1.0028211
a_track[467]	-5.153956e-02	0.30300066	-0.5334334180	0.435494854	1.0008417
a_track[468]	1.719799e-01	0.36710491	-0.4193451550	0.769279070	1.0011390
a_track[469]	-6.431244e-02	0.29884992	-0.5344788272	0.414697236	1.0004262
a_track[470]	-6.129498e-02	0.31386355	-0.5594063275	0.441105871	1.0018829
a_track[471]	-2.081097e-01	0.28162090	-0.6526219334	0.242219337	1.0013720
a_track[472]	2.258933e-01	0.32765177	-0.2853816367	0.748646101	1.0011591
a_track[473]	9.298711e-02	0.32270863	-0.4081161603	0.615776633	1.0021765
a_track[474]	7.089112e-02	0.33297707	-0.4587655190	0.606750924	1.0007087
a_track[475]	9.728188e-02	0.31211929	-0.3938971677	0.602615142	1.0026914
a_track[476]	-6.212970e-02	0.30340277	-0.5535620639	0.432203729	1.0000596
a_track[477]	1.799424e-01	0.35229111	-0.3825961096	0.741218707	1.0009067
a_track[478]	7.217036e-02	0.32393134	-0.4406995780	0.600456812	1.0001394
a_track[479]	7.564421e-02	0.33062774	-0.4456572249	0.593199743	1.0011723
a_track[480]	2.043160e-01	0.34505258	-0.3478952062	0.770538767	1.0016784
a_track[481]	-6.094913e-02	0.29296440	-0.5270288814	0.410677314	1.0028488
a_track[482]	8.897285e-02	0.31240228	-0.4070006549	0.586425750	1.0019317
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a_track[484]	-1.984827e-01	0.28341697	-0.6392065110	0.260583750	1.0010084
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a_track[486]	-1.858244e-01	0.29830235	-0.6619802946	0.285386409	0.9998388
a_track[487]	-4.837652e-02	0.31482685	-0.5506909901	0.457562382	1.0004524
a_track[488]	2.040143e-01	0.33262684	-0.3276707830	0.741942885	1.0005937
a_track[489]	-5.610720e-02	0.29374090	-0.5239982046	0.417583114	1.0037150
a_track[490]	-5.621389e-02	0.30152131	-0.5319904706	0.428013886	0.9999281
a_track[491]	7.462771e-02	0.32049149	-0.4280813014	0.573101804	1.0029263
a_track[492]	7.400231e-02	0.31550232	-0.4304858257	0.584423016	1.0019869
a_track[493]	-3.085226e-02	0.21739504	-0.3727410812	0.322157500	1.0027949
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a_track[496]	-3.253458e-02	0.21588068	-0.3767332531	0.310340409	1.0015326
a_track[497]	-3.319846e-02	0.22835360	-0.3961836338	0.349470822	0.9999875
a_track[498]	-2.036761e-01	0.20730833	-0.5326301180	0.134156381	1.0004591
a_track[499]	-2.968703e-02	0.22586295	-0.3779151462	0.340035676	1.0010044
a_track[500]	3.590762e-01	0.26045083	-0.0474760483	0.792610986	0.9997263
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a_track[502]	5.020036e-01	0.33576372	-0.0224501647	1.042272414	0.9996081
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a_track[504]	-1.057421e-01	0.24229113	-0.4841494470	0.277952705	1.0011543
a_track[505]	-5.752008e-01	0.21812975	-0.9170105426	-0.221396586	1.0011648
a_track[506]	-4.011208e-01	0.23688301	-0.7750915771	-0.019251654	1.0006177
a_track[507]	4.021019e-01	0.31913970	-0.1025474884	0.915762028	1.0009866
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a_track[513]	-3.007479e-01	0.21217515	-0.6459540405	0.045619187	1.0001003
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a_track[515]	-3.021333e-01	0.20573304	-0.6230027915	0.035773972	1.0015175
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a_track[518]	-7.556750e-02	0.22372036	-0.4324861048	0.289424929	0.9998973
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a_track[520]	-1.400220e-01	0.21566072	-0.4689388708	0.205768940	1.0024822
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a_track[523]	-3.885720e-01	0.22645140	-0.7426259309	-0.024151725	1.0004256
a_track[524]	3.849245e-01	0.25127709	-0.0131684842	0.800408475	1.0029755
a_track[525]	2.744830e-01	0.25616717	-0.1277593051	0.694322470	1.0001627
a_track[526]	-1.771994e-01	0.26183030	-0.6030371505	0.244642803	1.0006093
a_track[527]	-4.580315e-02	0.22824027	-0.4179873169	0.324222833	1.0016166
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a_track[535]	-1.685843e-01	0.20296890	-0.4881367220	0.160179884	1.0005335
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a_track[541]	-2.880148e-01	0.19057983	-0.5887551719	0.016990890	0.9996282
a_track[542]	-1.043617e-01	0.20370227	-0.4264606616	0.221687789	1.0001131
a_track[543]	-1.000723e-01	0.20807390	-0.4253014582	0.228981178	0.9999217
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a_track[545]	4.049924e-02	0.21176964	-0.2941200694	0.380311327	1.0040865
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a_track[547]	5.349609e-01	0.32086358	0.0309185359	1.054404526	1.0007870
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a_track[549]	1.123830e-01	0.21871960	-0.2336638512	0.465031450	1.0020195
a_track[550]	-1.608064e-01	0.20799513	-0.4874407321	0.167523840	1.0000176
a_track[551]	-3.537851e-02	0.21127101	-0.3693499765	0.309580812	1.0033672
a_track[552]	1.026194e-01	0.21778369	-0.2461066184	0.451690419	1.0007915
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a_track[554]	-1.106489e-01	0.22877230	-0.4728457207	0.253055183	1.0005322
a_track[555]	-2.868948e-02	0.22759368	-0.3908779450	0.334832868	1.0019037
a_track[556]	2.501303e-01	0.25130074	-0.1476491446	0.661163799	1.0017490
a_track[557]	-2.316854e-02	0.21446246	-0.3684299102	0.317036241	1.0046544
a_track[558]	6.215289e-01	0.32216757	0.1210195062	1.141847562	1.0031244
a_track[559]	-9.717354e-02	0.22639199	-0.4562023077	0.271523627	1.0011453
a_track[560]	3.576126e-01	0.26674439	-0.0589038226	0.787024085	1.0022144
a_track[561]	-4.502894e-01	0.19924364	-0.7666887490	-0.134069013	1.0006153
a_track[562]	4.409067e-01	0.27938878	0.0007947859	0.894691184	0.9995985
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a_track[564]	7.702740e-02	0.22532638	-0.2765053733	0.449228485	1.0009757
a_track[565]	-5.143509e-01	0.19874118	-0.8251858909	-0.191346886	1.0000573
a_track[566]	4.067564e-01	0.34634119	-0.1515242090	0.957070676	1.0010136
a_track[567]	2.915197e-01	0.25329078	-0.1083213862	0.703334280	1.0012358
a_track[568]	3.195101e-01	0.36762600	-0.2648611689	0.901423238	1.0036177
a_track[569]	-2.559279e-01	0.22629921	-0.6232902596	0.098474910	1.0005571
a_track[570]	-4.781565e-01	0.22474794	-0.8362123407	-0.121173133	1.0011940
a_track[571]	-1.342284e-01	0.24060870	-0.5131301245	0.252705508	1.0014957
a_track[572]	4.663707e-01	0.27470106	0.0380548129	0.901064735	1.0001850
a_track[573]	-2.265889e-02	0.23731185	-0.3988753611	0.356814066	1.0039458
a_track[574]	-2.532916e-01	0.21778189	-0.5973825669	0.097194587	1.0006503
a_track[575]	-3.342399e-01	0.20827476	-0.6616358625	0.005798563	1.0031288
a_track[576]	-1.841125e-01	0.21096085	-0.5191394485	0.159788328	1.0002495
a_track[577]	-4.081656e-01	0.19900786	-0.7245759329	-0.079179169	1.0007370
a_track[578]	6.293301e-02	0.22546059	-0.2940839973	0.424685212	1.0005273
a_track[579]	-1.150063e-01	0.22135619	-0.4647867878	0.232978253	1.0008183
a_track[580]	-2.303112e-02	0.23609302	-0.4001450432	0.350543853	1.0003519

a_track[581]	-4.614540e-01	0.20018393	-0.7844666123	-0.144970403	1.0002762
a_track[582]	5.030740e-01	0.30144731	0.0286925448	0.983254899	1.0013797
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a_track[584]	-3.218605e-01	0.20191823	-0.6438282053	0.002346290	1.0020403
a_track[585]	-1.808842e-01	0.21258594	-0.5159640210	0.168552255	1.0036157
a_track[586]	-3.299010e-01	0.20121527	-0.6435533931	-0.007742225	1.0012068
a_track[587]	-2.358094e-01	0.27522985	-0.6706745630	0.199983861	1.0000444
a_track[588]	-3.486993e-01	0.26722317	-0.7734872580	0.082805851	1.0000859
a_track[589]	3.200408e-01	0.29677927	-0.1536547973	0.786245186	1.0011474
a_track[590]	1.458902e-01	0.30712231	-0.3430222107	0.632926675	1.0009833
a_track[591]	2.802661e-01	0.31226114	-0.2147270099	0.767697492	1.0004917
a_track[592]	4.475136e-01	0.32072200	-0.0523911823	0.966786515	1.0019275
a_cell[1]	5.199304e-03	0.07165321	-0.1026587984	0.120507614	1.1001997
a_cell[2]	-3.063628e-03	0.06674867	-0.1091842715	0.102617939	1.0972147
a_cell[3]	-4.369418e-03	0.07165632	-0.1196023409	0.104175899	1.1003166
a_cell[4]	-1.408395e-03	0.07198599	-0.1153331033	0.114056167	1.1100544
a_cell[5]	-3.551273e-03	0.07065368	-0.1195992913	0.107131432	1.1031107
a_cell[6]	-5.363381e-03	0.07138159	-0.1250887145	0.104461019	1.1127869
a_cell[7]	1.022582e-02	0.07145635	-0.0948576008	0.127736082	1.1032656
a_cell[8]	5.283670e-03	0.06738530	-0.1034247881	0.113270360	1.0858759
a_cell[9]	2.238846e-02	0.07028744	-0.0766860819	0.142465830	1.0826179
a_cell[10]	-1.129470e-03	0.06878033	-0.1023316333	0.105380265	1.0931661
a_cell[11]	1.785528e-03	0.06993104	-0.1077815529	0.114142687	1.0936197
a_cell[12]	-3.216621e-03	0.07133947	-0.1179166717	0.107630732	1.1152801
a_cell[13]	-1.834570e-02	0.07033972	-0.1389267055	0.083484706	1.1022965
a_cell[14]	-1.335512e-02	0.07326436	-0.1353549326	0.091935003	1.0732921
a_cell[15]	3.498195e-02	0.07508010	-0.0616812135	0.177468037	1.0714659
a_cell[16]	-3.784342e-03	0.06787911	-0.1146415693	0.104369165	1.1081542
a_cell[17]	-4.076221e-03	0.07266521	-0.1211268648	0.105224028	1.1134398
a_cell[18]	2.621917e-02	0.07081639	-0.0665694921	0.153603632	1.0846531
a_cell[19]	1.028314e-02	0.06915544	-0.0919046280	0.123702307	1.1030995
a_cell[20]	-4.045970e-03	0.06818145	-0.1139822238	0.103482995	1.0906143
a_cell[21]	3.200361e-03	0.06687594	-0.1041167619	0.111865052	1.1089343
a_cell[22]	-1.616368e-02	0.06818721	-0.1308034006	0.082526082	1.0793002
a_cell[23]	-1.812203e-02	0.07049419	-0.1414631909	0.084490585	1.1062224
a_cell[24]	2.265367e-02	0.06758690	-0.0701176425	0.138980690	1.0778663
a_cell[25]	-1.189482e-02	0.06634630	-0.1225287239	0.087160414	1.0822672
a_cell[26]	1.089523e-02	0.06489293	-0.0875081569	0.122050530	1.0893492
a_cell[27]	1.144042e-03	0.06927254	-0.1057011157	0.110622574	1.1010941
a_cell[28]	2.071304e-03	0.06977858	-0.1083271790	0.110528476	1.0931955
a_cell[29]	4.220115e-04	0.07055582	-0.1131001960	0.111348003	1.0922773
a_cell[30]	-8.321524e-03	0.07006244	-0.1212337207	0.096395988	1.0947443
a_cell[31]	2.320060e-03	0.07078353	-0.1076378414	0.116501366	1.0972865

a_cell[32]	2.019138e-03	0.06889232	-0.1050448613	0.108283833	1.1059651
a_cell[33]	-8.113242e-03	0.07216780	-0.1248125880	0.100222896	1.0991107
a_cell[34]	-3.593687e-03	0.06829563	-0.1138935780	0.100277033	1.0989412
a_cell[35]	7.652269e-03	0.06934097	-0.0977304390	0.122152077	1.0988556
a_cell[36]	2.382154e-03	0.06936957	-0.1032849152	0.112762005	1.0957608
a_cell[37]	-1.248090e-03	0.06987349	-0.1135665345	0.107460122	1.1151013
a_cell[38]	-1.817294e-02	0.06738875	-0.1328633947	0.078302883	1.0894560
a_cell[39]	1.133642e-02	0.06533925	-0.0858469084	0.122419187	1.0912194
a_cell[40]	9.269018e-03	0.06383949	-0.0900286793	0.113054842	1.0992074
a_cell[41]	8.656246e-03	0.06217947	-0.0821319230	0.113650822	1.0945984
a_cell[42]	-3.374369e-02	0.06786218	-0.1573299905	0.054220491	1.0711794
a_cell[43]	1.471943e-02	0.06743279	-0.0831883455	0.132848544	1.0990383
a_cell[44]	-1.542407e-02	0.06608626	-0.1324335903	0.082553055	1.1050310
a_cell[45]	-4.582506e-03	0.06194703	-0.1063398037	0.092527460	1.0788705
a_cell[46]	-2.048353e-03	0.06662547	-0.1085653158	0.107375950	1.0946930
a_cell[47]	-2.933332e-04	0.06718590	-0.1096627969	0.106569301	1.0952156
a_cell[48]	1.479015e-02	0.07081954	-0.0902335511	0.136286699	1.0844399
a_cell[49]	-1.094216e-02	0.06677436	-0.1227821591	0.090418960	1.0825974
a_cell[50]	1.002833e-02	0.06484832	-0.0884369836	0.118929579	1.0992860
a_cell[51]	1.263614e-02	0.06976747	-0.0921517407	0.126764545	1.0857562
a_cell[52]	4.832534e-03	0.06688191	-0.0962921604	0.110503755	1.1015079
a_cell[53]	-3.392510e-03	0.06995013	-0.1144591403	0.104320522	1.0938221
a_cell[54]	3.578041e-03	0.06967483	-0.1041118176	0.118087071	1.0981669
a_cell[55]	-4.900822e-03	0.06665144	-0.1119238919	0.097156265	1.0852972
a_cell[56]	-1.124815e-02	0.06542707	-0.1235955255	0.086930625	1.0943266
a_cell[57]	-4.381069e-03	0.06607154	-0.1094608791	0.094502996	1.0898024
a_cell[58]	1.456563e-02	0.06661440	-0.0824647577	0.128722356	1.0893673
a_cell[59]	-3.099446e-02	0.06910482	-0.1551098214	0.062605070	1.0603960
a_cell[60]	2.268666e-02	0.06710699	-0.0723165021	0.138437664	1.0928817
a_cell[61]	-2.570799e-02	0.06914689	-0.1477538739	0.067352550	1.0926342
a_cell[62]	5.577116e-05	0.06283741	-0.1004828241	0.099674943	1.0856774
a_cell[63]	2.146926e-02	0.06653431	-0.0721495733	0.139415992	1.0940994
a_cell[64]	-3.477805e-02	0.07080295	-0.1655793191	0.057798825	1.0621792
a_cell[65]	-1.839522e-03	0.06748799	-0.1126045951	0.100546940	1.1106773
a_cell[66]	-1.030484e-02	0.06696178	-0.1238453354	0.088940118	1.0829486
a_cell[67]	3.034527e-03	0.06668722	-0.0995874519	0.109344801	1.0942683
a_cell[68]	3.891178e-03	0.06536906	-0.1004882163	0.112074413	1.0880960
a_cell[69]	-1.371931e-02	0.06713074	-0.1257418839	0.086128113	1.0815295
a_cell[70]	-9.767838e-04	0.06972121	-0.1164306959	0.111802466	1.1054500
a_cell[71]	7.873299e-03	0.06970899	-0.0975650668	0.127335163	1.1199142
a_cell[72]	-8.047829e-03	0.06827826	-0.1224745995	0.096975236	1.0874804
a_cell[73]	-4.869920e-03	0.07144829	-0.1182918192	0.103926154	1.1012225
a_cell[74]	-1.292622e-02	0.06838408	-0.1292548569	0.086854195	1.0855313

a_cell[75]	1.426035e-02	0.07046962	-0.0871958035	0.134228979	1.0897917
a_cell[76]	-8.842148e-03	0.06807996	-0.1206697121	0.097449418	1.1055799
a_cell[77]	-4.514557e-03	0.06780764	-0.1101392910	0.097805319	1.0859038
a_cell[78]	2.400249e-02	0.07026563	-0.0742059506	0.149877757	1.0818030
a_cell[79]	1.345846e-02	0.06726848	-0.0840735531	0.128505180	1.0859788
a_cell[80]	-1.825924e-02	0.06876204	-0.1375206703	0.081458315	1.0882890
a_cell[81]	-2.292516e-02	0.07083946	-0.1470367708	0.073722439	1.1060996
a_cell[82]	2.901439e-03	0.06655603	-0.0996369863	0.110580930	1.0928791
a_cell[83]	7.798380e-03	0.06385306	-0.0910157604	0.115543189	1.0951839
a_cell[84]	-3.980718e-02	0.07219633	-0.1737001145	0.052187331	1.0662149
a_cell[85]	-1.233976e-02	0.06593089	-0.1236038833	0.086305922	1.0894143
a_cell[86]	-6.511730e-03	0.06504660	-0.1155893318	0.097812232	1.0852510
a_cell[87]	2.135831e-02	0.06949358	-0.0796324966	0.143041807	1.0913485
a_cell[88]	-1.011687e-02	0.06753334	-0.1219801308	0.089943862	1.1006358
a_cell[89]	2.423853e-02	0.06996675	-0.0706457716	0.148425694	1.0783291
a_cell[90]	-1.847171e-02	0.06978126	-0.1342797260	0.081572265	1.1001873
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a_cell[92]	-6.777408e-03	0.06745249	-0.1137718861	0.096281484	1.1016531
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a_cell[94]	2.532087e-04	0.07206233	-0.1148221736	0.116770132	1.1025386
a_cell[95]	-1.040280e-02	0.07385075	-0.1309492779	0.097672114	1.1187292
a_cell[96]	6.170429e-03	0.06502998	-0.0898026965	0.114100322	1.0935706
a_cell[97]	-2.651834e-03	0.06654008	-0.1111989543	0.101567985	1.0811497
a_cell[98]	-1.456039e-02	0.06987521	-0.1315717550	0.086299987	1.0931841
a_cell[99]	4.287374e-02	0.07651101	-0.0538969337	0.184933063	1.0616734
a_cell[100]	-2.560728e-03	0.06816771	-0.1088909074	0.102507389	1.0939781
a_cell[101]	-1.038542e-02	0.06980232	-0.1284642098	0.094768167	1.0986237
a_cell[102]	1.137185e-02	0.06876483	-0.0879433232	0.127707221	1.0940581
a_cell[103]	-2.785974e-03	0.06884014	-0.1165239329	0.100261443	1.1013456
a_cell[104]	3.414488e-02	0.06851921	-0.0543333133	0.160969328	1.0611659
a_cell[105]	1.041672e-03	0.06241722	-0.0975913614	0.101683130	1.1004279
a_cell[106]	-9.990885e-03	0.06892798	-0.1291016663	0.094762410	1.0930107
a_cell[107]	-2.691944e-02	0.06923446	-0.1479304244	0.066103067	1.0770226
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a_cell[110]	-1.896587e-02	0.06748021	-0.1372993143	0.078055952	1.0733092
a_cell[111]	-2.715810e-02	0.06789449	-0.1467548962	0.064787867	1.0676200
a_cell[112]	-1.082750e-02	0.06752343	-0.1193187665	0.088391453	1.0805453
a_cell[113]	-1.370261e-02	0.06842397	-0.1318533763	0.086916105	1.0809168
a_cell[114]	1.503729e-03	0.06640814	-0.1045460144	0.109917745	1.0997911
a_cell[115]	-4.060688e-03	0.07118808	-0.1218663014	0.107989532	1.0972626
a_cell[116]	-1.390175e-02	0.06965734	-0.1293829307	0.088701953	1.1017972
a_cell[117]	-3.889852e-03	0.07250272	-0.1182791282	0.103470328	1.0954495

a_cell[118]	-4.958540e-03	0.06725648	-0.1135213875	0.099234994	1.0822433
a_cell[119]	3.937503e-02	0.07530248	-0.0552361895	0.172286539	1.0756334
a_cell[120]	1.744303e-03	0.06605528	-0.1043709367	0.105325923	1.1117303
a_cell[121]	7.462748e-03	0.06887442	-0.0934543398	0.121682596	1.0967216
a_cell[122]	2.081003e-02	0.06822479	-0.0751901855	0.141777104	1.0875178
a_cell[123]	-8.337231e-03	0.07003244	-0.1282837124	0.101007453	1.0982586
a_cell[124]	-9.368393e-03	0.06656399	-0.1236213519	0.091646020	1.0801595
a_cell[125]	-1.608618e-02	0.06832148	-0.1357292446	0.085944959	1.0897004
a_cell[126]	1.377754e-02	0.06740614	-0.0884122715	0.126862943	1.0853534
a_cell[127]	6.409765e-03	0.06629138	-0.0960640211	0.116754672	1.0994834
a_cell[128]	-1.559724e-02	0.06627972	-0.1289545671	0.083875772	1.0918130
a_cell[129]	1.850023e-02	0.06676416	-0.0764023519	0.133356920	1.0800190
a_cell[130]	9.454129e-03	0.06660933	-0.0902203996	0.122516505	1.0905111
a_cell[131]	-2.978390e-02	0.07163886	-0.1580264385	0.068700630	1.0830831
a_cell[132]	8.019923e-03	0.06468065	-0.0887628685	0.116894442	1.0834855
a_cell[133]	-6.529350e-03	0.06964432	-0.1226513827	0.099865539	1.1022625
a_cell[134]	-8.712161e-03	0.06515160	-0.1161841712	0.089106061	1.0967621
a_cell[135]	-2.436766e-02	0.06890715	-0.1463170911	0.070037723	1.0775635
a_cell[136]	1.878190e-02	0.06622613	-0.0751053717	0.133090572	1.0991602
a_cell[137]	1.094890e-02	0.06682019	-0.0870005230	0.121186744	1.0759836
a_cell[138]	2.836539e-03	0.06726570	-0.1016063453	0.114810749	1.0928636
a_cell[139]	-9.746989e-03	0.06649808	-0.1223349276	0.089032522	1.0830071
a_cell[140]	1.182359e-02	0.06863410	-0.0926159863	0.125724046	1.0956976
a_cell[141]	-1.290718e-02	0.07021902	-0.1335130655	0.090329679	1.0845716
a_cell[142]	-2.556098e-03	0.06952472	-0.1187394550	0.105147646	1.1109770
a_cell[143]	1.404551e-03	0.07091584	-0.1084345735	0.110575487	1.0897515
a_cell[144]	2.091404e-03	0.06661737	-0.1061657869	0.110748413	1.0882268
a_cell[145]	-1.351581e-02	0.06901076	-0.1283244334	0.088851266	1.0919058
a_cell[146]	5.989504e-03	0.06557098	-0.0948383448	0.111449763	1.0893776
a_cell[147]	1.074779e-03	0.07068424	-0.1066421637	0.113398517	1.1085472
a_cell[148]	-3.486526e-03	0.06805567	-0.1158650000	0.099108184	1.0930724
a_cell[149]	-9.166957e-03	0.07351939	-0.1287292791	0.097828874	1.1125650
a_cell[150]	1.439523e-02	0.06800297	-0.0853748172	0.131287400	1.0916819
a_cell[151]	-2.849497e-03	0.06731023	-0.1101403267	0.102128108	1.0947502
a_cell[152]	1.392321e-02	0.06632562	-0.0843832127	0.129011459	1.0793809
a_cell[153]	1.660009e-04	0.06667590	-0.1050452959	0.105570750	1.0956617
a_cell[154]	2.184697e-02	0.07243172	-0.0794996525	0.144098772	1.0864936
a_cell[155]	7.569545e-03	0.06467483	-0.0894558556	0.120552335	1.0785445
a_cell[156]	-1.972906e-02	0.06720282	-0.1328326780	0.076203024	1.0828992
a_cell[157]	-7.448714e-03	0.06393727	-0.1193687131	0.090836439	1.0825500
a_cell[158]	8.247663e-03	0.06597473	-0.0910455143	0.115160910	1.0940945
a_cell[159]	4.139974e-02	0.07183560	-0.0514023470	0.173719138	1.0690123
a_cell[160]	-2.633933e-03	0.06646575	-0.1133613138	0.103000863	1.1055708

a_cell[161]	-1.148312e-02	0.07119898	-0.1279909597	0.094879654	1.1055807
a_cell[162]	-3.893582e-03	0.06517339	-0.1081327822	0.097864016	1.0843679
a_cell[163]	-3.934661e-03	0.06623851	-0.1098294261	0.094509687	1.0872055
a_cell[164]	-5.978633e-03	0.06582821	-0.1113107595	0.092386194	1.0732796
a_cell[165]	-9.844400e-03	0.07169326	-0.1273242848	0.095912760	1.0920203
a_cell[166]	-1.544446e-03	0.06882011	-0.1110750049	0.109332746	1.0917318
a_cell[167]	-3.461860e-03	0.07189390	-0.1197680984	0.102327278	1.1076737
a_cell[168]	-1.648707e-02	0.07408447	-0.1423227097	0.091382840	1.0940847
a_cell[169]	5.182073e-02	0.07897650	-0.0450358745	0.196374907	1.0751911
a_cell[170]	6.104897e-03	0.06888083	-0.0977223941	0.117709116	1.0961645
a_cell[171]	8.426400e-03	0.06678913	-0.0937568851	0.121526399	1.0977138
a_cell[172]	-1.527008e-03	0.07049996	-0.1153863597	0.106893341	1.0963161
a_cell[173]	1.162195e-02	0.06773181	-0.0878439251	0.126096316	1.0929378
a_cell[174]	-1.489248e-03	0.07016434	-0.1123616559	0.108333290	1.1109392
a_cell[175]	-2.321530e-03	0.06939019	-0.1091688286	0.107747385	1.1056119
a_cell[176]	1.058503e-03	0.06888453	-0.1068048396	0.110440447	1.0953409
a_cell[177]	-6.775828e-03	0.06637857	-0.1178419360	0.096541187	1.0757031
a_cell[178]	-1.345647e-02	0.06585764	-0.1260298097	0.084152366	1.1077786
a_cell[179]	-6.847646e-03	0.06737236	-0.1124098177	0.091624230	1.1059627
a_cell[180]	1.481736e-02	0.06563491	-0.0796824709	0.127598233	1.0895007
a_cell[181]	1.186922e-02	0.06531363	-0.0866184547	0.120264325	1.0945615
a_cell[182]	-1.717390e-02	0.06974745	-0.1365583229	0.082151103	1.1047197
a_cell[183]	-1.371361e-02	0.06920748	-0.1336172161	0.089139410	1.0962157
a_cell[184]	7.373622e-03	0.07027767	-0.1000340979	0.125085414	1.0997775
a_cell[185]	6.503856e-03	0.06834088	-0.0992813755	0.117612552	1.0971899
a_cell[186]	-7.505248e-03	0.06384752	-0.1131142710	0.090434523	1.0806676
a_cell[187]	-3.592976e-03	0.06685415	-0.1092666111	0.099588607	1.1005146
a_cell[188]	-1.757414e-03	0.06968197	-0.1100489985	0.106516904	1.0963080
a_cell[189]	9.098665e-04	0.07107101	-0.1132121840	0.113485593	1.1008725
a_cell[190]	-2.706095e-03	0.07326799	-0.1198067540	0.109403198	1.1090423
a_cell[191]	2.453007e-03	0.07106374	-0.1090509931	0.120766839	1.0978300
a_cell[192]	1.115258e-02	0.06856162	-0.0911555426	0.125341913	1.0781327
a_cell[193]	-3.718315e-03	0.06852941	-0.1150028321	0.100414966	1.1102859
a_cell[194]	7.064946e-03	0.06942047	-0.0990110855	0.122731365	1.0961006
a_cell[195]	3.212323e-02	0.07056065	-0.0623549332	0.155486654	1.0682424
a_cell[196]	-1.016290e-02	0.07001839	-0.1301640226	0.098515255	1.0932909
a_cell[197]	2.970793e-02	0.07185810	-0.0672869405	0.156049329	1.0928859
a_cell[198]	2.044252e-03	0.06955403	-0.1043280126	0.109701605	1.0959951
a_cell[199]	7.920271e-05	0.06979816	-0.1093285727	0.105727744	1.0982657
a_cell[200]	-4.664073e-03	0.07270282	-0.1188078763	0.103772294	1.1120286
a_cell[201]	-5.882106e-03	0.06671292	-0.1179017926	0.095851639	1.0787763
a_cell[202]	1.630748e-02	0.06863466	-0.0806106560	0.134197322	1.0958136
a_cell[203]	1.560112e-02	0.06823457	-0.0866377585	0.131957873	1.1008819

a_cell[204]	5.590168e-02	0.07920367	-0.0383540722	0.205388058	1.0826162
a_cell[205]	2.859078e-02	0.06993993	-0.0688778342	0.150018728	1.0838526
a_cell[206]	-8.459864e-03	0.06862382	-0.1179604525	0.096545073	1.0774504
a_cell[207]	-1.084593e-02	0.06477541	-0.1188924272	0.088604366	1.0914012
a_cell[208]	-2.900720e-02	0.07155385	-0.1552409543	0.071327900	1.0738702
a_cell[209]	-5.429843e-03	0.06724928	-0.1173628294	0.101766731	1.0898159
a_cell[210]	7.106108e-03	0.06743489	-0.0938133334	0.116923112	1.0946908
a_cell[211]	5.741858e-02	0.08106303	-0.0374713763	0.212034343	1.0872080
a_cell[212]	-1.938442e-02	0.06905877	-0.1417799332	0.078955702	1.0807747
a_cell[213]	-5.222300e-03	0.06677047	-0.1143548735	0.099447657	1.1041859
a_cell[214]	2.126798e-02	0.07008865	-0.0732661040	0.145791611	1.0913342
a_cell[215]	-4.676900e-03	0.07302733	-0.1239106966	0.108117682	1.1113688
a_cell[216]	-4.318243e-03	0.06944592	-0.1177007979	0.104431124	1.0969000
a_cell[217]	-4.004380e-03	0.07117965	-0.1140764788	0.105598467	1.1011581
a_cell[218]	1.615353e-02	0.07050744	-0.0853228502	0.136988139	1.0959872
a_cell[219]	7.228363e-03	0.07135735	-0.0982884281	0.121877919	1.0914505
a_cell[220]	-6.179381e-03	0.07161093	-0.1265971040	0.103910620	1.1192588
a_cell[221]	8.679546e-03	0.06972023	-0.0969572951	0.123485074	1.0853587
a_cell[222]	1.476842e-03	0.06853119	-0.1030856365	0.110735164	1.0849472
a_cell[223]	1.858558e-03	0.06593607	-0.1007596858	0.110743445	1.0912904
a_cell[224]	7.585033e-04	0.06897512	-0.1065944693	0.111546688	1.1084177
a_cell[225]	1.509415e-02	0.07104214	-0.0894683585	0.135876231	1.0945656
a_cell[226]	5.176249e-05	0.07198221	-0.1120245278	0.116215185	1.1074477
a_cell[227]	1.430146e-02	0.06782158	-0.0846167241	0.124509463	1.0814605
a_cell[228]	-4.612494e-03	0.06937698	-0.1222454132	0.098768781	1.0906897
a_cell[229]	-1.017235e-02	0.06808184	-0.1231322926	0.089664470	1.0947535
a_cell[230]	-3.035015e-02	0.07313041	-0.1600086648	0.064749446	1.0770847
a_cell[231]	2.759883e-04	0.06789939	-0.1090373643	0.104599649	1.0899134
a_cell[232]	-2.743581e-02	0.07351799	-0.1580208136	0.072815038	1.0832127
a_cell[233]	-3.349297e-03	0.06619228	-0.1114127531	0.101452160	1.1037978
a_cell[234]	6.734632e-03	0.06654616	-0.0919067776	0.112459396	1.0803500
a_cell[235]	1.193081e-03	0.07231434	-0.1064040262	0.113070868	1.0969411
a_cell[236]	1.270549e-02	0.06774666	-0.0882731381	0.128373678	1.0719585
a_cell[237]	-3.741091e-03	0.06528254	-0.1110772338	0.099512376	1.0881004
a_cell[238]	-8.903947e-03	0.06758949	-0.1223490591	0.090753776	1.0966068
a_cell[239]	-7.202354e-03	0.07093355	-0.1229112149	0.094704777	1.1013991
a_cell[240]	-3.082504e-03	0.07092830	-0.1192782214	0.108642706	1.0927225
a_cell[241]	3.004419e-02	0.07376192	-0.0677333583	0.162094628	1.0778009
a_cell[242]	-8.958647e-03	0.06867913	-0.1245799171	0.091435890	1.0999239
a_cell[243]	-8.257053e-03	0.07320904	-0.1305219410	0.106488034	1.0952295
a_cell[244]	-7.458035e-03	0.07156207	-0.1272825545	0.099922653	1.0951309
a_cell[245]	-5.182350e-03	0.06808319	-0.1189959079	0.099787843	1.1063989
a_cell[246]	2.282601e-02	0.07288587	-0.0757863419	0.151713420	1.0853879

a_cell[247]	-9.213347e-03	0.07209844	-0.1328798044	0.095679875	1.0881641
a_cell[248]	-3.875740e-03	0.07138520	-0.1178750207	0.107580730	1.0926414
a_cell[249]	-7.396587e-04	0.07077214	-0.1101062102	0.112242873	1.0876628
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a_cell[253]	-1.839247e-02	0.07162849	-0.1409002895	0.088043190	1.1013541
a_cell[254]	-4.316985e-03	0.06990037	-0.1150514592	0.106365688	1.0939699
a_cell[255]	1.362058e-03	0.07025865	-0.1090533780	0.117078987	1.0928779
a_cell[256]	7.238886e-04	0.06754951	-0.1042915033	0.106654890	1.0929910
a_cell[257]	-1.349593e-02	0.07075333	-0.1306361220	0.090542706	1.0924138
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a_cell[260]	-1.271667e-02	0.07010734	-0.1254652095	0.089492003	1.0889668
a_cell[261]	7.635381e-03	0.06192955	-0.0864122216	0.106160939	1.0784120
a_cell[262]	5.860550e-03	0.06477207	-0.0948421765	0.110787409	1.0920390
a_cell[263]	-1.566146e-02	0.06721146	-0.1338185739	0.081818910	1.0769879
a_cell[264]	1.410571e-02	0.07027317	-0.0885431802	0.132943435	1.0931969
a_cell[265]	-4.132421e-03	0.07179368	-0.1198460830	0.104820167	1.1111113
a_cell[266]	8.657810e-03	0.07095946	-0.0983441506	0.127821189	1.0798562
a_cell[267]	-1.411339e-02	0.07073212	-0.1341143129	0.089591103	1.0904997
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a_cell[269]	-7.121048e-03	0.07452715	-0.1311643717	0.101935340	1.1027013
a_cell[270]	-1.685319e-02	0.07199575	-0.1408321556	0.089576134	1.0974298
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a_cell[274]	-4.626263e-03	0.07117606	-0.1187807427	0.106048909	1.0874115
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a_cell[280]	-9.613736e-04	0.07079799	-0.1103182405	0.112608215	1.1050362
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a_cell[297]	-1.363421e-02	0.07457077	-0.1419320400	0.094026764	1.1088551
a_cell[298]	-1.023952e-02	0.06899034	-0.1251841348	0.096410393	1.1057912
a_cell[299]	4.704013e-03	0.07263441	-0.1072634350	0.122087732	1.1112799
a_cell[300]	4.379857e-03	0.06807746	-0.1010876184	0.115614577	1.1038069
a_cell[301]	1.375080e-02	0.06813150	-0.0859549223	0.125886718	1.1069821
a_cell[302]	8.193849e-04	0.06805400	-0.1071597805	0.107424677	1.1069244
a_cell[303]	1.060509e-02	0.06983288	-0.0936028066	0.126580780	1.1041224
a_cell[304]	-1.506204e-02	0.07162912	-0.1323732970	0.090049235	1.0938021
a_cell[305]	-1.475555e-02	0.06900796	-0.1303865390	0.087654458	1.0738817
a_cell[306]	-2.115386e-02	0.06971901	-0.1466614927	0.075488926	1.0931444
a_cell[307]	-5.840516e-03	0.07078057	-0.1191192716	0.104286792	1.0894849
a_cell[308]	-2.100154e-04	0.06930027	-0.1100678978	0.111884713	1.1049466
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a_cell[310]	-1.154690e-02	0.06934609	-0.1228432714	0.088218835	1.1171979
a_cell[311]	3.163670e-03	0.06938143	-0.1050192418	0.118574331	1.0900750
a_cell[312]	-2.777065e-02	0.07419875	-0.1551054668	0.067617153	1.0859958
a_cell[313]	-5.030113e-03	0.07095110	-0.1219912208	0.105505873	1.1048303
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a_cell[315]	2.794607e-02	0.07054366	-0.0660132753	0.155851255	1.0770388
a_cell[316]	3.011994e-02	0.07127242	-0.0637966435	0.157732151	1.0776931
a_cell[317]	-2.264618e-02	0.07179109	-0.1461015251	0.075354086	1.0812198
a_cell[318]	2.215400e-03	0.06919318	-0.1013636095	0.109850501	1.0856357
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a_cell[320]	-1.562631e-03	0.06937127	-0.1115302980	0.109696919	1.1016565
a_cell[321]	1.144097e-02	0.06949215	-0.0882761443	0.128695185	1.0992799
a_cell[322]	2.256211e-03	0.07179124	-0.1088469025	0.116054633	1.0943121
a_cell[323]	2.682609e-03	0.07009592	-0.1039962915	0.115509410	1.1094800
a_cell[324]	4.930452e-03	0.06860691	-0.1031164981	0.113539069	1.1194908
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a_cell[326]	-7.291258e-04	0.07079484	-0.1130811656	0.109624659	1.1005713
a_cell[327]	3.968528e-03	0.06974096	-0.1031635051	0.115886373	1.0904659
a_cell[328]	4.292929e-03	0.06858560	-0.1011601080	0.114423552	1.1118759
a_cell[329]	-3.660755e-03	0.06991332	-0.1145205779	0.101420295	1.0943755
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a_cell[334]	3.204272e-02	0.07149846	-0.0610010802	0.158508346	1.0698640
a_cell[335]	2.279362e-02	0.06985585	-0.0721268660	0.147669006	1.0937935
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a_cell[339]	-9.082697e-03	0.06911342	-0.1214135012	0.094878316	1.0864731
a_cell[340]	-6.767731e-03	0.07083055	-0.1269503883	0.104496167	1.1059190
a_cell[341]	8.805337e-03	0.06907409	-0.0944360395	0.122109018	1.0939863
a_cell[342]	-1.648836e-03	0.07029943	-0.1119108794	0.109451410	1.1137206
a_cell[343]	-1.376361e-02	0.06958365	-0.1332329064	0.091942871	1.1000767
a_cell[344]	-1.757817e-02	0.06674399	-0.1309513189	0.081414301	1.0821948
a_cell[345]	-4.797759e-03	0.06678716	-0.1133061451	0.100522088	1.1061329
a_cell[346]	6.101883e-03	0.07320970	-0.1040901379	0.129305918	1.0900312
a_cell[347]	6.727859e-03	0.06959673	-0.1011033817	0.121545963	1.1298042
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a_cell[356]	-2.244409e-03	0.07313763	-0.1147113539	0.109173097	1.0943254
a_cell[357]	-5.180797e-03	0.06963839	-0.1180816092	0.100453492	1.0942423
a_cell[358]	-1.167006e-02	0.07032032	-0.1310110745	0.094019808	1.0876000
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a_cell[231]	5777.002629
a_cell[232]	208.880382
a_cell[233]	5769.270153
a_cell[234]	4439.346159
a_cell[235]	6243.877397
a_cell[236]	3969.232827
a_cell[237]	5131.045865
a_cell[238]	5489.825779
a_cell[239]	3545.459781
a_cell[240]	5431.087389
a_cell[241]	115.063165
a_cell[242]	5381.109140
a_cell[243]	5643.017466
a_cell[244]	4667.602770
a_cell[245]	3713.439101
a_cell[246]	323.175332
a_cell[247]	3600.833435
a_cell[248]	5958.558597
a_cell[249]	5804.995775
a_cell[250]	6105.832407

a_cell[251]	1457.160881
a_cell[252]	197.750045
a_cell[253]	1122.815638
a_cell[254]	5653.479291
a_cell[255]	5060.000909
a_cell[256]	5716.128146
a_cell[257]	2034.232746
a_cell[258]	5300.695079
a_cell[259]	6545.729942
a_cell[260]	2887.844664
a_cell[261]	5153.287190
a_cell[262]	4710.826290
a_cell[263]	1336.692473
a_cell[264]	3360.088862
a_cell[265]	4772.354561
a_cell[266]	4726.190753
a_cell[267]	2336.630683
a_cell[268]	5358.110068
a_cell[269]	5807.586512
a_cell[270]	1506.716426
a_cell[271]	3466.146772
a_cell[272]	4397.220408
a_cell[273]	5468.359956
a_cell[274]	5638.807091
a_cell[275]	6069.773400
a_cell[276]	4291.118702
a_cell[277]	2350.416168
a_cell[278]	5454.468273
a_cell[279]	4787.084727
a_cell[280]	5519.013094
a_cell[281]	3928.694107
a_cell[282]	4947.512999
a_cell[283]	5378.938156
a_cell[284]	246.688610
a_cell[285]	5142.097909
a_cell[286]	174.055665
a_cell[287]	4248.646320
a_cell[288]	891.827969
a_cell[289]	6214.514918
a_cell[290]	4789.950395
a_cell[291]	6157.246036
a_cell[292]	5787.658623
a_cell[293]	5660.591510

a_cell[294]	4519.103460
a_cell[295]	5544.399511
a_cell[296]	4727.479621
a_cell[297]	2244.909869
a_cell[298]	4745.257448
a_cell[299]	5283.676314
a_cell[300]	5675.746770
a_cell[301]	4286.913434
a_cell[302]	5515.786037
a_cell[303]	3708.867653
a_cell[304]	4021.487844
a_cell[305]	3527.107734
a_cell[306]	732.463901
a_cell[307]	4267.996250
a_cell[308]	4204.257215
a_cell[309]	4578.018196
a_cell[310]	5390.878271
a_cell[311]	4936.322065
a_cell[312]	286.658783
a_cell[313]	3839.939839
a_cell[314]	5848.730317
a_cell[315]	158.285407
a_cell[316]	113.191233
a_cell[317]	304.187895
a_cell[318]	5777.043474
a_cell[319]	4988.454195
a_cell[320]	5599.059368
a_cell[321]	3955.564327
a_cell[322]	5860.136011
a_cell[323]	5870.615477
a_cell[324]	5873.754597
a_cell[325]	228.365339
a_cell[326]	7028.440462
a_cell[327]	4961.744442
a_cell[328]	5018.819700
a_cell[329]	5168.898289
a_cell[330]	5709.154908
a_cell[331]	5963.155581
a_cell[332]	5616.994969
a_cell[333]	5468.304401
a_cell[334]	98.118024
a_cell[335]	464.789981
a_cell[336]	5404.382005

a_cell[337]	4168.052879
a_cell[338]	5277.145921
a_cell[339]	4677.719996
a_cell[340]	5626.165882
a_cell[341]	4370.101116
a_cell[342]	5669.949865
a_cell[343]	4252.154963
a_cell[344]	1848.309441
a_cell[345]	6016.998361
a_cell[346]	4555.404398
a_cell[347]	4421.247769
a_cell[348]	4946.709497
a_cell[349]	5264.487108
a_cell[350]	5684.966746
a_cell[351]	3557.093952
a_cell[352]	1848.939878
a_cell[353]	5516.284133
a_cell[354]	5216.180888
a_cell[355]	5450.431263
a_cell[356]	5406.330573
a_cell[357]	4831.834073
a_cell[358]	3985.160317
a_cell[359]	3307.567562
a_cell[360]	6253.235980
a_cell[361]	4973.004771
a_cell[362]	5013.216974
a_cell[363]	735.101266
a_cell[364]	4702.765212
a_cell[365]	5266.168391
a_cell[366]	4857.088073
a_cell[367]	3037.196673
a_cell[368]	770.360496
a_cell[369]	6337.314365
a_cell[370]	5864.758040
a_cell[371]	136.035099
a_cell[372]	3892.125993
a_cell[373]	4533.462865
a_cell[374]	1009.426462
a_cell[375]	5849.163655
a_cell[376]	5113.164378
a_cell[377]	5477.766393
a_cell[378]	1048.335258
a_cell[379]	3877.105113

a_cell[380]	5480.148503
a_cell[381]	5135.639990
a_cell[382]	115.792080
a_cell[383]	3729.561597
a_cell[384]	4526.283485
a_cell[385]	1021.098806
a_cell[386]	4281.629475
a_cell[387]	4891.466883
a_cell[388]	2996.437459
a_cell[389]	5145.142034
a_cell[390]	1452.399389
a_cell[391]	5652.536198
a_cell[392]	5295.623371
a_cell[393]	5443.399470
a_cell[394]	4234.813421
a_cell[395]	1785.671859
a_cell[396]	3428.947168
a_cell[397]	5875.434200
a_cell[398]	5192.809273
a_cell[399]	5638.824165
a_cell[400]	5597.061106
a_cell[401]	4869.551163
a_cell[402]	5151.157044
a_cell[403]	5759.894139
a_cell[404]	4873.578137
a_cell[405]	5622.366922
a_cell[406]	4316.118520
a_cell[407]	4636.450768
a_cell[408]	93.701719
a_cell[409]	5385.446017
a_cell[410]	4603.459067
a_cell[411]	5040.220597
a_cell[412]	4112.578297
a_cell[413]	290.249702
a_cell[414]	4895.268343
a_cell[415]	4118.620452
a_cell[416]	2860.375615
a_cell[417]	4283.134383
a_cell[418]	3552.351748
a_cell[419]	4774.281839
a_cell[420]	5516.298243
a_cell[421]	5239.557259
a_cell[422]	5862.942874

```

a_cell[423] 5387.823487
a_cell[424] 5754.246852
a_cell[425] 4384.450185
a_cell[426] 19.812132
a_cell[427] 2964.587070
a_cell[428] 6072.386365
a_cell[429] 5111.984325
a_cell[430] 3368.392042
a_cell[431] 4609.311900
a_cell[432] 4689.748452
a_cell[433] 4596.217801
a_cell[434] 4445.881647
a_cell[435] 4360.842460
a_cell[436] 5733.620809
a_cell[437] 5562.269903
a_cell[438] 4121.811013
a_cell[439] 4364.768914
a_cell[440] 6617.039688
a_cell[441] 5621.877922
a_cell[442] 5223.700887
a_cell[443] 4283.120379
a_cell[444] 941.904677
a_cell[445] 5332.987652
a_cell[446] 6418.107029
a_cell[447] 4969.273141
a_cell[448] 4766.690084
a_cell[449] 3750.192436
a_cell[450] 2359.989651
a_cell[451] 3823.098379
a_cell[452] 4991.383879
a_cell[453] 6080.655377
a_cell[454] 5674.484258
b_tide      2699.834886
b_moon      2149.208096
sigma_month 1939.591836
sigma_year  2365.847965
sigma_track 1033.536912
sigma_cell  7.709065
phi         1344.589308

```

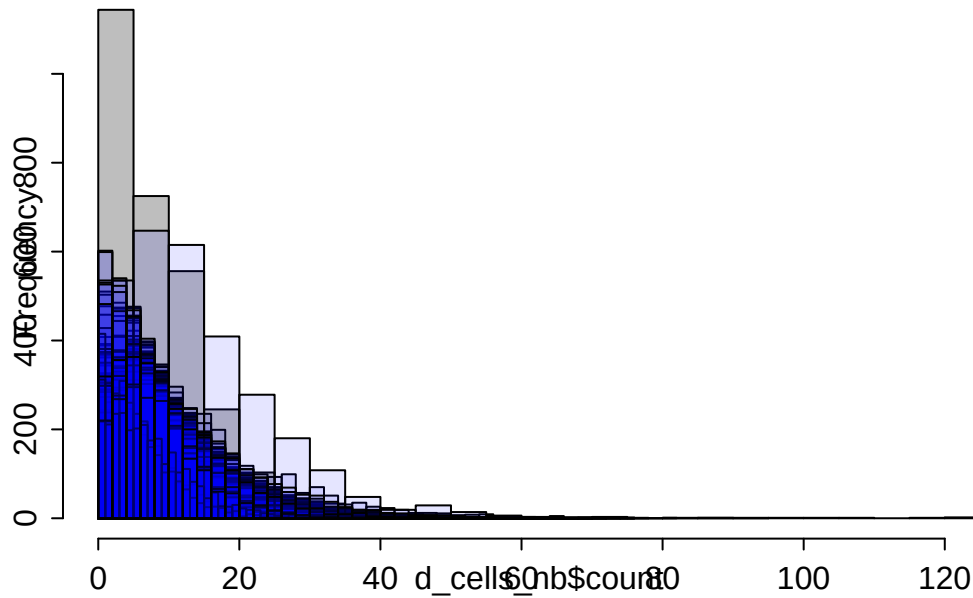
```

## posterior predictive check
x_sim <- sim(m_space_use, n = 1000)

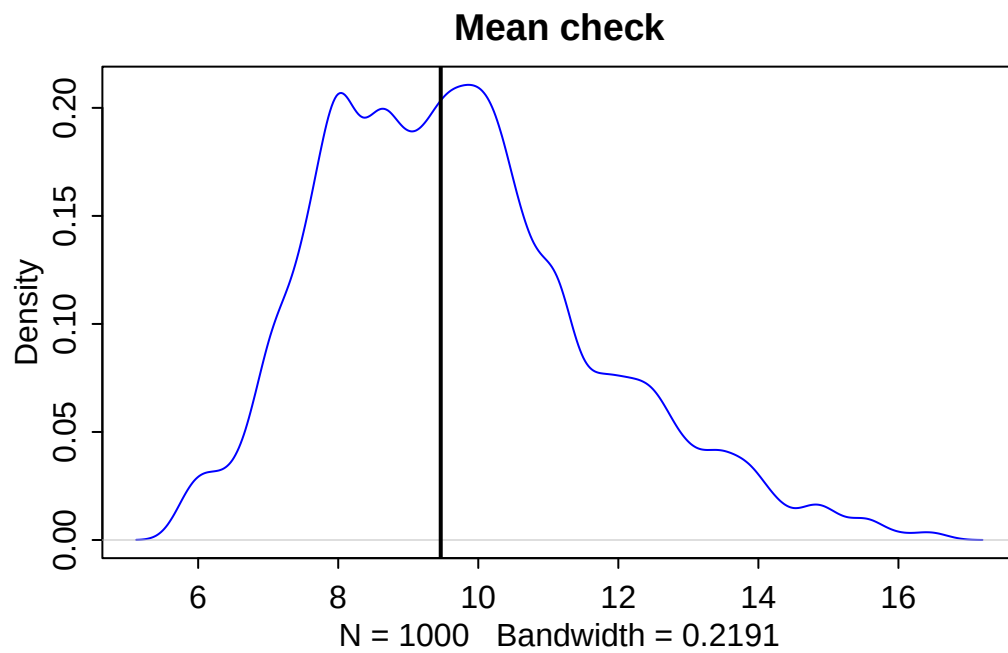
```

```
hist(d_cells_nb$count, breaks = 30, col = "grey",
     main = "Observed vs simulated counts")
for(i in 1:50) {
  hist(x_sim[i,], breaks = 30,
       col = col.alpha("blue", 0.1), add = TRUE)
}
```

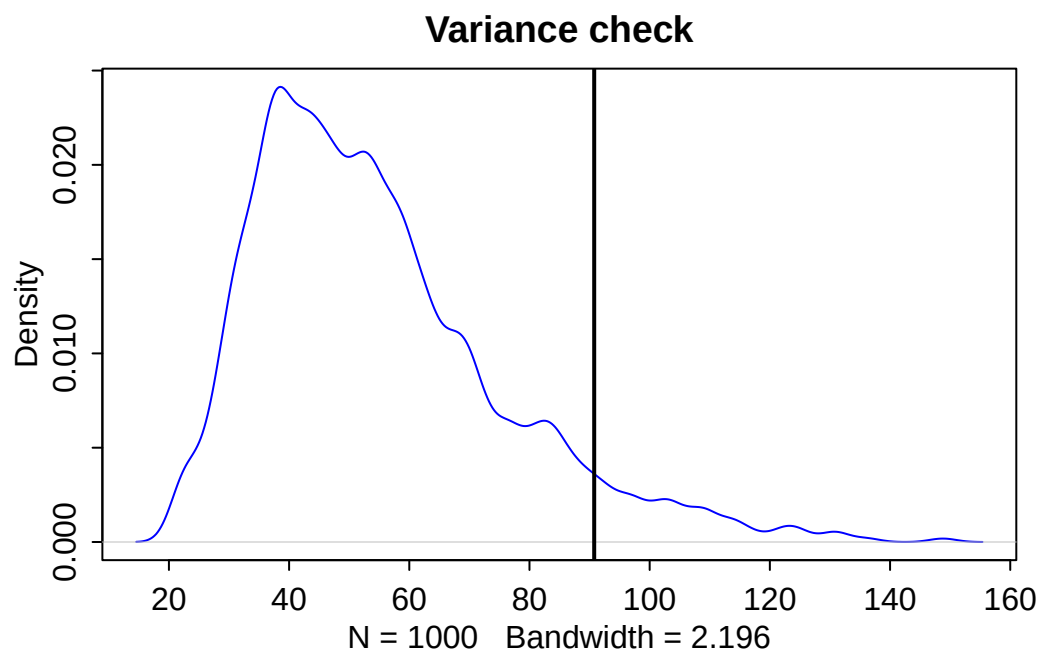
Observed vs simulated counts



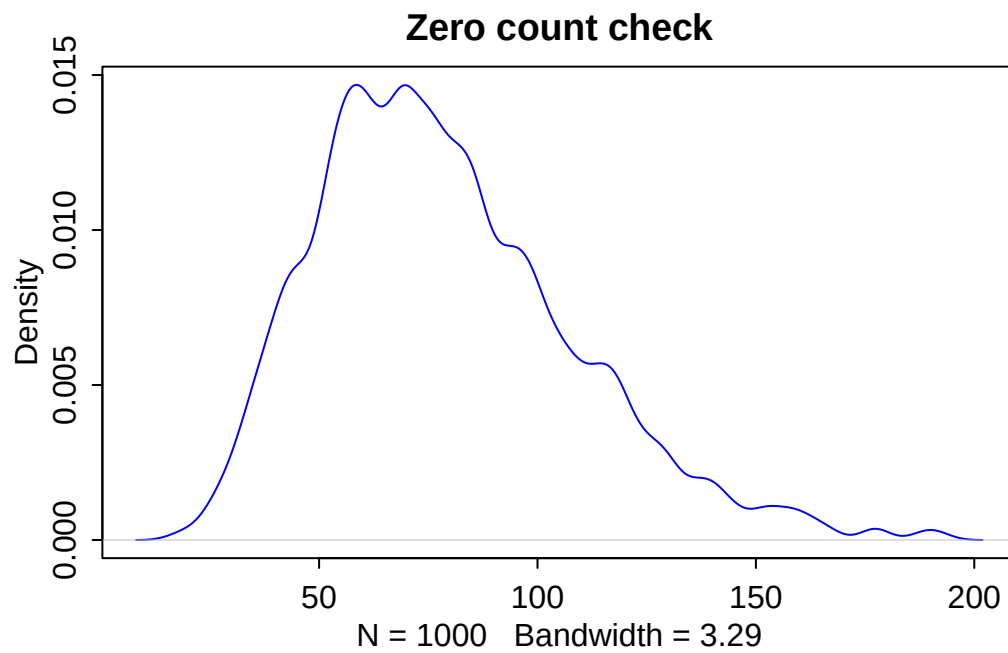
```
mean_sim <- apply(x_sim, 1, mean)
dens(mean_sim, col = "blue")
abline(v = mean(d_cells_nb$count), lwd = 2)
title("Mean check")
```



```
var_sim <- apply(x_sim, 1, var)
dens(var_sim, col = "blue")
abline(v = var(d_cells_nb$count), lwd = 2)
title("Variance check")
```

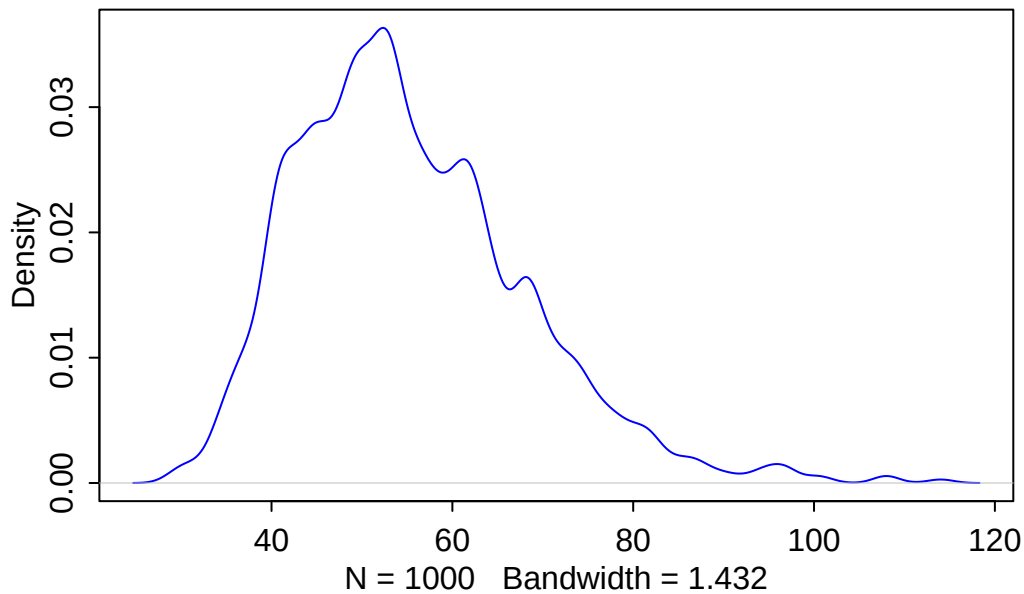


```
zero_sim <- apply(x_sim, 1, function(x) sum(x == 0))  
dens(zero_sim, col = "blue")  
abline(v = sum(d_cells_nb$count == 0), lwd = 2)  
title("Zero count check")
```



```
max_sim <- apply(x_sim, 1, max)
dens(max_sim, col = "blue")
abline(v = max(d_cells_nb$count), lwd = 2)
title("Maximum count check")
```

Maximum count check



```
post <- extract.samples(m_space_use)

cell_hotspots <- gps_cells %>%
  distinct(cell_id, x_cell, y_cell) %>%
  arrange(cell_id) %>%
  mutate(
    mean_cell_effect = apply(post$a_cell, 2, mean),
    lpi_cell_effect = apply(post$a_cell, 2, PI, prob = 0.89)[1, ],
    upi_cell_effect = apply(post$a_cell, 2, PI, prob = 0.89)[2, ]
  ) %>%
  arrange(desc(mean_cell_effect))

head(cell_hotspots, 20)
```

A tibble: 20 x 6

	cell_id	x_cell	y_cell	mean_cell_effect	lpi_cell_effect	upi_cell_effect
	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	426	22566	382894	0.0796	-0.0251	0.263
2	211	22557	382903	0.0574	-0.0375	0.212
3	204	22557	382896	0.0559	-0.0384	0.205
4	169	22556	382886	0.0518	-0.0450	0.196
5	99	22553	382888	0.0429	-0.0539	0.185

6	159	22555	382899	0.0414	-0.0514	0.174
7	119	22554	382884	0.0394	-0.0552	0.172
8	15	22548	382903	0.0350	-0.0617	0.177
9	382	22564	382892	0.0346	-0.0574	0.168
10	104	22553	382893	0.0341	-0.0543	0.161
11	408	22565	382895	0.0340	-0.0576	0.164
12	195	22557	382887	0.0321	-0.0624	0.155
13	334	22562	382894	0.0320	-0.0610	0.159
14	371	22563	382905	0.0303	-0.0654	0.159
15	316	22561	382905	0.0301	-0.0638	0.158
16	241	22559	382883	0.0300	-0.0677	0.162
17	197	22557	382889	0.0297	-0.0673	0.156
18	205	22557	382897	0.0286	-0.0689	0.150
19	315	22561	382904	0.0279	-0.0660	0.156
20	252	22559	382894	0.0276	-0.0721	0.156

```
### Extreme hotspot
```

```
# Extract posterior samples
```

```
post <- extract.samples(m_space_use)
```

```
# compute mean cell effect
```

```
cell_effect <- apply(post$a_cell, 2, mean)
```

```
cell_low <- apply(post$a_cell, 2, PI, prob = 0.89)[1, ]
```

```
cell_high <- apply(post$a_cell, 2, PI, prob = 0.89)[2, ]
```

```
# attach back to your spatial data
```

```
cell_map <- gps_cells %>%
```

```
  distinct(cell_id, x_cell, y_cell) %>%
```

```
  arrange(cell_id) %>%
```

```
  mutate(
```

```
    effect = cell_effect,
```

```
    low = cell_low,
```

```
    high = cell_high
```

```
)
```

```
cell_map <- cell_map %>%
```

```
  mutate(
```

```
    intensity = exp(effect)
```

```
)
```

```
library(ggplot2)
```

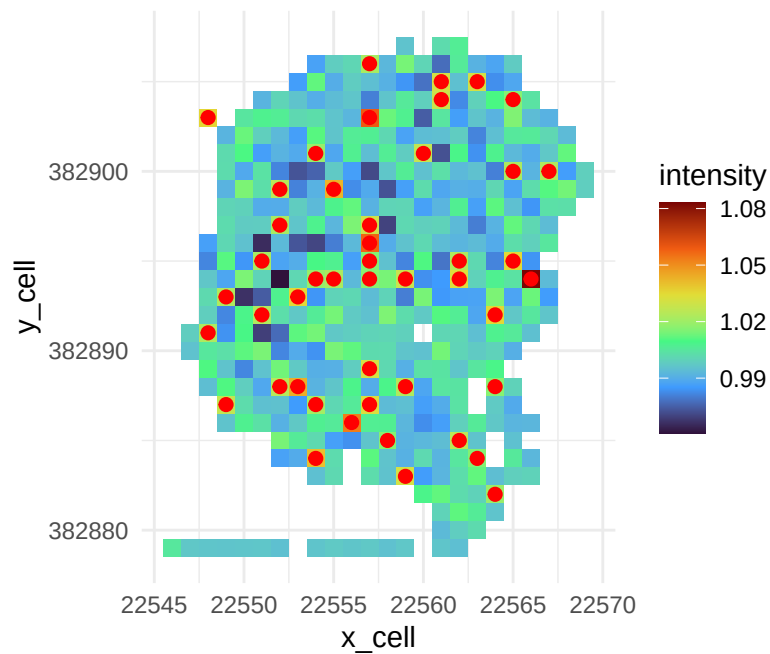
```

threshold <- quantile(cell_map$intensity, 0.9)

cell_map <- cell_map %>%
  mutate(
    hotspot = intensity > threshold
  )

ggplot(cell_map, aes(x = x_cell, y = y_cell)) +
  geom_tile(aes(fill = intensity)) +
  geom_point(data = subset(cell_map, hotspot),
            color = "red", size = 2) +
  scale_fill_viridis_c(option = "turbo") +
  coord_equal() +
  theme_minimal()

```



```

### Consecutive points in the same cell

# create bouts
library(dplyr)

cell_size <- 50

```

```

gps_bouts <- gps_env %>%
  arrange(file_name, datetime_local) %>% # IMPORTANT: time order
  mutate(
    x_cell = floor(x_true_m / cell_size),
    y_cell = floor(y_true_m / cell_size),
    cell_id = paste(x_cell, y_cell, sep = "_")
  ) %>%
  group_by(file_name) %>%
  mutate(
    new_bout = cell_id != lag(cell_id, default = first(cell_id)),
    bout_id = cumsum(new_bout)
  ) %>%
  ungroup()

#calculate bout duration
bouts_summary <- gps_bouts %>%
  group_by(file_name, cell_id, bout_id) %>%
  summarise(
    n_points = n(),
    duration_sec = n_points * 2,
    .groups = "drop"
  )

# summarise per track-cell
cell_behavior <- bouts_summary %>%
  group_by(file_name, cell_id) %>%
  summarise(
    n_visits = n(), # number of bouts
    total_time = sum(duration_sec), # total time in cell
    mean_duration = mean(duration_sec), # average stay
    max_duration = max(duration_sec), # longest stay
    .groups = "drop"
  )

View(cell_behavior)

```

Map

```

library(tidyr)

# Attach spatial coordinates
cell_behavior <- cell_behavior %>%

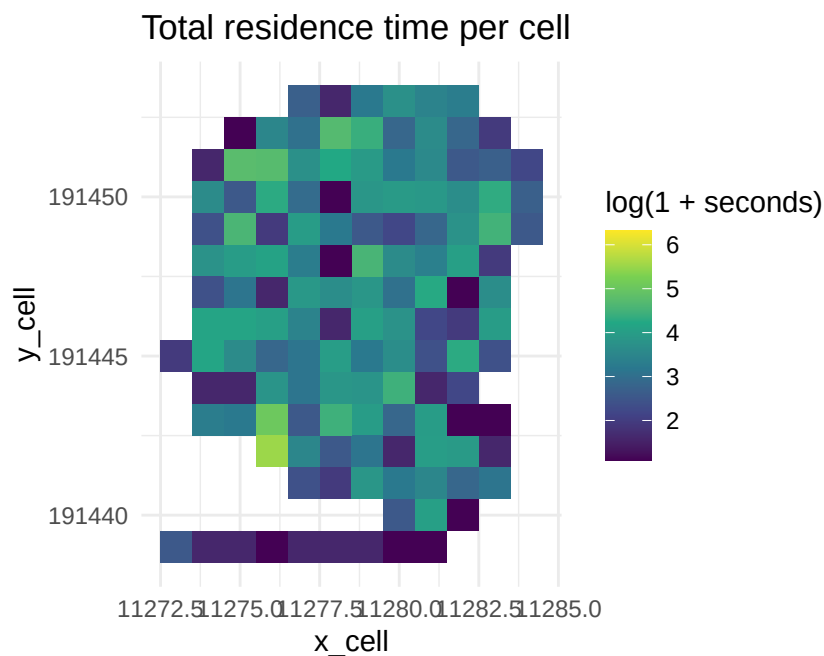
```

```

separate(cell_id, into = c("x_cell", "y_cell"), sep = "_") %>%
mutate(
  x_cell = as.numeric(x_cell),
  y_cell = as.numeric(y_cell)
)

## Total time (log scale)
ggplot(cell_behavior, aes(x = x_cell, y = y_cell, fill = log1p(total_time))) +
  geom_tile() +
  scale_fill_viridis_c() +
  coord_equal() +
  labs(
    title = "Total residence time per cell",
    fill = "log(1 + seconds)"
  ) +
  theme_minimal()

```



```

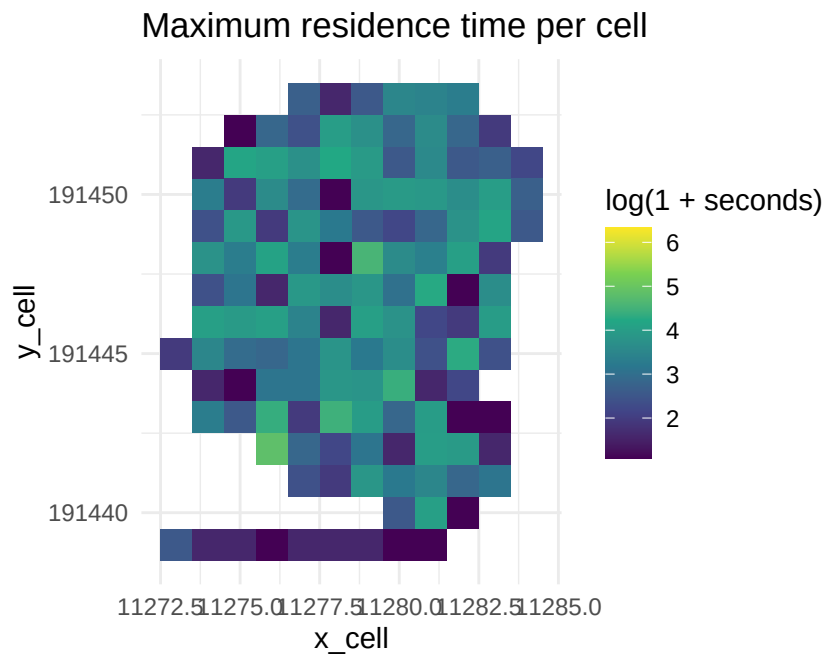
## Maximum duration (log scale)
ggplot(cell_behavior, aes(x = x_cell, y = y_cell, fill = log1p(max_duration))) +
  geom_tile() +
  scale_fill_viridis_c() +
  coord_equal() +

```

```

labs(
  title = "Maximum residence time per cell",
  fill = "log(1 + seconds)"
) +
theme_minimal()

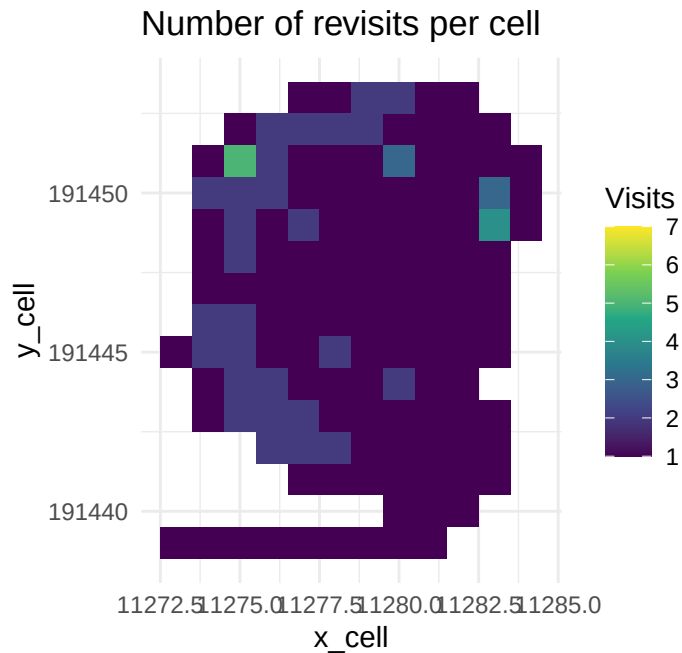
```



```

## number of visits
ggplot(cell_behavior, aes(x = x_cell, y = y_cell, fill = n_visits)) +
  geom_tile() +
  scale_fill_viridis_c() +
  coord_equal() +
  labs(
    title = "Number of revisits per cell",
    fill = "Visits"
  ) +
  theme_minimal()

```



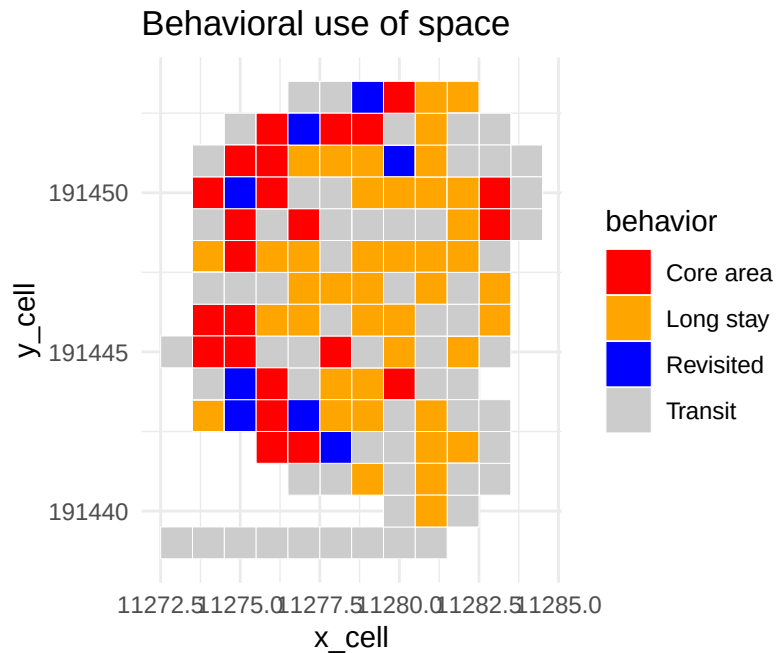
```
### Classify behavior per cell
```

```
cell_behavior <- cell_behavior %>%
  mutate(
    high_time = total_time > median(total_time),
    high_visit = n_visits > median(n_visits),

    behavior = case_when(
      high_time & high_visit ~ "Core area",
      high_time & !high_visit ~ "Long stay",
      !high_time & high_visit ~ "Revisited",
      TRUE ~ "Transit"
    )
  )

ggplot(cell_behavior, aes(x = x_cell, y = y_cell, fill = behavior)) +
  geom_tile(color = "white", size = 0.1) +
  scale_fill_manual(values = c(
    "Core area" = "red",
    "Long stay" = "orange",
    "Revisited" = "blue",
    "Transit" = "grey80"
  )) +
```

```
coord_equal() +
theme_minimal() +
labs(title = "Behavioral use of space")
```



```
## Aggregate Bayesian intensity map into 50m cells
cell_map_agg <- cell_map %>%
  mutate(
    x_cell_agg = floor(x_cell / 2),
    y_cell_agg = floor(y_cell / 2)
  ) %>%
  group_by(x_cell_agg, y_cell_agg) %>%
  summarise(
    intensity = mean(intensity, na.rm = TRUE),
    effect = mean(effect, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  rename(
    x_cell = x_cell_agg,
    y_cell = y_cell_agg
  )
```

```

### compare datasets and test agreement
compare_map <- cell_behavior %>%
  left_join(cell_map_agg, by = c("x_cell", "y_cell"))

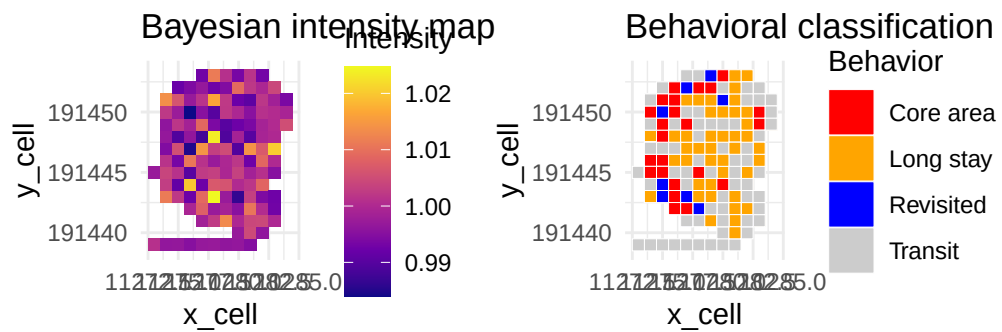
## side by side comparison
library(ggplot2)
library(patchwork)

p1 <- ggplot(cell_map_agg, aes(x = x_cell, y = y_cell, fill = intensity)) +
  geom_tile() +
  scale_fill_viridis_c(option = "plasma") +
  coord_equal() +
  theme_minimal() +
  labs(
    title = "Bayesian intensity map",
    fill = "Intensity"
  )

p2 <- ggplot(cell_behavior, aes(x = x_cell, y = y_cell, fill = behavior)) +
  geom_tile(color = "white", linewidth = 0.1) +
  scale_fill_manual(values = c(
    "Core area" = "red",
    "Long stay" = "orange",
    "Revisited" = "blue",
    "Transit" = "grey80"
  )) +
  coord_equal() +
  theme_minimal() +
  labs(
    title = "Behavioral classification",
    fill = "Behavior"
  )

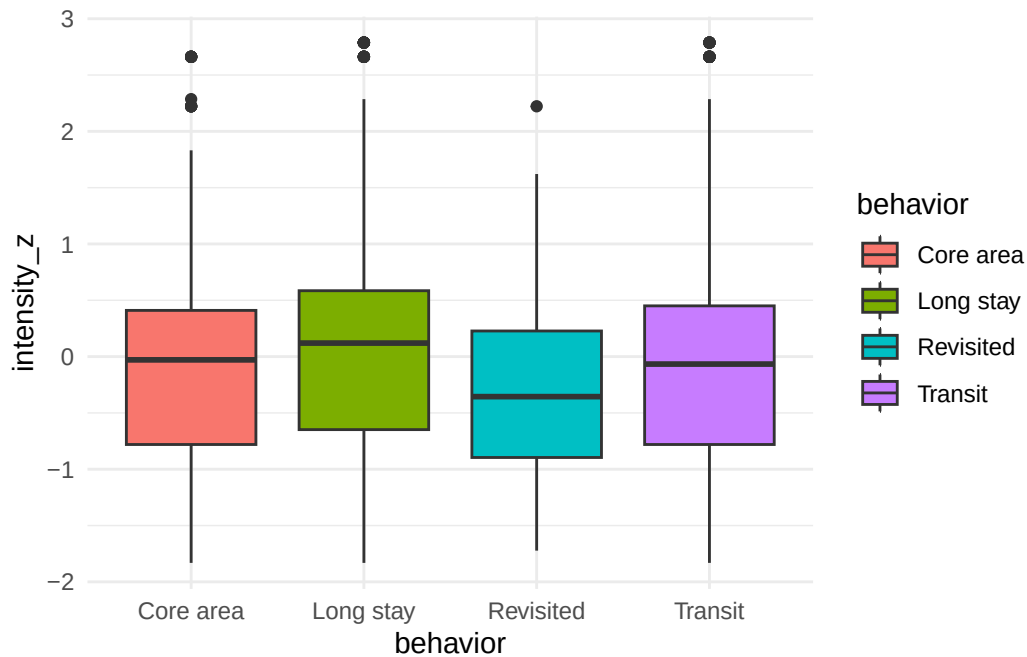
p1 + p2

```



```
## comparison map
compare_map <- compare_map %>%
  mutate(
    intensity_z = scale(intensity)
  )

ggplot(compare_map, aes(x = behavior, y = intensity_z, fill = behavior)) +
  geom_boxplot() +
  theme_minimal()
```



```
### Bout duration Gaussian model

# First extract one row per bout with environmental data
bout_covariates <- gps_bouts %>%
  group_by(file_name, bout_id) %>%
  summarise(
    tide_height_m = first(tide_height_m),
    moon_illumination = first(moon_illumination),
    month = first(month),
    year = first(year),
    .groups = "drop"
  )

# Join with bout durations
bout_data <- bouts_summary %>%
  left_join(bout_covariates, by = c("file_name", "bout_id"))

# prepare variables
bout_data <- bout_data %>%
  filter(duration_sec > 0) %>% # just in case
  mutate(
    log_duration = log(duration_sec),
```

```

# IDs for hierarchical effects
month_id = as.integer(factor(month)),
year_id = as.integer(factor(year)),
track_id = as.integer(factor(file_name)),

# standardize predictors
tide_s = as.numeric(scale(tide_height_m)),
moon_s = as.numeric(scale(moon_illumination))
)

# build data luist for ulam
d_bout <- list(
  N = nrow(bout_data),
  Y = bout_data$log_duration,
  tide = bout_data$tide_s,
  moon = bout_data$moon_s,
  month = bout_data$month_id,
  year = bout_data$year_id,
  track = bout_data$track_id,

  n_month = length(unique(bout_data$month_id)),
  n_year = length(unique(bout_data$year_id)),
  n_track = length(unique(bout_data$track_id))
)

# Full bout model
library(rethinking)

m_bout <- ulam(
  alist(
    Y ~ dnorm(mu, sigma),

    mu <- a +
      a_month[month] +
      a_year[year] +
      a_track[track] +
      b_tide * tide +
      b_moon * moon,

    # priors
    a ~ dnorm(0, 1),

```

```

a_month[month] ~ dnorm(0, sigma_month),
a_year[year] ~ dnorm(0, sigma_year),
a_track[track] ~ dnorm(0, sigma_track),

b_tide ~ dnorm(0, 1),
b_moon ~ dnorm(0, 1),

sigma ~ dexp(1),
sigma_month ~ dexp(1),
sigma_year ~ dexp(1),
sigma_track ~ dexp(1)
),
data = d_bout,
chains = 4,
cores = 4,
iter = 2000
)

```

Running MCMC with 4 parallel chains, with 1 thread(s) per chain...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)

Chain 2 Iteration: 1 / 2000 [0%] (Warmup)

Chain 3 Iteration: 1 / 2000 [0%] (Warmup)

Chain 4 Iteration: 1 / 2000 [0%] (Warmup)

Chain 1 Iteration: 100 / 2000 [5%] (Warmup)

Chain 2 Iteration: 100 / 2000 [5%] (Warmup)

Chain 3 Iteration: 100 / 2000 [5%] (Warmup)

Chain 4 Iteration: 100 / 2000 [5%] (Warmup)

Chain 3 Iteration: 200 / 2000 [10%] (Warmup)

Chain 1 Iteration: 200 / 2000 [10%] (Warmup)

Chain 2 Iteration: 200 / 2000 [10%] (Warmup)

Chain 4 Iteration: 200 / 2000 [10%] (Warmup)

Chain 3 Iteration: 300 / 2000 [15%] (Warmup)

Chain 2 Iteration: 300 / 2000 [15%] (Warmup)

Chain 1 Iteration: 300 / 2000 [15%] (Warmup)

Chain 4 Iteration: 300 / 2000 [15%] (Warmup)

Chain 3 Iteration: 400 / 2000 [20%] (Warmup)
 Chain 1 Iteration: 400 / 2000 [20%] (Warmup)
 Chain 4 Iteration: 400 / 2000 [20%] (Warmup)
 Chain 3 Iteration: 500 / 2000 [25%] (Warmup)
 Chain 1 Iteration: 500 / 2000 [25%] (Warmup)
 Chain 2 Iteration: 400 / 2000 [20%] (Warmup)
 Chain 3 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 4 Iteration: 500 / 2000 [25%] (Warmup)
 Chain 1 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 3 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 1 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 4 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 1 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 1 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 4 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 3 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 1 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 1 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 4 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 3 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 2 Iteration: 500 / 2000 [25%] (Warmup)
 Chain 1 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 2 Iteration: 600 / 2000 [30%] (Warmup)
 Chain 3 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 3 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 4 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 2 Iteration: 700 / 2000 [35%] (Warmup)
 Chain 1 Iteration: 1200 / 2000 [60%] (Sampling)
 Chain 3 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 2 Iteration: 800 / 2000 [40%] (Warmup)
 Chain 4 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 4 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 3 Iteration: 1200 / 2000 [60%] (Sampling)
 Chain 1 Iteration: 1300 / 2000 [65%] (Sampling)
 Chain 2 Iteration: 900 / 2000 [45%] (Warmup)
 Chain 4 Iteration: 1100 / 2000 [55%] (Sampling)
 Chain 3 Iteration: 1300 / 2000 [65%] (Sampling)
 Chain 4 Iteration: 1200 / 2000 [60%] (Sampling)
 Chain 1 Iteration: 1400 / 2000 [70%] (Sampling)
 Chain 3 Iteration: 1400 / 2000 [70%] (Sampling)
 Chain 2 Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 2 Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 4 Iteration: 1300 / 2000 [65%] (Sampling)

```

Chain 3 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 39.1 seconds.
Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 39.5 seconds.
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 41.2 seconds.
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 48.2 seconds.

```

All 4 chains finished successfully.
 Mean chain execution time: 42.0 seconds.
 Total execution time: 48.2 seconds.

```

# predicted effect of tide
tide_seq_m <- seq(min(bout_data$tide_height_m),
                  max(bout_data$tide_height_m),
                  length.out = 100)

```

```

tide_seq_s <- (tide_seq_m - mean(bout_data$tide_height_m)) /
              sd(bout_data$tide_height_m)

pred_data <- list(
  tide = tide_seq_s,
  moon = rep(0, length(tide_seq_s)),
  month = rep(1, length(tide_seq_s)),
  year = rep(1, length(tide_seq_s)),
  track = rep(1, length(tide_seq_s))
)

mu <- link(m_bout, data = pred_data)

mu_mean <- apply(mu, 2, function(x) mean(x, na.rm = TRUE))

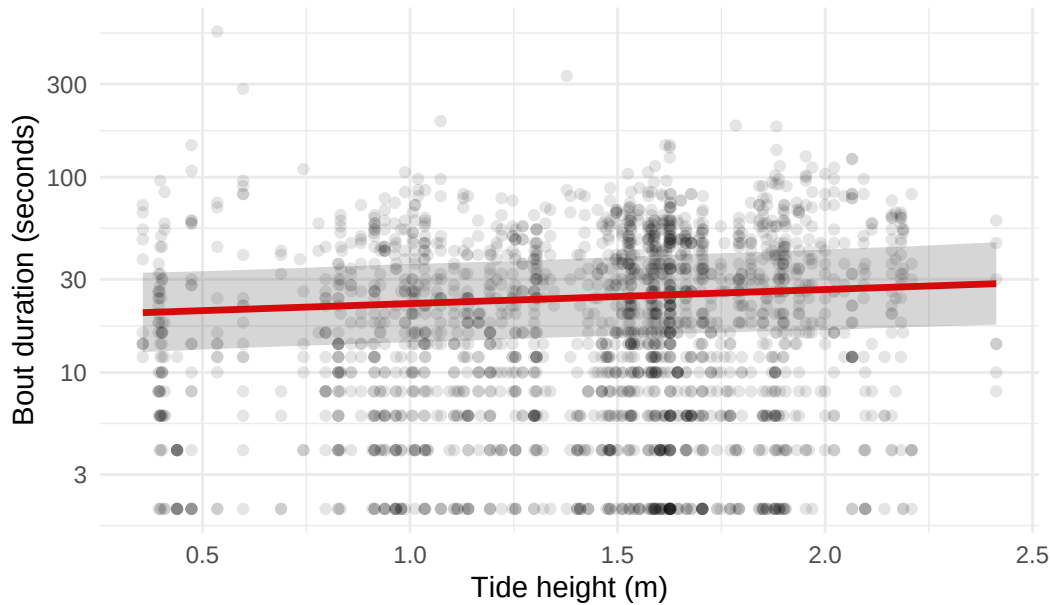
mu_PI <- apply(mu, 2, function(x)
  quantile(x, probs = c(0.055, 0.945), na.rm = TRUE)
)

pred_df <- data.frame(
  tide_m = tide_seq_m,
  duration = exp(mu_mean),
  low = exp(mu_PI[1, ]),
  high = exp(mu_PI[2, ])
)

ggplot(bout_data, aes(x = tide_height_m, y = duration_sec)) +
  geom_point(alpha = 0.1) +
  geom_line(data = pred_df,
    aes(x = tide_m, y = duration),
    color = "red", size = 1.2) +
  geom_ribbon(data = pred_df,
    aes(x = tide_m, ymin = low, ymax = high),
    alpha = 0.2,
    inherit.aes = FALSE) +
  scale_y_log10() +
  labs(
    x = "Tide height (m)",
    y = "Bout duration (seconds)",
    title = "Effect of tide on shark residence time"
  ) +
  theme_minimal()

```

Effect of tide on shark residence time



```
precis(m_bout, depth = 2)
```

	mean	sd	5.5%	94.5%	rhat
a	2.7793960418	0.22754108	2.384055838	3.055138110	1.0054144
a_month[1]	-0.0374253093	0.05414361	-0.136537641	0.028551589	1.0048916
a_month[2]	0.0133926764	0.05221273	-0.061470899	0.108758751	1.0074696
a_month[3]	0.0052105443	0.06233504	-0.088202890	0.106688812	1.0131542
a_month[4]	0.0132006006	0.06464990	-0.078567581	0.121325499	1.0061034
a_month[5]	-0.0152927709	0.06654229	-0.129244616	0.075475215	1.0099272
a_month[6]	-0.0315376352	0.07252964	-0.170370500	0.053741101	1.0087786
a_month[7]	0.0235513733	0.06397166	-0.059162077	0.138229415	1.0062622
a_month[8]	0.0254396528	0.06604761	-0.058813222	0.147663803	1.0105051
a_month[9]	0.0071186198	0.06656918	-0.092424477	0.117831606	1.0164485
a_month[10]	0.0079210408	0.05160551	-0.069936197	0.093919364	1.0092185
a_month[11]	0.0019476990	0.05956022	-0.087934348	0.097170410	1.0102531
a_year[1]	0.3622544536	0.27001622	0.020683220	0.838380460	1.0076187
a_year[2]	0.0270623147	0.22884830	-0.250562112	0.422151661	1.0048731
a_year[3]	0.1026385688	0.23113306	-0.180214641	0.496998158	1.0071150
a_year[4]	-0.2560140234	0.22729637	-0.533533067	0.127893519	1.0050951
a_year[5]	0.2836442842	0.23774945	-0.006383018	0.691305489	1.0064751

a_track[1]	0.0809156707	0.24340394	-0.318889165	0.465696977	1.0012972
a_track[2]	0.0425697057	0.23593031	-0.327327399	0.420440210	1.0024180
a_track[3]	0.1598744967	0.24379705	-0.221078155	0.553252008	1.0007130
a_track[4]	0.1274685162	0.25263277	-0.275156969	0.520713296	1.0015534
a_track[5]	0.1735326745	0.26336686	-0.239376447	0.600343417	1.0070853
a_track[6]	-0.4022415028	0.25809518	-0.831136063	0.001075610	1.0019662
a_track[7]	-0.0375612222	0.23174657	-0.402590771	0.327754044	1.0005160
a_track[8]	0.0082833748	0.22401812	-0.344412167	0.360370913	1.0004273
a_track[9]	0.0208980393	0.22380332	-0.337915800	0.387807876	0.9993751
a_track[10]	-0.3015887010	0.23313357	-0.677875210	0.066571525	1.0021336
a_track[11]	0.1852550999	0.23397420	-0.173574866	0.562659158	1.0006286
a_track[12]	0.1300820672	0.22884821	-0.231538327	0.500856986	1.0016261
a_track[13]	0.0349469515	0.22930185	-0.335481918	0.398765551	1.0054163
a_track[14]	0.2245575816	0.24491861	-0.172477614	0.615478250	1.0019740
a_track[15]	0.1785599104	0.24372521	-0.200776662	0.574580558	1.0012053
a_track[16]	0.1106273151	0.24698139	-0.277145633	0.508924540	1.0011027
a_track[17]	-0.1346616517	0.24257110	-0.530102910	0.242160610	1.0007028
a_track[18]	0.1392715556	0.23508133	-0.223463733	0.516747291	1.0018935
a_track[19]	-0.1092531342	0.23369484	-0.485383783	0.263346402	1.0006321
a_track[20]	-0.2521407486	0.20943150	-0.587382213	0.086919842	1.0029568
a_track[21]	-0.0041143961	0.23411431	-0.380069181	0.368852020	1.0003170
a_track[22]	-0.2027891375	0.22257048	-0.557451394	0.148161923	1.0001894
a_track[23]	0.1870945031	0.24489854	-0.201571583	0.585971665	1.0003194
a_track[24]	-0.2687101808	0.25956061	-0.687485054	0.134852005	1.0020547
a_track[25]	-0.2659088405	0.24219354	-0.666289096	0.113180196	1.0007310
a_track[26]	-0.0519565591	0.24694635	-0.442427484	0.347130211	1.0016341
a_track[27]	-0.0336734056	0.27458166	-0.464577329	0.399054733	1.0006519
a_track[28]	-0.1341503563	0.27101906	-0.568955829	0.290304187	1.0016051
a_track[29]	0.1123505123	0.27747811	-0.331536445	0.549519166	1.0002151
a_track[30]	0.2043410665	0.25605286	-0.193589439	0.620484898	0.9994432
a_track[31]	0.0763098232	0.23695784	-0.291846947	0.446823998	1.0012632
a_track[32]	0.0906801085	0.23014973	-0.274686943	0.462139311	1.0008279
a_track[33]	0.2014460781	0.24349613	-0.192240822	0.582941617	1.0012283
a_track[34]	0.1975609381	0.24645979	-0.183345550	0.588962913	1.0008162
a_track[35]	-0.1482257598	0.23052936	-0.508878161	0.222378742	0.9994140
a_track[36]	0.0707182858	0.24649239	-0.315629857	0.470724023	1.0011625
a_track[37]	-0.1293443475	0.22503132	-0.494751546	0.229098743	1.0028851
a_track[38]	-0.0910294088	0.23018863	-0.454539034	0.279282710	1.0001328
a_track[39]	-0.5052099887	0.21694416	-0.864251798	-0.174226926	1.0023158
a_track[40]	0.0005647859	0.23160789	-0.374567041	0.376478229	1.0014354
a_track[41]	-0.0096051263	0.23332552	-0.386073417	0.372244280	1.0019354
a_track[42]	-0.1085134522	0.26606400	-0.540079388	0.312408296	1.0053632
a_track[43]	-0.0048095962	0.22959270	-0.362454937	0.370514095	1.0000248

a_track[44]	0.1016447552	0.23703101	-0.278218687	0.476764824	1.0011350
a_track[45]	0.1402582893	0.27744447	-0.297623685	0.580984077	1.0028220
a_track[46]	0.0323921246	0.23644912	-0.348674227	0.407847336	1.0006176
a_track[47]	0.1511760340	0.24715432	-0.240427423	0.553732630	1.0008828
a_track[48]	0.1839798827	0.25724878	-0.223659542	0.598600638	1.0005413
a_track[49]	0.0535033078	0.25904452	-0.360550716	0.456984811	1.0017908
a_track[50]	-0.2686710902	0.24905600	-0.679562513	0.123804091	1.0037833
a_track[51]	0.0644022469	0.24160253	-0.319527349	0.441316665	1.0006233
a_track[52]	-0.1953776596	0.25008064	-0.597251930	0.186989873	1.0000239
a_track[53]	-0.0471901507	0.27772066	-0.484033393	0.392793027	1.0058141
a_track[54]	0.0863728629	0.25477583	-0.315768658	0.495699172	1.0012544
a_track[55]	0.0523561112	0.27896879	-0.389683715	0.500792679	0.9997904
a_track[56]	-0.1173701172	0.23408747	-0.492359024	0.252095296	1.0015367
a_track[57]	-0.1247896192	0.22489889	-0.490146937	0.228638444	1.0036201
a_track[58]	0.0787685704	0.24386606	-0.315262783	0.470731500	1.0027524
a_track[59]	0.1924910767	0.24194398	-0.190229397	0.589535269	0.9996445
a_track[60]	0.0573024872	0.28698765	-0.398863664	0.511174634	1.0024126
a_track[61]	0.1775267953	0.25470919	-0.224998960	0.595239302	1.0018781
a_track[62]	0.0687198485	0.28179917	-0.377779623	0.511312131	1.0002690
a_track[63]	-0.0071395755	0.25862871	-0.432933486	0.405089037	1.0016862
a_track[64]	-0.0854084152	0.22712374	-0.450710669	0.281926919	1.0009916
a_track[65]	-0.0871311785	0.27944775	-0.530443475	0.350805145	1.0009716
a_track[66]	-0.3507225419	0.21583407	-0.704484959	-0.011932341	1.0024097
a_track[67]	-0.2135823503	0.26997673	-0.651245271	0.207727521	1.0020148
a_track[68]	-0.2992437130	0.23008351	-0.673187124	0.055545749	1.0012563
a_track[69]	-0.0062362324	0.23808076	-0.386424624	0.367027187	1.0002237
a_track[70]	0.1578265245	0.24364940	-0.223046873	0.550939639	1.0032194
a_track[71]	0.0771666281	0.23472241	-0.298487876	0.460868117	1.0014154
a_track[72]	-0.0321060222	0.23386787	-0.406243903	0.334706386	1.0002753
a_track[73]	0.0357885202	0.27228276	-0.392941966	0.474968122	1.0007496
a_track[74]	-0.1022448473	0.23412763	-0.486950501	0.268167396	1.0014254
a_track[75]	-0.1079770656	0.27004246	-0.544689252	0.316627652	1.0058151
a_track[76]	0.1467288446	0.25380159	-0.261667638	0.555986651	1.0035787
a_track[77]	0.1021653135	0.24376182	-0.275531994	0.488568069	1.0012316
a_track[78]	0.0639479338	0.24065823	-0.319199428	0.432453698	1.0008081
a_track[79]	0.0418031922	0.23905150	-0.342924180	0.426148368	1.0075286
a_track[80]	0.2055494294	0.26427145	-0.213068697	0.628794681	1.0002243
a_track[81]	0.0828578508	0.25962372	-0.329067737	0.495355647	1.0036881
a_track[82]	0.1331212399	0.27710696	-0.303197848	0.584736102	1.0024400
a_track[83]	-0.1882889407	0.26456910	-0.613510624	0.230079781	0.9997068
a_track[84]	0.0731449647	0.28175693	-0.376819414	0.520227255	1.0010184
a_track[85]	-0.0453012269	0.24559644	-0.433383667	0.344271416	1.0037665
a_track[86]	0.0743145197	0.26592360	-0.330211048	0.499881454	1.0023135

a_track[87]	-0.0139843737	0.23088273	-0.379351517	0.348603224	1.0036231
a_track[88]	-0.0231709072	0.27026995	-0.447964993	0.395308051	1.0017133
a_track[89]	0.0347180433	0.27745819	-0.412294744	0.486342281	1.0001848
a_track[90]	0.0212655570	0.27255357	-0.413807061	0.459290156	1.0004745
a_track[91]	0.0448370537	0.25825706	-0.368036445	0.464125968	1.0039994
a_track[92]	0.0435733350	0.24177761	-0.341877182	0.431236159	1.0007941
a_track[93]	0.1266642853	0.25945617	-0.284242627	0.544401983	1.0032555
a_track[94]	-0.0970489899	0.27569634	-0.537975579	0.325198913	1.0007438
a_track[95]	-0.0749560134	0.26641808	-0.493421458	0.348169506	1.0024867
a_track[96]	-0.0258688448	0.25949507	-0.430805288	0.379240789	1.0012101
a_track[97]	0.1051345250	0.24064954	-0.280418253	0.491065288	1.0031589
a_track[98]	-0.0445146072	0.26229379	-0.476207212	0.370149157	1.0028221
a_track[99]	0.0644752158	0.25642325	-0.345366454	0.477057072	1.0019811
a_track[100]	0.0561978360	0.26408896	-0.355103261	0.491575632	1.0012893
a_track[101]	-0.0031927934	0.27540959	-0.439679097	0.440010324	1.0010925
a_track[102]	-0.0889898157	0.28472126	-0.535913643	0.353580896	1.0006075
a_track[103]	0.0932875648	0.26848643	-0.328882641	0.529091483	1.0019893
a_track[104]	0.1157477912	0.25391991	-0.283103736	0.528857620	1.0016058
a_track[105]	0.0826642726	0.25334667	-0.316404462	0.486737271	1.0026435
a_track[106]	0.0051284540	0.24359869	-0.381443674	0.389352635	1.0037511
a_track[107]	0.1274157432	0.28827681	-0.327194399	0.589380307	1.0018712
a_track[108]	-0.1660849715	0.22612377	-0.523288491	0.186112493	1.0033140
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a_track[125]	-0.0858404909	0.28696707	-0.546870708	0.360548355	1.0015801
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a_track[304]	0.0604740816	0.27430730	-0.380138268	0.488487059	1.0053886
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a_track[308]	0.0634360370	0.27647423	-0.377541473	0.493275680	0.9997675
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```

a_track[589] 4770.6545
a_track[590] 6239.5605
a_track[591] 4895.3408
a_track[592] 4694.5141
b_tide       2889.1423
b_moon       4041.7610
sigma        2156.2291
sigma_month   188.4371
sigma_year    1096.1589
sigma_track   173.5294

```

```
precis(m_bout, pars = c("b_tide", "b_moon"))
```

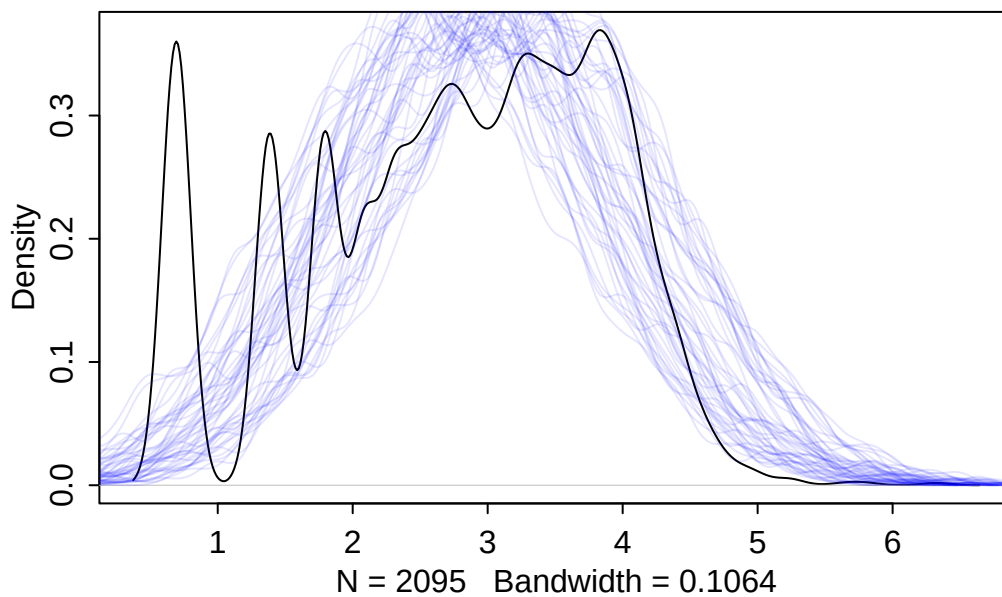
	mean	sd	5.5%	94.5%	rhat	ess_bulk
b_tide	0.07234120	0.0311051	0.02247358	0.12147784	1.000243	2889.142
b_moon	-0.03058374	0.0287960	-0.07633442	0.01636454	1.000322	4041.761

```

y_sim <- sim(m_bout, n = 1000)

dens(d_bout$Y)
for(i in 1:50) dens(y_sim[i,], add = TRUE, col = col.alpha("blue", 0.1))

```



```
post <- extract.samples(m_bout)

mean(exp(post$b_tide))
```

```
[1] 1.075542
```

```
mean(exp(post$b_moon))
```

```
[1] 0.9702813
```

```
## Probability of long-stay vs transit

# response variable (bout level)
threshold <- quantile(bout_data$duration_sec, 0.75, na.rm = TRUE)

bout_data <- bout_data %>%
  mutate(
    long_stay = ifelse(duration_sec >= threshold, 1, 0)
  )

# build data
d_longstay <- list(
  N = nrow(bout_data),
  Y = bout_data$long_stay,

  tide = bout_data$tide_s,
  moon = bout_data$moon_s,
  month = bout_data$month_id,
  year = bout_data$year_id,
  track = bout_data$track_id,

  n_month = length(unique(bout_data$month_id)),
  n_year = length(unique(bout_data$year_id)),
  n_track = length(unique(bout_data$track_id))
)

d_longstay <- list(
  N = nrow(bout_data),
  Y = bout_data$long_stay,
```

```

tide = bout_data$tide_s,
moon = bout_data$moon_s,
month = bout_data$month_id,
year = bout_data$year_id,
track = bout_data$track_id,

n_month = length(unique(bout_data$month_id)),
n_year = length(unique(bout_data$year_id)),
n_track = length(unique(bout_data$track_id))
)

# Bayesian logistic model
library(rethinking)

m_longstay <- ulam(
  alist(
    Y ~ dbern(p),

    logit(p) <- a +
      a_year[year] +
      a_track[track] +
      b_tide * tide +
      b_moon * moon,

    a ~ dnorm(0, 1),

    a_year[year] ~ dnorm(0, sigma_year),
    a_track[track] ~ dnorm(0, sigma_track),

    b_tide ~ dnorm(0, 1),
    b_moon ~ dnorm(0, 1),

    sigma_year ~ dexp(1),
    sigma_track ~ dexp(1)
  ),
  data = d_longstay,
  chains = 4,
  cores = 4,
  iter = 4000,
  control = list(adapt_delta = 0.95)
)

```

Running MCMC with 4 parallel chains, with 1 thread(s) per chain...

Chain 1 Iteration: 1 / 4000 [0%] (Warmup)

Chain 2 Iteration: 1 / 4000 [0%] (Warmup)

Chain 3 Iteration: 1 / 4000 [0%] (Warmup)

Chain 4 Iteration: 1 / 4000 [0%] (Warmup)

Chain 3 Iteration: 100 / 4000 [2%] (Warmup)

Chain 1 Iteration: 100 / 4000 [2%] (Warmup)

Chain 3 Iteration: 200 / 4000 [5%] (Warmup)

Chain 4 Iteration: 100 / 4000 [2%] (Warmup)

Chain 2 Iteration: 100 / 4000 [2%] (Warmup)

Chain 1 Iteration: 200 / 4000 [5%] (Warmup)

Chain 3 Iteration: 300 / 4000 [7%] (Warmup)

Chain 4 Iteration: 200 / 4000 [5%] (Warmup)

Chain 3 Iteration: 400 / 4000 [10%] (Warmup)

Chain 2 Iteration: 200 / 4000 [5%] (Warmup)

Chain 1 Iteration: 300 / 4000 [7%] (Warmup)

Chain 4 Iteration: 300 / 4000 [7%] (Warmup)

Chain 3 Iteration: 500 / 4000 [12%] (Warmup)

Chain 2 Iteration: 300 / 4000 [7%] (Warmup)

Chain 4 Iteration: 400 / 4000 [10%] (Warmup)

Chain 3 Iteration: 600 / 4000 [15%] (Warmup)

Chain 1 Iteration: 400 / 4000 [10%] (Warmup)

Chain 2 Iteration: 400 / 4000 [10%] (Warmup)

Chain 2 Iteration: 500 / 4000 [12%] (Warmup)

Chain 4 Iteration: 500 / 4000 [12%] (Warmup)

Chain 3 Iteration: 700 / 4000 [17%] (Warmup)

Chain 1 Iteration: 500 / 4000 [12%] (Warmup)

Chain 2 Iteration: 600 / 4000 [15%] (Warmup)

Chain 3 Iteration: 800 / 4000 [20%] (Warmup)

Chain 2 Iteration: 700 / 4000 [17%] (Warmup)

Chain 4 Iteration: 600 / 4000 [15%] (Warmup)

Chain 3 Iteration: 900 / 4000 [22%] (Warmup)

Chain 2 Iteration: 800 / 4000 [20%] (Warmup)

Chain 1 Iteration: 600 / 4000 [15%] (Warmup)

Chain 4 Iteration: 700 / 4000 [17%] (Warmup)

Chain 2 Iteration: 900 / 4000 [22%] (Warmup)

Chain 4 Iteration: 800 / 4000 [20%] (Warmup)

Chain 3 Iteration: 1000 / 4000 [25%] (Warmup)

Chain 4 Iteration: 900 / 4000 [22%] (Warmup)

Chain 1 Iteration: 700 / 4000 [17%] (Warmup)
 Chain 2 Iteration: 1000 / 4000 [25%] (Warmup)
 Chain 3 Iteration: 1100 / 4000 [27%] (Warmup)
 Chain 3 Iteration: 1200 / 4000 [30%] (Warmup)
 Chain 1 Iteration: 800 / 4000 [20%] (Warmup)
 Chain 2 Iteration: 1100 / 4000 [27%] (Warmup)
 Chain 4 Iteration: 1000 / 4000 [25%] (Warmup)
 Chain 3 Iteration: 1300 / 4000 [32%] (Warmup)
 Chain 3 Iteration: 1400 / 4000 [35%] (Warmup)
 Chain 1 Iteration: 900 / 4000 [22%] (Warmup)
 Chain 3 Iteration: 1500 / 4000 [37%] (Warmup)
 Chain 1 Iteration: 1000 / 4000 [25%] (Warmup)
 Chain 4 Iteration: 1100 / 4000 [27%] (Warmup)
 Chain 2 Iteration: 1200 / 4000 [30%] (Warmup)
 Chain 1 Iteration: 1100 / 4000 [27%] (Warmup)
 Chain 3 Iteration: 1600 / 4000 [40%] (Warmup)
 Chain 1 Iteration: 1200 / 4000 [30%] (Warmup)
 Chain 3 Iteration: 1700 / 4000 [42%] (Warmup)
 Chain 1 Iteration: 1300 / 4000 [32%] (Warmup)
 Chain 2 Iteration: 1300 / 4000 [32%] (Warmup)
 Chain 4 Iteration: 1200 / 4000 [30%] (Warmup)
 Chain 3 Iteration: 1800 / 4000 [45%] (Warmup)
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 Chain 4 Iteration: 1300 / 4000 [32%] (Warmup)
 Chain 2 Iteration: 1400 / 4000 [35%] (Warmup)
 Chain 1 Iteration: 1500 / 4000 [37%] (Warmup)
 Chain 4 Iteration: 1400 / 4000 [35%] (Warmup)
 Chain 2 Iteration: 1500 / 4000 [37%] (Warmup)
 Chain 3 Iteration: 1900 / 4000 [47%] (Warmup)
 Chain 1 Iteration: 1600 / 4000 [40%] (Warmup)
 Chain 4 Iteration: 1500 / 4000 [37%] (Warmup)
 Chain 3 Iteration: 2000 / 4000 [50%] (Warmup)
 Chain 1 Iteration: 1700 / 4000 [42%] (Warmup)
 Chain 3 Iteration: 2001 / 4000 [50%] (Sampling)
 Chain 2 Iteration: 1600 / 4000 [40%] (Warmup)
 Chain 4 Iteration: 1600 / 4000 [40%] (Warmup)
 Chain 3 Iteration: 2100 / 4000 [52%] (Sampling)
 Chain 1 Iteration: 1800 / 4000 [45%] (Warmup)
 Chain 2 Iteration: 1700 / 4000 [42%] (Warmup)
 Chain 1 Iteration: 1900 / 4000 [47%] (Warmup)
 Chain 3 Iteration: 2200 / 4000 [55%] (Sampling)
 Chain 4 Iteration: 1700 / 4000 [42%] (Warmup)
 Chain 2 Iteration: 1800 / 4000 [45%] (Warmup)

Chain 3 Iteration: 2300 / 4000 [57%] (Sampling)
 Chain 2 Iteration: 1900 / 4000 [47%] (Warmup)
 Chain 3 Iteration: 2400 / 4000 [60%] (Sampling)
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 Chain 1 Iteration: 2001 / 4000 [50%] (Sampling)
 Chain 4 Iteration: 1800 / 4000 [45%] (Warmup)
 Chain 2 Iteration: 2000 / 4000 [50%] (Warmup)
 Chain 2 Iteration: 2001 / 4000 [50%] (Sampling)
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 Chain 1 Iteration: 2100 / 4000 [52%] (Sampling)
 Chain 2 Iteration: 2200 / 4000 [55%] (Sampling)
 Chain 4 Iteration: 1900 / 4000 [47%] (Warmup)
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 Chain 2 Iteration: 2300 / 4000 [57%] (Sampling)
 Chain 1 Iteration: 2200 / 4000 [55%] (Sampling)
 Chain 2 Iteration: 2400 / 4000 [60%] (Sampling)
 Chain 4 Iteration: 2000 / 4000 [50%] (Warmup)
 Chain 4 Iteration: 2001 / 4000 [50%] (Sampling)
 Chain 2 Iteration: 2500 / 4000 [62%] (Sampling)
 Chain 1 Iteration: 2300 / 4000 [57%] (Sampling)
 Chain 3 Iteration: 2600 / 4000 [65%] (Sampling)
 Chain 4 Iteration: 2100 / 4000 [52%] (Sampling)
 Chain 2 Iteration: 2600 / 4000 [65%] (Sampling)
 Chain 1 Iteration: 2400 / 4000 [60%] (Sampling)
 Chain 1 Iteration: 2500 / 4000 [62%] (Sampling)
 Chain 2 Iteration: 2700 / 4000 [67%] (Sampling)
 Chain 3 Iteration: 2700 / 4000 [67%] (Sampling)
 Chain 4 Iteration: 2200 / 4000 [55%] (Sampling)
 Chain 2 Iteration: 2800 / 4000 [70%] (Sampling)
 Chain 1 Iteration: 2600 / 4000 [65%] (Sampling)
 Chain 2 Iteration: 2900 / 4000 [72%] (Sampling)
 Chain 3 Iteration: 2800 / 4000 [70%] (Sampling)
 Chain 4 Iteration: 2300 / 4000 [57%] (Sampling)
 Chain 1 Iteration: 2700 / 4000 [67%] (Sampling)
 Chain 2 Iteration: 3000 / 4000 [75%] (Sampling)
 Chain 3 Iteration: 2900 / 4000 [72%] (Sampling)
 Chain 2 Iteration: 3100 / 4000 [77%] (Sampling)
 Chain 4 Iteration: 2400 / 4000 [60%] (Sampling)
 Chain 1 Iteration: 2800 / 4000 [70%] (Sampling)
 Chain 2 Iteration: 3200 / 4000 [80%] (Sampling)
 Chain 3 Iteration: 3000 / 4000 [75%] (Sampling)
 Chain 2 Iteration: 3300 / 4000 [82%] (Sampling)
 Chain 2 Iteration: 3400 / 4000 [85%] (Sampling)

Chain 2 Iteration: 3500 / 4000 [87%] (Sampling)
 Chain 1 Iteration: 2900 / 4000 [72%] (Sampling)
 Chain 4 Iteration: 2500 / 4000 [62%] (Sampling)
 Chain 2 Iteration: 3600 / 4000 [90%] (Sampling)
 Chain 3 Iteration: 3100 / 4000 [77%] (Sampling)
 Chain 2 Iteration: 3700 / 4000 [92%] (Sampling)
 Chain 2 Iteration: 3800 / 4000 [95%] (Sampling)
 Chain 2 Iteration: 3900 / 4000 [97%] (Sampling)
 Chain 3 Iteration: 3200 / 4000 [80%] (Sampling)
 Chain 1 Iteration: 3000 / 4000 [75%] (Sampling)
 Chain 4 Iteration: 2600 / 4000 [65%] (Sampling)
 Chain 2 Iteration: 4000 / 4000 [100%] (Sampling)
 Chain 2 finished in 83.8 seconds.
 Chain 3 Iteration: 3300 / 4000 [82%] (Sampling)
 Chain 1 Iteration: 3100 / 4000 [77%] (Sampling)
 Chain 1 Iteration: 3200 / 4000 [80%] (Sampling)
 Chain 3 Iteration: 3400 / 4000 [85%] (Sampling)
 Chain 4 Iteration: 2700 / 4000 [67%] (Sampling)
 Chain 1 Iteration: 3300 / 4000 [82%] (Sampling)
 Chain 1 Iteration: 3400 / 4000 [85%] (Sampling)
 Chain 3 Iteration: 3500 / 4000 [87%] (Sampling)
 Chain 4 Iteration: 2800 / 4000 [70%] (Sampling)
 Chain 1 Iteration: 3500 / 4000 [87%] (Sampling)
 Chain 1 Iteration: 3600 / 4000 [90%] (Sampling)
 Chain 3 Iteration: 3600 / 4000 [90%] (Sampling)
 Chain 4 Iteration: 2900 / 4000 [72%] (Sampling)
 Chain 1 Iteration: 3700 / 4000 [92%] (Sampling)
 Chain 3 Iteration: 3700 / 4000 [92%] (Sampling)
 Chain 1 Iteration: 3800 / 4000 [95%] (Sampling)
 Chain 1 Iteration: 3900 / 4000 [97%] (Sampling)
 Chain 4 Iteration: 3000 / 4000 [75%] (Sampling)
 Chain 3 Iteration: 3800 / 4000 [95%] (Sampling)
 Chain 1 Iteration: 4000 / 4000 [100%] (Sampling)
 Chain 1 finished in 110.2 seconds.
 Chain 3 Iteration: 3900 / 4000 [97%] (Sampling)
 Chain 3 Iteration: 4000 / 4000 [100%] (Sampling)
 Chain 3 finished in 111.2 seconds.
 Chain 4 Iteration: 3100 / 4000 [77%] (Sampling)
 Chain 4 Iteration: 3200 / 4000 [80%] (Sampling)
 Chain 4 Iteration: 3300 / 4000 [82%] (Sampling)
 Chain 4 Iteration: 3400 / 4000 [85%] (Sampling)
 Chain 4 Iteration: 3500 / 4000 [87%] (Sampling)
 Chain 4 Iteration: 3600 / 4000 [90%] (Sampling)

```
Chain 4 Iteration: 3700 / 4000 [ 92%] (Sampling)
Chain 4 Iteration: 3800 / 4000 [ 95%] (Sampling)
Chain 4 Iteration: 3900 / 4000 [ 97%] (Sampling)
Chain 4 Iteration: 4000 / 4000 [100%] (Sampling)
Chain 4 finished in 139.4 seconds.
```

```
All 4 chains finished successfully.
Mean chain execution time: 111.1 seconds.
Total execution time: 139.6 seconds.
```

```
precis(m_longstay, pars = c("b_tide", "b_moon"))
```

	mean	sd	5.5%	94.5%	rhat	ess_bulk
b_tide	0.07712872	0.06140261	-0.02162223	0.17610937	1.000606	6127.603
b_moon	-0.04901872	0.05617357	-0.13972459	0.04045971	1.001265	6439.563

```
# convert to odds ratios
post <- extract.samples(m_longstay)

mean(exp(post$b_tide))
```

```
[1] 1.082219
```

```
mean(exp(post$b_moon))
```

```
[1] 0.953666
```